Zigaton Consulting Ltd

Zigaton

Zigaton VectorSight UAS: Modular Tactical UAS for Precision Laser Rangefinding, Target Geolocation, and Real-Time Cueing

Submitted to:

Department of National Defence / Canadian Armed Forces

Challenge: True North Precision: Low Cost Drones with Laser Ranging

Solicitation: W7714-248676/012

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Submission Date	June 10, 2026

Confidentiality Notice

Submission Identifiers

Item	Information
Proposal Title	Zigaton VectorSight UAS: Modular Tactical UAS for Precision Laser Rangefinding, Target Geolocation, and Real-Time Cueing
Challenge Name	True North Precision: Low Cost Drones with Laser Ranging
Solicitation Number	W7714-248676/012
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Operating Name	Zigaton
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Acronyms and Definitions

Acronym / Term	Definition
ATAK	Android Tactical Assault Kit; a tactical situational-awareness platform used for mapping, coordination, and data sharing.
BOM	Bill of Materials.
C2	Command and Control.
CAF	Canadian Armed Forces.
CoT	Cursor-on-Target; a data format used for tactical position, event, and target information exchange.
DND	Department of National Defence.
EO Camera	Electro-optical camera.
EO	Essential Outcome, when used in the challenge compliance matrix.
EO/IR	Electro-optical / infrared sensor payload.
FEA	Finite Element Analysis.
FMV	Full-motion video.
GCS	Ground Control Station.
GNSS	Global Navigation Satellite System.
IMU	Inertial Measurement Unit.
ISR	Intelligence, Surveillance, and Reconnaissance.
MGRS	Military Grid Reference System.
PDB	Power Distribution Board.
RTK	Real-Time Kinematic positioning.
STANAG	NATO Standardization Agreement.
TRL	Technology Readiness Level.
UAS	Uncrewed Aerial System.
UAV	Uncrewed Aerial Vehicle.
VectorSight UAS	Zigaton's modular tactical UAS family proposed for precision laser rangefinding, target geolocation, and real-time cueing.

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1 Executive Summary

1.1 Proposed Capability

Zigaton Consulting Ltd proposes **Zigaton VectorSight UAS**, a modular tactical uncrewed aerial system designed to provide precision laser rangefinding, target geolocation, tactical intelligence, surveillance, and reconnaissance (ISR), and real-time target cueing for platoon- and company-level operations.

VectorSight UAS addresses the True North Precision challenge by closing the gap between low-cost commercial drones that provide imagery without reliable targeting data and higher-cost ISR platforms that are less scalable for distributed tactical use. The proposed system is not only an air vehicle; it is an integrated mission system combining the drone, stabilized sensor payload, Class 1 eye-safe laser rangefinder, navigation stack, onboard geolocation logic, communications links, ground-control interface, and target-cueing outputs.

The primary prototype configuration, **VectorSight-M**, is a tactical multirotor UAS designed for rapid deployment, precision range measurement, target coordinate generation, operator cueing, and field validation. The prototype is structured as the baseline demonstrator for this proposal, while the underlying payload, avionics, data, and ground-control architecture supports extension to heavier-lift and longer-endurance UAS configurations.

1.2 Operational Problem Addressed

Tactical units need timely, accurate, and actionable target information to support battlefield awareness, reconnaissance, and indirect fire-support workflows. Commercial small UAS platforms have improved access to aerial observation, but many systems do not provide integrated target range measurement, reliable target geolocation, low-latency target updates, or navigation resilience when GNSS performance is degraded.

Larger ISR platforms can provide advanced sensing and targeting functions, but they are typically more expensive, less available at lower tactical echelons, and harder to scale across distributed units. This creates a clear operational need for a low-cost, modular, rapidly deployable UAS that can provide precision rangefinding and target location outputs while remaining practical for field use, operator transport, and scalable procurement.

VectorSight UAS is designed to meet that need by combining practical tactical UAS design with stabilized sensing, laser rangefinding, target geolocation, resilient navigation, real-time data transfer, and operator-facing cueing outputs.

1.3 Primary Prototype Configuration

The proposed primary prototype is **VectorSight-M**, a tactical multirotor UAS configuration designed to remain within the challenge limit of ≤ 25 kg for the integrated drone and payload system. VectorSight-M is sized and architected around the payload, power, avionics, navigation, communications, and ground-control functions required for precision target cueing.

The prototype configuration includes:

- Tactical multirotor air vehicle with modular payload integration and field-maintainable design.
- EO/IR payload architecture integrated with a Class 1 eye-safe laser rangefinder.

- Stabilized payload interface for sensor pointing, boresight alignment, and vibration reduction.
- Onboard target geolocation using range measurement, aircraft position, aircraft attitude, payload pointing angle, and timestamped sensor data.
- Navigation architecture using GNSS with a defined path for RTK enhancement, inertial fallback, and visual odometry support.
- Real-time telemetry, video, and mission-data communication to the ground-control system.
- Ground-control interface for aircraft telemetry, payload information, target range, target coordinates, and operator cueing outputs.
- Structured logging for verification, test-data review, performance analysis, and final capability assessment support.

VectorSight-M is positioned as an integrated tactical UAS mission system with a defined development path, supplier-supported component ecosystem, verification plan, and scalable architecture for future operational variants.

1.4 Essential Outcome Compliance Summary

The VectorSight UAS architecture is requirements-driven and aligned to the essential outcomes of the True North Precision challenge. The proposed system is designed to support precision laser rangefinding, target geolocation, operational radius, endurance, low-latency target updates, GNSS-degraded operation, environmental robustness, safety, data handling, and integrated system weight control.

At the proposal level, VectorSight-M addresses the essential outcomes as follows:

- **Precision rangefinding:** Class 1 eye-safe laser rangefinder integrated with payload stabilization, boresight calibration, timestamping, and target-geolocation processing to support ±2 m range accuracy at 1 km against a vehicle-sized target.
- **Target geolocation:** fusion of measured range, aircraft position, aircraft attitude, payload pointing angle, and mission timestamp to generate tactical target-location outputs, including MGRS-compatible coordinate reporting.
- **Moving-target updates:** onboard processing and operator cueing workflow designed to support continuous updates for targets moving up to 10 m/s with low-latency transmission to the ground-control system.
- **Operational radius and endurance:** airframe, propulsion, battery, payload, and mission-profile sizing to support the required 3–4 km operational radius and minimum 30-minute endurance target.
- **GNSS-degraded resilience:** navigation architecture based on GNSS, inertial sensing, and planned enhancement paths for RTK, visual odometry, and degraded-mode operation.
- **Environmental robustness:** airframe, payload mounting, power system, and test plan designed around wind, temperature, obscurant, precipitation, vibration, and field-maintenance considerations.
- **Safety and data handling:** Class 1 eye-safe laser implementation, flight safety procedures, battery safety controls, mission logging, access control, and ITSP.10.171-aware data-handling processes.

Detailed compliance mapping is provided in Section 2 and Appendix A. Verification methods are expanded in Section 12 and Appendix G, including rangefinding accuracy testing, target geolocation validation, moving-target testing, GNSS degradation testing, endurance/radius testing, environmental testing, and safety verification.

1.5 Desired Outcome Differentiation Summary

Zigaton treats the desired outcomes as competitive differentiators, not optional add-ons. VectorSight UAS is architected to support a scalable tactical ISR and target-cueing capability beyond the minimum essential outcomes.

The VectorSight architecture includes:

- **Modular payload bay:** mechanical, electrical, and software interfaces supporting EO, EO/IR, laser rangefinding, and future payload variants.
- **Target-designation growth path:** payload bay, power, compute, gimbal, and software architecture designed to support future integration of advanced target-designation workflows where applicable.
- **Higher-precision rangefinding pathway:** boresight calibration, sensor stabilization, range-data validation, and geolocation error-budget control to support improved accuracy objectives.
- **Interoperability roadmap:** target-cueing data architecture designed for ATAK/Cursor-on-Target integration, structured target outputs, and STANAG-aligned interoperability concepts.
- **GNSS-resilient navigation:** architecture supporting RTK enhancement, inertial fallback, visual odometry pathway, and degraded-mode mission continuity.
- **Moving-target support:** timestamped measurement logic, update-rate control, and operator-facing target cueing for moving objects.
- **Supply-chain transparency:** country-of-origin tracking, alternate supplier planning, configuration control, and vendor-supported procurement readiness.
- **Scalable tactical UAS architecture:** a common mission-system approach that supports multiple UAS classes while keeping VectorSight-M focused, testable, and directly relevant to the challenge.

This approach positions VectorSight UAS as both a near-term prototype for the True North Precision challenge and a scalable Canadian tactical UAS capability relevant to ISR modernization, target acquisition, and distributed battlefield sensing.

1.6 Funding Request and Delivery Timeline

Zigaton Consulting Ltd proposes a phased development, integration, and demonstration program to deliver the VectorSight-M prototype and associated technical data package. The work plan covers final design, procurement, airframe and payload integration, bench testing, ground testing, flight testing, target-geolocation validation, system refinement, and final capability assessment readiness.

The proposed project will deliver:

- One integrated VectorSight-M prototype system.

- Integrated EO/IR and Class 1 eye-safe laser rangefinding payload architecture.
- Target geolocation and real-time cueing workflow.
- Ground-control configuration for telemetry, video, range, target-coordinate display, and mission logging.
- Technical data package including system architecture, preliminary BOM, wiring/interface definition, payload integration notes, and configuration records.
- Verification and validation test report covering rangefinding, geolocation, endurance, radius, navigation resilience, safety, and operational testing.
- Operator and maintenance documentation for prototype demonstration and evaluation.
- Final capability assessment readiness package.

The funding request, labour allocation, material and equipment budget, vendor and subcontractor costs, test costs, contingency, and cost-to-milestone mapping are provided in Section 15. The proposed schedule is structured around milestone deliverables and acceptance criteria, as detailed in Section 11.

2 Compliance Snapshot

This section provides a consolidated compliance snapshot for the True North Precision challenge. It summarizes how the proposed **Zigaton VectorSight UAS** addresses the Essential Outcomes (EO1–EO7), aligns with the Desired Outcomes (DO1–DO9), and maps key performance requirements to the verification methods used throughout this proposal.

The detailed technical approach is provided in Sections 5 through 9. The full verification and validation approach is provided in Section 12 and Appendix G.

2.1 Essential Outcomes EO1–EO7 Table

EO	Capability Area	VectorSight Compliance Approach	Verification Method
EO1	Precision rangefinding	VectorSight-M integrates a Class 1 eye-safe laser rangefinder with stabilized payload mounting, boresight calibration, timestamped measurements, and target-geolocation processing. The system is designed to support ±2 m range accuracy at 1 km to a vehicle-sized target, 10-figure MGRS-compatible output, continuous updates for targets moving up to 10 m/s, $\leq$ 1 s latency, and $\leq$ 2 m positional error at 1 km.	Datasheet review, boresight calibration, static range test, moving-target test, latency measurement, geolocation comparison to surveyed ground truth.
EO2	Platform performance	The primary prototype is configured as a tactical multirotor UAS designed around a 3–4 km operational radius under LOS or authorized BVLOS conditions and a minimum 30-minute endurance target under standard conditions. Propulsion, battery, payload, and power architecture are sized against mission profile, payload mass, and operating reserve requirements.	Weight budget, propulsion sizing analysis, hover test, endurance flight test, operational-radius test, power-log review.

EO	Capability Area	VectorSight Compliance Approach	Verification Method
EO3	Resilience and navigation	The navigation architecture combines GNSS, inertial sensing, onboard flight-control estimation, and enhancement paths for RTK correction, inertial fallback, and visual odometry. The objective is to maintain the rangefinding and target-geolocation workflow during GNSS degradation through degraded-mode navigation and sensor-fusion support.	GNSS degradation scenario test, inertial fallback evaluation, telemetry/log review, degraded-mode ground or flight test.
EO4	Environmental and operational robustness	The airframe, payload bay, mounting interfaces, enclosure approach, and test plan are structured for operation at 0 °C and above, sustained/gusting winds up to 10 m/s, and target-acquisition assessment in dust, smoke, or obscurant conditions.	Cold-operation check, wind flight test, payload stability review, obscurant target-acquisition test, environmental test report.
EO5	Data handling and safety	The rangefinder implementation is based on Class 1 eye-safe laser operation. Mission data, telemetry, video, logs, and test records are handled through controlled storage, access control, configuration tracking, and an ITSP.10.171-compliance-oriented data-handling process.	Laser classification evidence, safety procedure review, data-handling checklist, access-control review, logging and export test.
EO6	Size / integrated system weight	VectorSight-M is designed to remain below the 25 kg integrated drone-plus-payload limit through mass budgeting, component selection, payload-bay control, power-system sizing, and configuration management.	Mass properties review, component weigh-in, final integrated system weigh-in, configuration-control record.

EO	Capability Area	VectorSight Compliance Approach	Verification Method
EO7	Operational testing readiness	The project plan is structured to deliver a prototype ready for the May 2027 final capability assessment activity. Work packages include final design, procurement, integration, bench testing, ground testing, flight testing, refinement, documentation, and assessment readiness.	Milestone reviews, test-readiness review, flight-test records, final capability assessment readiness package.

2.2 Desired Outcomes DO1–DO9 Table

DO	Capability Area	VectorSight Alignment / Growth Path	Evidence or Verification Path
DO1	Target designation	The VectorSight architecture preserves a payload, power, compute, gimbal, and software pathway for future target-designation payload integration. The proposed baseline remains focused on Class 1 eye-safe rangefinding while maintaining growth space for advanced payload variants.	Payload bay design review, power/interface budget, vendor data, growth-path assessment.
DO2	Advanced rangefinding	The system uses stabilization, boresight calibration, timestamp synchronization, and geolocation error-budget control to support an advanced rangefinding pathway toward ± 1 m accuracy at 1 km on smaller target signatures.	Static range test, target-size sensitivity test, calibration report, geolocation error-budget review.
DO3	Enhanced environmental robustness	The design approach supports growth toward $-20\,^{\circ}$C operation, 15 m/s wind tolerance, precipitation exposure, and obscurant performance evaluation through airframe hardening, payload protection, mounting design, and environmental test planning.	Environmental growth test plan, enclosure review, payload protection review, field-test data.

DO	Capability Area	VectorSight Alignment / Growth Path	Evidence or Verification Path
DO4	Sensor-to-shooter integration	VectorSight is designed to output structured target cueing information through the ground-control workflow. The data architecture supports an ATAK/Cursor-on-Target integration pathway, STANAG 4586/4609 alignment path, FMV handling, and future fires-adjustment workflow support.	GCS integration test, target-message export test, FMV handling test, interoperability demonstration.
DO5	High-speed target tracking	The moving-target architecture is designed around timestamped measurements, update-rate control, predictive smoothing pathway, and operator-facing target cueing.	Moving-target test, update-rate measurement, latency measurement, track-continuity review.
DO6	Modular payload design	The platform is designed around a modular payload bay with defined mechanical, electrical, and data interfaces to support replacement or upgrade of sensors, batteries, and mission payloads without major structural redesign.	CAD/interface review, payload swap demonstration, connector/interface checklist, maintainability review.
DO7	Autonomous navigation	The system supports waypoint and semi-autonomous operation with a navigation growth path toward GNSS-denied operation using inertial fusion, visual odometry, and obstacle-awareness features.	Waypoint mission test, degraded-navigation assessment, visual-odometry integration review, flight-log analysis.
DO8	Origin transparency	Zigaton will maintain a country-of-origin matrix for critical systems including flight controller, data transmission devices, onboard computers, software elements, payload sensors, radios, and power electronics.	Vendor declarations, datasheets, procurement records, country-of-origin matrix.

DO	Capability Area	VectorSight Alignment / Growth Path	Evidence or Verification Path
DO9	EM resilience	The communications architecture includes EM-resilience considerations such as remote GCS transmitter extension, antenna placement planning, link-budget review, alternate data-link options, and assessment of non-EM control/data-transfer pathways where applicable.	Link-budget analysis, range test, antenna placement review, remote transmitter configuration test.

2.3 Weight, Endurance, Range, Latency, Safety, and Data-Handling Summary

Performance Area	VectorSight Proposal Position	Primary Evidence Location
Integrated system weight	Designed to remain below the 25 kg integrated drone-plus-payload limit through mass budgeting, configuration control, and final weigh-in verification.	Section 7; Appendix E
Endurance	Sized around the challenge target of at least 30 minutes under standard conditions, verified through flight testing and power-log review.	Section 7; Section 12
Operational radius	Designed around a 3–4 km operational radius under appropriate LOS or authorized BVLOS operating conditions.	Section 7; Section 8
Rangefinding accuracy	Designed to validate precision rangefinding at 1 km using known-distance targets and ground-truth comparison.	Section 6; Section 12; Appendix D
Target update latency	Designed to support target-update latency at or below the required threshold using timestamped measurements, onboard processing, and real-time telemetry.	Section 6; Section 9; Section 12
Moving-target support	Designed to support continuous target updates through stabilized sensing, update-rate control, predictive smoothing pathway, and operator-facing cueing outputs.	Section 6; Section 12

Performance Area	VectorSight Proposal Position	Primary Evidence Location
Laser safety	Based on Class 1 eye-safe laser rangefinder operation, supported by vendor documentation, safety procedures, and operational controls.	Section 6; Section 13
Data handling	Uses an ITSP.10.171-compliance-oriented approach to mission data, telemetry, logs, access control, encryption pathway, configuration records, and exportable test data.	Section 9; Section 13
Environmental operation	Test plan addresses cold operation, wind performance, obscurant target acquisition, precipitation growth testing, vibration, and field maintainability.	Section 7; Section 12; Section 13
Final assessment readiness	Schedule is structured to reach a test-ready prototype state for final capability assessment through phased design, procurement, integration, test, refinement, and documentation.	Section 11; Section 16

2.4 Key Verification Methods

Zigaton will use a requirements-based verification approach in which each challenge outcome is mapped to one or more verification methods. Compliance will be supported by measurements, logs, inspection records, calibration procedures, vendor evidence, configuration records, and test reports.

Verification Method	Purpose	Applies To
Datasheet and vendor evidence review	Confirms vendor-stated capability, laser class, sensor interfaces, environmental limits, communication interfaces, country-of-origin information, and integration constraints.	Payload, rangefinder, radios, flight controller, compute, power electronics
Engineering analysis	Validates expected system performance before prototype testing, including mass budget, endurance estimate, power budget, link budget, geolocation error budget, and thermal considerations.	Weight, endurance, radius, latency, geolocation, power, thermal

Verification Method	Purpose	Applies To
Bench testing	Verifies power rails, payload interfaces, sensor communication, logging, GCS connection, software configuration, and ground-control workflow before flight.	Power, avionics, payload, data handling, GCS
Ground testing	Validates rangefinder operation, boresight alignment, telemetry continuity, antenna configuration, operator workflow, and safety procedures under controlled non-flight conditions.	Rangefinding, payload, GCS, safety, communications
Static range test	Uses known-distance targets to evaluate range accuracy, repeatability, alignment, and target-geolocation processing.	EO1, DO2, Appendix D
Moving-target test	Uses controlled target movement to evaluate update continuity, latency, positional error, tracking behavior, and operator cueing outputs.	EO1, DO5
Flight testing	Verifies integrated performance in representative conditions, including endurance, radius, payload stability, target acquisition, operator workflow, and logging.	EO2, EO4, EO7
GNSS degradation test	Evaluates degraded-mode navigation behavior, range/geolocation continuity, inertial fallback, and navigation-resilience growth path.	EO3, DO7
Environmental and operational testing	Evaluates performance under wind, temperature, precipitation, dust/smoke/obscurant, vibration, and field-maintenance scenarios.	EO4, DO3
Safety verification	Confirms laser safety documentation, flight safety procedures, battery safety controls, emergency procedures, and operational risk mitigations.	EO5, Section 13
Configuration audit	Verifies that BOM, firmware, software, calibration files, payload configuration, test logs, and supplier records are controlled throughout the project.	EO6, EO7, DO8, Section 13

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EO1	Precision rangefinding	VectorSight-M integrates a Class 1 eye-safe laser rangefinder with stabilized payload mounting, boresight calibration, timestamped measurements, and target-geolocation processing. The system is designed to support ±2 m range accuracy at 1 km to a vehicle-sized target, 10-figure MGRS-compatible output, continuous updates for targets moving up to 10 m/s, ≤ 1 s latency, and ≤ 2 m positional error at 1 km.	Datasheet review, boresight calibration, static range test, moving-target test, latency measurement, geolocation comparison to surveyed ground truth.
EO2	Platform performance	The primary prototype is configured as a tactical multirotor UAS designed around a 3–4 km operational radius under LOS or authorized BVLOS conditions and a minimum 30-minute endurance target under standard conditions. Propulsion, battery, payload, and power architecture are sized against mission profile, payload mass, and operating reserve requirements.	Weight budget, propulsion sizing analysis, hover test, endurance flight test, operational-radius test, power-log review.

EO	Capability Area	VectorSight Compliance Approach	Verification Method
EO3	Resilience and navigation	The navigation architecture combines GNSS, inertial sensing, onboard flight-control estimation, and enhancement paths for RTK correction, inertial fallback, and visual odometry. The objective is to maintain the rangefinding and target-geolocation workflow during GNSS degradation through degraded-mode navigation and sensor-fusion support.	GNSS degradation scenario test, inertial fallback evaluation, telemetry/log review, degraded-mode ground or flight test.
EO4	Environmental and operational robustness	The airframe, payload bay, mounting interfaces, enclosure approach, and test plan are structured for operation at 0 °C and above, sustained/gusting winds up to 10 m/s, and target-acquisition assessment in dust, smoke, or obscurant conditions.	Cold-operation check, wind flight test, payload stability review, obscurant target-acquisition test, environmental test report.
EO5	Data handling and safety	The rangefinder implementation is based on Class 1 eye-safe laser operation. Mission data, telemetry, video, logs, and test records are handled through controlled storage, access control, configuration tracking, and an ITSP.10.171-compliance-oriented data-handling process.	Laser classification evidence, safety procedure review, data-handling checklist, access-control review, logging and export test.
EO6	Size / integrated system weight	VectorSight-M is designed to remain below the 25 kg integrated drone-plus-payload limit through mass budgeting, component selection, payload-bay control, power-system sizing, and configuration management.	Mass properties review, component weigh-in, final integrated system weigh-in, configuration-control record.

EO	Capability Area	VectorSight Compliance Approach	Verification Method
EO7	Operational testing readiness	The project plan is structured to deliver a prototype ready for the May 2027 final capability assessment activity. Work packages include final design, procurement, integration, bench testing, ground testing, flight testing, refinement, documentation, and assessment readiness.	Milestone reviews, test-readiness review, flight-test records, final capability assessment readiness package.

3.2 Desired Outcomes DO1–DO9 Table

DO	Capability Area	VectorSight Alignment / Growth Path	Evidence or Verification Path
DO1	Target designation	The VectorSight architecture preserves a payload, power, compute, gimbal, and software pathway for future target-designation payload integration. The proposed baseline remains focused on Class 1 eye-safe rangefinding while maintaining growth space for advanced payload variants.	Payload bay design review, power/interface budget, vendor data, growth-path assessment.
DO2	Advanced rangefinding	The system uses stabilization, boresight calibration, timestamp synchronization, and geolocation error-budget control to support an advanced rangefinding pathway toward ± 1 m accuracy at 1 km on smaller target signatures.	Static range test, target-size sensitivity test, calibration report, geolocation error-budget review.
DO3	Enhanced environmental robustness	The design approach supports growth toward $-20\,^{\circ}$C operation, 15 m/s wind tolerance, precipitation exposure, and obscurant performance evaluation through airframe hardening, payload protection, mounting design, and environmental test planning.	Environmental growth test plan, enclosure review, payload protection review, field-test data.

DO	Capability Area	VectorSight Alignment / Growth Path	Evidence or Verification Path
DO4	Sensor-to-shooter integration	VectorSight is designed to output structured target cueing information through the ground-control workflow. The data architecture supports an ATAK/Cursor-on-Target integration pathway, STANAG 4586/4609 alignment path, FMV handling, and future fires-adjustment workflow support.	GCS integration test, target-message export test, FMV handling test, interoperability demonstration.
DO5	High-speed target tracking	The moving-target architecture is designed around timestamped measurements, update-rate control, predictive smoothing pathway, and operator-facing target cueing.	Moving-target test, update-rate measurement, latency measurement, track-continuity review.
DO6	Modular payload design	The platform is designed around a modular payload bay with defined mechanical, electrical, and data interfaces to support replacement or upgrade of sensors, batteries, and mission payloads without major structural redesign.	CAD/interface review, payload swap demonstration, connector/interface checklist, maintainability review.
DO7	Autonomous navigation	The system supports waypoint and semi-autonomous operation with a navigation growth path toward GNSS-denied operation using inertial fusion, visual odometry, and obstacle-awareness features.	Waypoint mission test, degraded-navigation assessment, visual-odometry integration review, flight-log analysis.
DO8	Origin transparency	Zigaton will maintain a country-of-origin matrix for critical systems including flight controller, data transmission devices, onboard computers, software elements, payload sensors, radios, and power electronics.	Vendor declarations, datasheets, procurement records, country-of-origin matrix.

DO	Capability Area	VectorSight Alignment / Growth Path	Evidence or Verification Path
DO9	EM resilience	The communications architecture includes EM-resilience considerations such as remote GCS transmitter extension, antenna placement planning, link-budget review, alternate data-link options, and assessment of non-EM control/data-transfer pathways where applicable.	Link-budget analysis, range test, antenna placement review, remote transmitter configuration test.

3.3 Weight, Endurance, Range, Latency, Safety, and Data-Handling Summary

Performance Area	VectorSight Proposal Position	Primary Evidence Location
Integrated system weight	Designed to remain below the 25 kg integrated drone-plus-payload limit through mass budgeting, configuration control, and final weigh-in verification.	Section 7; Appendix E
Endurance	Sized around the challenge target of at least 30 minutes under standard conditions, verified through flight testing and power-log review.	Section 7; Section 12
Operational radius	Designed around a 3–4 km operational radius under appropriate LOS or authorized BVLOS operating conditions.	Section 7; Section 8
Rangefinding accuracy	Designed to validate precision rangefinding at 1 km using known-distance targets and ground-truth comparison.	Section 6; Section 12; Appendix D
Target update latency	Designed to support target-update latency at or below the required threshold using timestamped measurements, onboard processing, and real-time telemetry.	Section 6; Section 9; Section 12
Moving-target support	Designed to support continuous target updates through stabilized sensing, update-rate control, predictive smoothing pathway, and operator-facing cueing outputs.	Section 6; Section 12

Performance Area	VectorSight Proposal Position	Primary Evidence Location
Laser safety	Based on Class 1 eye-safe laser rangefinder operation, supported by vendor documentation, safety procedures, and operational controls.	Section 6; Section 13
Data handling	Uses an ITSP.10.171-compliance-oriented approach to mission data, telemetry, logs, access control, encryption pathway, configuration records, and exportable test data.	Section 9; Section 13
Environmental operation	Test plan addresses cold operation, wind performance, obscurant target acquisition, precipitation growth testing, vibration, and field maintainability.	Section 7; Section 12; Section 13
Final assessment readiness	Schedule is structured to reach a test-ready prototype state for final capability assessment through phased design, procurement, integration, test, refinement, and documentation.	Section 11; Section 16

3.4 Key Verification Methods

Zigaton will use a requirements-based verification approach in which each challenge outcome is mapped to one or more verification methods. Compliance will be supported by measurements, logs, inspection records, calibration procedures, vendor evidence, configuration records, and test reports.

Verification Method	Purpose	Applies To
Datasheet and vendor evidence review	Confirms vendor-stated capability, laser class, sensor interfaces, environmental limits, communication interfaces, country-of-origin information, and integration constraints.	Payload, rangefinder, radios, flight controller, compute, power electronics
Engineering analysis	Validates expected system performance before prototype testing, including mass budget, endurance estimate, power budget, link budget, geolocation error budget, and thermal considerations.	Weight, endurance, radius, latency, geolocation, power, thermal

Verification Method	Purpose	Applies To
Bench testing	Verifies power rails, payload interfaces, sensor communication, logging, GCS connection, software configuration, and ground-control workflow before flight.	Power, avionics, payload, data handling, GCS
Ground testing	Validates rangefinder operation, boresight alignment, telemetry continuity, antenna configuration, operator workflow, and safety procedures under controlled non-flight conditions.	Rangefinding, payload, GCS, safety, communications
Static range test	Uses known-distance targets to evaluate range accuracy, repeatability, alignment, and target-geolocation processing.	EO1, DO2, Appendix D
Moving-target test	Uses controlled target movement to evaluate update continuity, latency, positional error, tracking behavior, and operator cueing outputs.	EO1, DO5
Flight testing	Verifies integrated performance in representative conditions, including endurance, radius, payload stability, target acquisition, operator workflow, and logging.	EO2, EO4, EO7
GNSS degradation test	Evaluates degraded-mode navigation behavior, range/geolocation continuity, inertial fallback, and navigation-resilience growth path.	EO3, DO7
Environmental and operational testing	Evaluates performance under wind, temperature, precipitation, dust/smoke/obscurant, vibration, and field-maintenance scenarios.	EO4, DO3
Safety verification	Confirms laser safety documentation, flight safety procedures, battery safety controls, emergency procedures, and operational risk mitigations.	EO5, Section 13
Configuration audit	Verifies that BOM, firmware, software, calibration files, payload configuration, test logs, and supplier records are controlled throughout the project.	EO6, EO7, DO8, Section 13

4 Proposed Solution

4.1 Solution Name

The proposed solution is **Zigaton VectorSight UAS**, a modular tactical uncrewed aerial system designed for precision laser rangefinding, target geolocation, tactical ISR, and real-time target cueing.

The name **VectorSight** reflects the system's core mission: converting aerial observation into range-based, directionally referenced, and coordinate-supported target information. The system is designed for tactical users who require more than video awareness. VectorSight integrates the air vehicle, stabilized payload, Class 1 eye-safe laser rangefinder, navigation stack, onboard processing, communications links, ground-control interface, and mission-data outputs into one tactical mission system.

For this proposal, the primary deliverable is **VectorSight-M**, a tactical multirotor prototype optimized for the True North Precision challenge. The broader VectorSight architecture also supports future growth configurations, including heavier-lift and longer-endurance variants, without diluting the funded prototype focus.

The platform path is:

- **VectorSight-M:** Primary multirotor prototype for this proposal, optimized for tactical ISR, precision rangefinding, target geolocation, and field validation.
- **VectorSight-CX:** Coaxial/heavy-lift growth configuration for larger payloads, increased payload margin, and expanded mission equipment options.
- **VectorSight-VTOL:** Future VTOL/fixed-wing growth path for longer-radius and longer-endurance ISR missions.

4.2 Primary Prototype Configuration

VectorSight-M is proposed as a tactical multirotor UAS configuration designed to remain within the integrated drone-plus-payload weight limit of ≤ 25 kg while supporting the payload, avionics, navigation, communications, power, and ground-control functions required for precision target cueing.

The primary prototype is configured around the following major subsystems:

- **Air vehicle:** Tactical multirotor platform designed for vertical takeoff and landing, rapid deployment, modular payload integration, and field maintainability.
- **Payload architecture:** EO/IR payload option integrated with a Class 1 eye-safe laser rangefinder and stabilized sensor interface.
- **Target geolocation:** Onboard processing path that combines measured range, aircraft position, aircraft attitude, payload pointing angle, and timestamped sensor data.
- **Navigation architecture:** GNSS-based navigation with defined paths for RTK enhancement, inertial fallback, visual odometry, and degraded-mode operation.
- **Communications:** Real-time command/control, telemetry, video, mission-data transfer, and target-cueing output to the ground-control system.

- **Ground-control workflow:** Operator interface for aircraft status, payload status, target range, target-coordinate output, mission logging, and cueing information.
- **Verification data:** Structured logs and test records to support rangefinding validation, geolocation accuracy review, endurance testing, safety review, and final assessment readiness.

VectorSight-M is structured as a complete tactical mission-system prototype. The airframe provides mobility and payload carriage, while the core capability is created by the integration of sensing, rangefinding, geolocation, operator workflow, and verification-ready data handling.

4.3 System-Level Description

VectorSight UAS is designed as an integrated system-of-systems. The proposed capability depends on disciplined integration across mechanical, electrical, software, payload, communications, safety, and test domains.

At system level, the VectorSight-M prototype includes:

- A tactical multirotor air vehicle with payload, battery, propulsion, avionics, and enclosure integration.
- A stabilized EO/IR payload architecture with Class 1 eye-safe laser rangefinding.
- A power architecture supporting flight-critical systems, payload electronics, onboard compute, communications, and regulated subsystem rails.
- A navigation architecture using GNSS, inertial sensing, flight-controller estimation, and enhancement paths for degraded-navigation resilience.
- Onboard processing for range capture, aircraft pose association, payload pointing association, timestamping, target-coordinate generation, and mission logging.
- Communications links for command/control, telemetry, video, and target-cueing information.
- A ground-control system that presents live aircraft and payload information to the operator and supports cueing, data export, and test-data review.

Major interfaces will be controlled through configuration records, interface definitions, wiring documentation, payload integration notes, calibration records, and test procedures. This approach keeps the prototype testable and maintainable while creating a technical foundation for future variants.

4.4 Concept of Operations

The VectorSight concept of operations is built around rapid tactical deployment, target observation, precision range measurement, coordinate generation, and operator cueing.

A representative mission sequence is:

1. **Pre-mission preparation:** Operator checks aircraft status, battery state, payload configuration, laser safety controls, communications links, mission plan, and logging configuration.
2. **Launch and transit:** VectorSight-M launches vertically and transits to the observation area using operator-directed or waypoint-assisted flight.

3. **Target search and observation:** The operator uses the EO/IR payload architecture to observe the area, identify a target, and stabilize the payload view.

4. **Range measurement:** The Class 1 eye-safe laser rangefinder measures range to the selected target.

5. **Target geolocation:** The system associates measured range with aircraft position, aircraft attitude, payload pointing angle, and timestamped sensor data to generate target-location output.

6. **Cueing output:** The ground-control workflow displays target range, target coordinates, aircraft telemetry, payload status, and mission data to the operator.

7. **Moving-target update:** For moving targets, the system supports repeated range and target-update cycles with timestamped logs for review.

8. **Recovery and data review:** The aircraft returns, lands, and mission logs are reviewed for verification, reporting, and system refinement.

This concept of operations keeps the system focused on the challenge requirement: low-cost tactical UAS support for precision rangefinding, target geolocation, and real-time cueing.

4.5 Operator Workflow

The operator workflow is designed to reduce the gap between aerial observation and actionable target information. Instead of requiring the operator to manually interpret video, estimate range, and separately calculate target position, VectorSight presents target-relevant information through an integrated ground-control workflow.

The proposed operator workflow includes:

- Confirm aircraft readiness, payload readiness, communications status, battery state, mission configuration, and logging setup.
- Launch and navigate VectorSight-M to the mission area.
- Use the EO/IR payload architecture to observe terrain and locate the target.
- Stabilize the payload view and select or confirm the target of interest.
- Trigger or receive laser range measurement from the Class 1 eye-safe rangefinder.
- View target range, aircraft telemetry, payload status, and generated coordinate output.
- Maintain target updates where required, including moving-target cueing support.
- Export or review mission logs, target data, and verification records after the mission.

The workflow prioritizes clear operator information, timestamped measurement records, and repeatable test procedures so that system performance can be verified through controlled trials.

4.6 Key Differentiators

VectorSight UAS is differentiated by its mission-system architecture, modularity, and verification-driven design. The proposed system is not presented as a commercial drone with a sensor attached. It is presented as a tactical UAS architecture designed around integrated precision rangefinding, target geolocation, and operator cueing.

Key differentiators include:

- **Integrated target-cueing architecture:** Combines range measurement, aircraft pose, payload pointing, timestamping, and ground-control display into a structured target-cueing workflow.
- **Focused prototype with scalable architecture:** Uses VectorSight-M as the primary prototype while preserving a common architecture for heavier-lift and longer-endurance variants.
- **Class 1 eye-safe rangefinding baseline:** Supports safe prototype operation while preserving growth space for advanced payload variants.
- **Modular payload approach:** Uses defined mechanical, electrical, and data interfaces for current and future payload options.
- **GNSS-resilience growth path:** Includes GNSS, inertial sensing, and planned enhancement paths for RTK, visual odometry, and degraded-mode continuity.
- **Interoperability pathway:** Supports structured target outputs, ATAK/Cursor-on-Target integration pathway, FMV handling, and STANAG-aligned growth.
- **Verification-first execution:** Ties compliance to test methods, logs, calibration procedures, configuration records, and final capability assessment readiness.
- **Canadian defence capability growth:** Supports domestic tactical UAS integration capability, supply-chain transparency, vendor readiness, and scalable production considerations.

4.7 Defence Impact

VectorSight UAS is intended to improve the ability of tactical users to generate timely and actionable target information at lower echelons. By combining small UAS deployment with precision rangefinding and target geolocation, the system supports improved situational awareness, reconnaissance, target cueing, and mission-data collection.

The expected defence impact includes:

- Improved tactical ISR capability for platoon- and company-level operations.
- Faster transition from visual observation to target-location output.
- Reduced dependence on scarce or higher-cost ISR assets for local target cueing.
- Increased scalability through a modular, lower-cost UAS architecture.
- Improved verification confidence through structured testing, logging, and compliance mapping.

- Growth path toward interoperability, advanced payloads, resilient navigation, and future target-designation workflows.
- Contribution to Canadian defence industrial capability in tactical UAS integration, payload systems, and prototype-to-field-test execution.

VectorSight UAS therefore provides both immediate prototype value and strategic growth value. The immediate value is a field-testable tactical UAS prototype for precision rangefinding and target geolocation. The broader value is a scalable Canadian mission-system architecture that can support future tactical ISR, target acquisition, and distributed sensing requirements.

5 Technical Architecture

5.1 System Architecture Diagram

VectorSight UAS is designed as an integrated tactical mission system connecting the air vehicle, payload, navigation stack, mission compute layer, communications links, ground-control workflow, and verification data environment. The architecture is organized to support precision rangefinding, target geolocation, real-time cueing, operator awareness, and controlled test-data collection.

The system architecture is organized around five primary flows:

- **Command and control flow:** operator commands from the ground-control system to the aircraft flight controller, mission computer, and payload control interfaces.
- **Payload sensing flow:** EO/IR imagery and laser range measurements from the stabilized payload to onboard processing, operator display, and logging.
- **Navigation and pose flow:** GNSS, inertial, flight-controller, and payload pointing data used to support aircraft state estimation and target geolocation.
- **Target cueing flow:** processed range, target-coordinate, timestamp, confidence, and cueing outputs displayed to the operator and logged for verification.
- **Power and health-monitoring flow:** battery, power distribution, current/voltage sensing, payload power, avionics power, and system-health telemetry.

VectorSight UAS Preliminary System Architecture

Air Vehicle	propulsion, battery, frame, landing gear, payload bay, avionics bay
Payload Layer	EO/IR sensor, Class 1 laser rangefinder, stabilized mount/gimbal, boresight references
Navigation Layer	GNSS, IMU, barometer, compass, flight-controller state estimate, RTK/visual-odometry growth path
Mission Compute	range association, pose association, timestamping, geolocation logic, cueing output, logging
Communications	command/control, telemetry, video, payload status, target-data link
Ground Control	live video, aircraft telemetry, payload status, target range, target coordinates, mission logs

Final Appendix C diagram will show subsystem boundaries, data paths, power paths, and verification logging interfaces.

Figure 1: Preliminary VectorSight UAS system architecture.

The final architecture diagram in Appendix C will show subsystem boundaries, command/control paths, telemetry paths, video paths, payload data paths, geolocation processing, power distribution, and test-data logging interfaces.

5.2 Air Vehicle Architecture

The primary prototype, **VectorSight-M**, is configured as a tactical multirotor UAS. The multirotor configuration is selected for vertical takeoff and landing, rapid deployment, hover capability, stable target observation, practical payload pointing, and controlled field-test execution.

The air vehicle architecture includes:

- **Frame structure:** modular airframe sized for payload, avionics, battery, propulsion, landing gear, and field-service access.
- **Propulsion system:** motor, propeller, ESC, and battery architecture sized against payload mass, endurance, operational radius, and reserve requirements.
- **Avionics bay:** protected installation area for flight controller, GNSS receiver, inertial sensors, telemetry hardware, payload interface electronics, and mission compute hardware.
- **Payload mounting zone:** controlled mechanical interface for EO/IR sensor, laser rangefinder, gimbal/stabilization hardware, vibration isolation, and boresight alignment.
- **Battery and power bay:** battery installation and power-distribution architecture designed for maintainability, cable management, service access, and configuration control.
- **Landing and handling features:** geometry supporting payload clearance, launch/recovery handling, transportability, ground stability, and field maintenance.

The air vehicle is the mobility and payload-carriage layer of the VectorSight mission system. Mechanical, electrical, and software interfaces are controlled to support repeatable integration, maintenance, test readiness, and future payload variants.

5.3 Payload Bay and Modular Integration

The payload bay is a central design feature of VectorSight UAS. It enables the baseline rangefinding mission while preserving integration space for future EO, EO/IR, laser rangefinding, target-designation, and mission-payload variants.

The payload bay design approach includes:

- **Mechanical interface control:** defined mounting points, payload envelope, mass limit, center-of-gravity control, access provisions, and alignment references.
- **Electrical interface control:** regulated power outputs, grounding approach, connector definition, current monitoring, protection features, and harness routing.
- **Data interface control:** payload communication link, sensor data path, timestamp association, mission-compute interface, and logging interface.
- **Vibration and alignment control:** isolation strategy, stabilized mounting, boresight reference surfaces, and calibration procedure.
- **Field maintainability:** payload access, replacement procedure, inspection points, wiring protection, configuration tracking, and spare-parts planning.

This modular payload approach supports the proposed prototype while allowing future payload substitutions without major redesign of the air vehicle or mission-data architecture.

5.4 EO/IR and Laser Rangefinding Payload

The VectorSight-M payload architecture integrates EO/IR observation capability with Class 1 eye-safe laser rangefinding. The payload is designed to support target search, operator confirmation, stabilized pointing, range measurement, and target-geolocation processing.

The EO/IR and laser rangefinding payload architecture includes:

- **EO/IR observation:** visual and thermal observation pathway for target search, operator confirmation, and mission awareness.
- **Class 1 eye-safe laser rangefinder:** range measurement capability selected for safe prototype testing and compliance with laser safety requirements.
- **Stabilized sensor interface:** gimbal or stabilized mounting approach to reduce pointing error, image disturbance, and range-measurement instability.
- **Boresight alignment:** calibrated relationship between camera line of sight, laser rangefinder axis, payload frame, and aircraft reference frame.
- **Payload health monitoring:** payload status, operating mode, communication status, measurement state, and operator feedback.

The payload architecture is designed to produce target-relevant data rather than imagery alone. EO/IR observation supports acquisition and confirmation, while the laser rangefinder provides measured distance for coordinate generation and cueing.

5.5 Target Geolocation Architecture

The target geolocation architecture converts observation, range, aircraft pose, and payload pointing data into target-location output. This function is central to the VectorSight mission system.

The target geolocation process uses the following inputs:

- Laser-measured range to the selected target.
- Aircraft GNSS position.
- Aircraft attitude from the flight controller and inertial sensors.
- Payload pointing angle or gimbal orientation.
- Timestamp associated with range measurement, aircraft pose, and payload pointing state.
- Calibration data for boresight alignment and payload-to-airframe reference relationship.

The mission compute layer associates these inputs to estimate target position and generate operator-facing target-location output. The architecture supports MGRS-compatible reporting, mission logging, and post-test comparison against surveyed ground-truth data.

Data Element	Purpose in Target Geolocation
Range measurement	Establishes distance from aircraft/payload line of sight to target.
Aircraft position	Provides the starting reference point for target-location calculation.
Aircraft attitude	Defines aircraft orientation at the measurement timestamp.
Payload pointing angle	Defines line-of-sight direction from the payload to the target.
Timestamp	Aligns range, pose, and pointing data to reduce synchronization error.
Boresight calibration	Corrects physical alignment offsets between sensor axes and aircraft reference frame.

The geolocation architecture will be verified through known-distance range tests, surveyed target locations, static target trials, moving-target trials, timestamp review, and error-budget analysis.

5.6 Navigation and GNSS-Degraded Resilience

VectorSight-M uses a navigation architecture designed around GNSS-based operation with enhancement paths for degraded-navigation resilience. The objective is to preserve mission safety and maintain the rangefinding/geolocation workflow when GNSS quality is reduced or uncertain.

The navigation architecture includes:

- **GNSS navigation:** baseline aircraft positioning for mission navigation, aircraft state estimation, and target-geolocation reference.
- **RTK enhancement path:** higher-accuracy GNSS correction pathway for improved position accuracy where operationally available.
- **Inertial sensing:** IMU-based attitude and motion estimation for aircraft control, pose estimation, and short-duration fallback support.
- **Flight-controller estimation:** onboard state estimation combining GNSS, IMU, barometric, compass, and flight-control data.
- **Visual odometry pathway:** camera-based motion-estimation growth path for GNSS-degraded navigation support.
- **Degraded-mode operation:** operator and software workflow to preserve mission safety and range/geolocation continuity under reduced navigation confidence.

The navigation design will be evaluated through flight logs, GNSS quality monitoring, controlled degradation scenarios, inertial fallback review, and target-geolocation comparison under varying navigation conditions.

5.7 Compute and Software Architecture

The compute and software architecture supports flight control, payload interface, target-geolocation processing, communications, logging, configuration control, and operator workflow.

The architecture separates flight-critical control from mission processing. The flight controller is responsible for aircraft stabilization, navigation, failsafe behavior, telemetry, and flight logging. The mission compute layer supports payload data handling, range measurement association, target-geolocation processing, target-cueing outputs, and mission-data logging.

The software architecture includes:

- **Flight-control software:** aircraft stabilization, navigation, telemetry, failsafe configuration, and flight logs.
- **Payload interface software:** communication with EO/IR payload, rangefinder, gimbal/stabilized mount, and payload health data.
- **Target-geolocation logic:** association of range, aircraft pose, payload pointing, timestamp, and calibration data.
- **Operator workflow logic:** presentation of target range, target coordinates, aircraft telemetry, payload status, cueing data, and mission state.
- **Logging and replay:** mission data recording for verification, post-test analysis, and final reporting.
- **Configuration control:** tracking of firmware, software versions, calibration files, payload configuration, and test configuration.

This separation reduces integration risk, supports maintainability, and allows the target-cueing function to evolve without destabilizing flight-critical systems.

5.8 Communications and Data Links

VectorSight-M requires communications links for command/control, telemetry, video, payload status, target cueing, and mission-data transfer. The communications architecture is designed to support the 3–4 km operational radius requirement under appropriate line-of-sight or authorized BVLOS operating conditions.

The communications architecture includes:

- **Command/control link:** operator command path for aircraft control, mission execution, failsafe behavior, and flight supervision.
- **Telemetry link:** aircraft status, position, attitude, battery state, payload state, link status, and mission state.
- **Video link:** EO/IR video feed for target search, confirmation, operator awareness, and recording.
- **Target-data link:** range, target coordinates, target update status, cueing messages, and mission-data outputs.
- **Ground-system link management:** antenna placement, link budget, remote transmitter extension option, and operator station layout.
- **EM-resilience considerations:** alternate link planning, antenna diversity growth path, remote GCS transmitter placement, and degraded-link procedures.

The communications system will be verified through bench link testing, ground range testing, flight-link monitoring, video continuity review, telemetry-log review, and operational-radius testing.

5.9 Ground Control System

The ground-control system is the operator-facing component of the VectorSight mission system. It provides aircraft control, payload awareness, mission monitoring, target cueing, and test-data access.

The ground-control system will support:

- Aircraft position, altitude, speed, attitude, battery state, flight mode, and link status.
- EO/IR video display for target search and observation.
- Payload status and rangefinder measurement status.
- Target range and target-coordinate output.
- MGRS-compatible target reporting pathway.
- Mission logging, test-data export, and post-test review.
- Growth path for ATAK/Cursor-on-Target integration and structured target-message output.

The ground-control workflow is designed to reduce operator workload by consolidating the information required for target cueing into a structured interface. The operator can observe the target, confirm range measurement, view target-location output, monitor aircraft status, and review mission logs without relying on separate manual calculations during the mission.

5.10 Power, Thermal, and Maintainability Architecture

The power architecture supports propulsion, flight-critical avionics, payload electronics, mission compute, communications, and regulated subsystem loads. The design separates high-current propulsion loads from regulated avionics and payload loads while maintaining current/voltage monitoring, protection, and configuration control.

The power architecture includes:

- **Main battery system:** primary energy source sized against mission endurance, payload load, operational radius, and reserve requirements.
- **Power distribution:** controlled distribution to propulsion, avionics, compute, payload, communications, and accessory subsystems.
- **Regulated rails:** regulated voltage outputs for payload electronics, flight-control peripherals, communications, mission compute, and auxiliary loads.
- **Current and voltage sensing:** monitoring for power-log review, endurance validation, diagnostics, and safety.
- **Protection features:** fusing, transient suppression, grounding strategy, connector control, strain relief, and fault isolation.
- **Thermal management:** layout and airflow approach for regulators, mission compute, payload electronics, batteries, and communication devices.

- **Maintainability:** accessible connectors, modular replacement, labeled harnesses, inspection points, spare-parts strategy, and configuration records.

Thermal and maintainability considerations are included from the start because payload integration, power regulation, compute load, and communications hardware can become limiting factors during field use. The system will be validated through bench power tests, current logging, thermal inspection, flight-load review, and maintenance procedure checks.

6 Precision Rangefinding and Target Geolocation

6.1 Rangefinding Method

VectorSight-M measures target range using an integrated Class 1 eye-safe laser rangefinder aligned with the stabilized EO/IR payload architecture. The range measurement is treated as a mission-critical input because it supports target geolocation, operator cueing, moving-target updates, and verification against known-distance test references.

The rangefinding operating sequence is:

1. The operator observes and confirms the target through the EO/IR payload architecture.
2. The payload line of sight is stabilized and associated with the aircraft reference frame.
3. The Class 1 laser rangefinder measures distance to the selected target.
4. The measured range is timestamped and associated with aircraft position, aircraft attitude, payload pointing angle, and payload calibration data.
5. The mission compute layer uses the associated data to generate target-location output and operator cueing.
6. The measured range, system state, payload state, and target-cueing output are logged for verification and post-test review.

The rangefinding architecture is designed to support the challenge requirement for ±2 m range accuracy at 1 km to a vehicle-sized target. The design also preserves a growth path toward higher-precision rangefinding objectives through payload stabilization, boresight calibration, timestamp control, sensor selection, and geolocation error-budget management.

Rangefinding performance will be verified using known-distance targets, repeated measurements, static target tests, target-size variation, moving-target tests, and comparison against surveyed or independently measured ground-truth references.

6.2 Class 1 Eye-Safe Laser Implementation

The baseline VectorSight-M rangefinding implementation uses a **Class 1 eye-safe laser rangefinder**. This supports safe prototype operation, controlled field testing, and compliance with the challenge laser safety requirement.

The Class 1 laser implementation approach includes:

- Vendor-supported laser rangefinder selection with documented Class 1 eye-safe classification.
- Integration of the rangefinder within the stabilized payload architecture.
- Defined electrical power, data, mounting, and thermal interfaces.
- Operator workflow controls for rangefinder status, measurement command, and measurement feedback.

- Laser safety procedures for bench testing, ground testing, flight testing, range testing, and final assessment preparation.
- Configuration records linking installed rangefinder model, serial number, vendor documentation, firmware/configuration state, and test results.

Laser safety will be treated as both a design requirement and a test-readiness requirement. Vendor documentation, safety procedures, pre-test checklists, operational controls, and final configuration records will be maintained as part of the technical data package.

6.3 Boresight Calibration

Boresight calibration ensures that the EO/IR sensor line of sight, laser rangefinder axis, payload frame, stabilized mount or gimbal frame, and aircraft reference frame are consistently related. Without controlled boresight calibration, range measurements may not align with the visual target reference, degrading target-geolocation accuracy.

The VectorSight boresight calibration approach includes:

- Mechanical alignment references between the EO/IR payload, rangefinder, stabilized mount, and aircraft frame.
- Calibration procedure to identify alignment offsets between the EO/IR line of sight and the rangefinder axis.
- Calibration record for payload-to-airframe reference relationships.
- Verification using known-distance and known-location targets.
- Re-check procedure after payload removal, transport, hard landing, maintenance activity, or configuration change.

Boresight calibration will be controlled through configuration records and test procedures. The calibration state used during each rangefinding or geolocation test will be recorded so that measurement results can be traced to the installed payload configuration.

6.4 Gimbal Stabilization

Payload stabilization supports target observation, range measurement, and geolocation consistency. The stabilized payload interface reduces pointing disturbance caused by aircraft motion, vibration, wind, and operator input.

The stabilization approach supports:

- Improved target observation through reduced image disturbance.
- More consistent laser rangefinder pointing during measurement.
- Improved association between payload pointing angle and aircraft reference frame.
- Reduced sensitivity to short-duration aircraft attitude changes during range capture.
- Repeatable boresight calibration and verification.

The stabilized payload architecture will be evaluated through ground testing, hover testing, payload video review, range-measurement repeatability checks, and flight-test data review. Payload stability is treated as a contributor to the overall geolocation error budget, not as an isolated feature.

6.5 Timestamp Synchronization

Timestamp synchronization is required because target geolocation depends on associating the correct range measurement with the correct aircraft position, aircraft attitude, payload pointing angle, and mission state. Misalignment between these data sources can increase target-location error, especially during aircraft motion, payload movement, or moving-target observation.

The VectorSight timestamping approach includes:

- Timestamping of laser range measurements.
- Association of range measurements with aircraft position and attitude data.
- Association of payload pointing state with the same measurement event.
- Logging of measurement time, aircraft state, payload state, and computed target output.
- Post-test review of timing consistency during static and moving-target tests.

Timestamp synchronization will be verified through bench testing, software logging checks, ground target trials, and flight-test log review. The objective is to ensure that range, aircraft pose, and payload pointing data are aligned closely enough to support reliable target-geolocation output and repeatable verification.

6.6 Coordinate Transformation

The coordinate transformation function converts range and pointing information into a target-location estimate. VectorSight-M uses measured range, aircraft position, aircraft attitude, payload pointing angle, and calibration data to generate operator-facing target-location output.

At proposal level, the coordinate transformation workflow is:

1. Determine aircraft position from the navigation solution.
2. Determine aircraft attitude from the flight controller and inertial sensors.
3. Determine payload line-of-sight direction from payload pointing angle and boresight calibration.
4. Associate measured range with the timestamped aircraft and payload state.
5. Project the measured line of sight from the aircraft reference position toward the selected target.
6. Convert the resulting target-location estimate into the operator reporting format.
7. Log input data, output data, and confidence/error indicators for verification.

The coordinate transformation implementation will be tested against surveyed ground targets and known-distance references. Verification will compare generated target-location output against independent ground truth and quantify the effects of range error, aircraft position error, attitude error, payload pointing error, boresight error, timestamp error, and target-definition error.

6.7 10-Figure MGRS Output

VectorSight-M is designed to support MGRS-compatible target reporting as part of the operator-facing cueing workflow. The purpose of this output pathway is to provide target-location information in a format relevant to tactical users and compatible with future interoperability pathways.

The proposed MGRS output workflow includes:

- Generation of target-location estimate from range, pose, pointing, and calibration data.
- Conversion of target-location estimate into an MGRS-compatible coordinate output.
- Display of target coordinate information in the ground-control workflow.
- Logging of coordinate output, measurement timestamp, source inputs, and system state.
- Export pathway for later integration with structured target-message formats.

The 10-figure MGRS output pathway will be verified by comparing generated coordinates against known target locations during controlled tests. Output formatting, coordinate consistency, and operator readability will be evaluated as part of ground-control workflow validation.

6.8 Geolocation Error Budget

The geolocation error budget identifies the primary contributors to target-location error and links them to mitigation and verification methods. VectorSight treats geolocation accuracy as a system-level outcome, not only as a rangefinder specification.

Primary error contributors include:

- **Range measurement error:** difference between measured and true target range.
- **Aircraft position error:** uncertainty in the aircraft navigation solution.
- **Aircraft attitude error:** uncertainty in roll, pitch, and yaw at the measurement timestamp.
- **Payload pointing error:** uncertainty in gimbal or stabilized mount pointing state.
- **Boresight alignment error:** residual misalignment between sensor axes and reference frames.
- **Timestamp error:** mismatch between range measurement time, aircraft pose time, and payload pointing time.
- **Target definition error:** uncertainty caused by target size, surface geometry, reflectivity, obscuration, or operator selection.
- **Environmental error:** wind, vibration, visibility, precipitation, dust, smoke, and other field conditions affecting sensing and pointing.

The preliminary error-budget control strategy is summarized below:

Error Contributor	Control Approach	Verification Method
Range error	Vendor-supported rangefinder selection, known-distance testing, repeated measurements.	Static range test, datasheet review.
Aircraft position error	GNSS quality monitoring, RTK growth path, flight-log review.	Ground truth comparison, GNSS log review.
Aircraft attitude error	Flight-controller calibration, IMU checks, stable flight profile during measurement.	Flight log review, hover/stability test.
Payload pointing error	Stabilized mount/gimbal control, payload pointing feedback, mechanical interface control.	Payload stability test, pointing repeatability review.
Boresight error	Calibration procedure, alignment references, post-maintenance re-check.	Boresight calibration report.
Timestamp error	Measurement timestamping, sensor-state association, log review.	Bench timing test, flight log review.
Target definition error	Operator confirmation, target-size test cases, repeated observations.	Static and moving-target trials.
Environmental error	Test condition logging, wind/visibility recording, vibration review.	Environmental test report.

A detailed geolocation error budget will be provided in Appendix D. The error budget will be refined during integration and testing using payload data, aircraft logs, calibration results, and field measurements.

6.9 Moving-Target Update Method

VectorSight-M is designed to support continuous target updates for moving targets through repeated measurement, timestamped data association, and operator-facing cueing updates. The moving-target update method supports the challenge requirement for targets moving up to 10 m/s and preserves a growth path toward higher-speed target tracking.

The moving-target update method includes:

- Operator or payload-assisted target observation through the EO/IR payload architecture.
- Repeated range measurements associated with timestamped aircraft and payload state.
- Update-rate control to maintain a useful stream of target cueing information.
- Target-location output updates based on latest range, pose, and pointing association.
- Logging of measurement sequence, target update timing, aircraft state, payload state, and output coordinates.
- Post-test review of update continuity, latency, positional consistency, and operator workflow.

The moving-target function will be verified using controlled target-motion trials. Test cases will assess update continuity, latency, range consistency, target-location variation, and operator readability of the cueing output. Results will be used to refine update-rate logic, payload stabilization settings, operator workflow, and geolocation error-budget assumptions.

7 Platform Performance

7.1 Weight Budget

VectorSight-M is designed to remain within the True North Precision integrated system weight limit of ≤ 25 kg for the complete drone-plus-payload system. The weight budget is treated as a controlled engineering requirement because mass directly affects endurance, operational radius, propulsion margin, payload stability, handling, transportability, and final assessment readiness.

The VectorSight-M weight budget is organized by major subsystem:

- **Airframe and structure:** frame, arms, fasteners, landing gear, payload bay, enclosures, and structural mounting features.
- **Propulsion system:** motors, propellers, ESCs, motor wiring, mounting hardware, and propulsion accessories.
- **Energy storage:** main flight battery, battery mounting hardware, retention system, and battery wiring.
- **Avionics and navigation:** flight controller, GNSS receiver, inertial sensors, telemetry hardware, antennas, and supporting electronics.
- **Mission payload:** EO/IR payload architecture, Class 1 eye-safe laser rangefinder, stabilized mount or gimbal, payload interface hardware, and payload enclosure elements.
- **Mission compute and communications:** onboard compute, video link, target-data link, command/control interfaces, and related cabling.
- **Power distribution and harnessing:** power distribution hardware, regulated rails, current/voltage sensing, connectors, fusing, transient protection, and wire harnesses.
- **Contingency mass:** reserve allocation for brackets, wiring growth, protection hardware, fasteners, environmental sealing, test instrumentation, and integration changes.

The mass budget will be refined through component selection, CAD integration, supplier datasheets, measured component weights, and final integrated system weigh-in. Mass properties will be controlled through configuration management so that any payload, battery, wiring, or structural change is reflected in the final weight record.

Subsystem	Weight-Control Approach	Verification Method
Airframe and structure	CAD-based mass tracking, structural simplification, controlled payload-bay design, material selection.	CAD mass estimate, component weigh-in.
Propulsion system	Motor/propeller/ESC sizing matched to payload mass, endurance, and thrust margin requirements.	Datasheet review, propulsion sizing, bench/hover test.
Energy storage	Battery selected against endurance, discharge capability, reserve margin, and integration constraints.	Datasheet review, battery weigh-in, flight power log.
Payload	Payload envelope, mounting, stabilization, and rangefinder integration controlled through interface design.	Payload weigh-in, integration inspection.
Avionics and compute	Flight-critical and mission electronics selected for capability, mass, power draw, and integration compatibility.	BOM review, component weigh-in.
Harnessing and power	Wire gauge, connectors, protection, and routing controlled to prevent unmanaged mass growth.	Harness inspection, final weigh-in.
Contingency	Reserved mass allocation for integration hardware, test instrumentation, and field-maintenance provisions.	Mass budget review, configuration audit.

7.2 Payload Capacity

Payload capacity is defined by lift capability, mass distribution, center-of-gravity control, power availability, vibration environment, data interface compatibility, thermal limits, landing clearance, and field maintainability. VectorSight-M is designed around a payload architecture that supports the baseline EO/IR and Class 1 laser rangefinding mission while preserving growth space for alternate payload configurations.

The payload capacity approach includes:

- **Payload mass allocation:** reserved mass envelope for EO/IR payload, rangefinder, stabilized mount, payload enclosure, and interface hardware.
- **Center-of-gravity control:** payload placement managed relative to battery, avionics, airframe geometry, and propulsion layout.
- **Mechanical interface margin:** mounting points designed to carry payload mass, vibration loads, and landing/handling loads.
- **Electrical power capacity:** regulated power outputs sized for payload electronics, rangefinder, stabilization hardware, and mission compute support.

- **Data interface capacity:** payload communication pathways defined for imagery, range data, payload state, and timestamped mission data.
- **Thermal and access provisions:** payload bay layout designed for heat rejection, inspection, connector access, and field replacement.

The payload design philosophy is to avoid a one-off payload installation. VectorSight-M uses a modular payload-bay concept that supports baseline evaluation and future payload substitution through controlled mechanical, electrical, and data interfaces.

7.3 Endurance Model

The VectorSight-M endurance model accounts for aircraft mass, propulsion efficiency, battery energy, payload power draw, mission profile, wind conditions, reserve energy, and operator workflow. The platform is designed around the challenge target of at least 30 minutes of endurance under standard operating conditions.

Endurance will be estimated and verified using the following process:

1. Establish the integrated mass estimate for airframe, propulsion, battery, payload, avionics, compute, communications, harnessing, and contingency.
2. Select motor, propeller, ESC, and battery combinations using thrust, current, voltage, efficiency, and thermal data.
3. Estimate hover power, transit power, payload power, avionics power, communications power, and mission-compute power.
4. Define the mission profile, including takeoff, transit, hover/observation, target ranging, moving-target update, return, landing, and reserve.
5. Compare predicted power use against usable battery energy and discharge constraints.
6. Validate the estimate through hover testing, mission-profile flight testing, current logging, battery usage review, and post-flight thermal inspection.

The endurance model will not rely on a single ideal hover estimate. It will account for payload operation, wind exposure, operational reserve, thermal limits, battery voltage sag, and field-use conditions. The final endurance claim will be supported by flight-test logs and power-system data.

7.4 3–4 km Operational Radius

VectorSight-M is designed around the required 3–4 km operational radius under appropriate line-of-sight or authorized BVLOS operating conditions. The operational radius requirement is treated as a combined air vehicle, energy, navigation, communications, and operator workflow requirement.

The radius performance approach includes:

- **Energy margin:** battery and propulsion sizing sufficient to support outbound transit, observation/ranging time, return transit, and reserve.

- **Communications margin:** command/control, telemetry, video, and target-data links assessed against range, antenna placement, terrain, interference, and operator station configuration.
- **Navigation confidence:** GNSS and inertial state estimation monitored during transit, observation, and return.
- **Mission timing:** flight plan structured to preserve endurance margin while allowing target observation and rangefinding.
- **Return and failsafe behavior:** procedures configured for link degradation, battery threshold, navigation uncertainty, and operator abort.

Operational radius will be verified progressively, starting with bench communications checks, ground range tests, short-range flight tests, expanded-radius flights, and final mission-profile validation. Flight logs, link-quality data, battery usage, operator notes, and mission-data logs will support the radius assessment.

7.5 Propulsion and Battery Sizing Logic

The propulsion and battery sizing logic is structured to provide lift margin, endurance, thermal control, controllability, payload stability, and operational reserve. The propulsion system will be selected using engineering analysis, vendor data, bench testing, and flight validation.

The sizing logic includes:

- **Total mass estimate:** integrated weight of the airframe, payload, battery, avionics, communications, power system, harnessing, and contingency.
- **Thrust margin:** motor/propeller configuration selected to provide control authority for wind response, maneuvering, and safe recovery.
- **Hover efficiency:** propeller diameter, motor KV, voltage, current, and operating point selected to improve efficiency at mission-relevant throttle settings.
- **Battery voltage and capacity:** selected to support required power, discharge rate, usable energy, voltage stability, and endurance target.
- **ESC capability:** current rating and thermal capability matched to motor demand, transient load, and operational reserve.
- **Thermal margin:** motors, ESCs, batteries, and power distribution reviewed for heat generation during hover, climb, wind hold, and return flight.
- **Flight reserve:** battery usage controlled to preserve reserve energy for return, landing, and contingency events.

The propulsion design will be verified through supplier datasheets, propulsion analysis, static thrust or bench data where available, hover tests, flight current logs, thermal inspection, and mission-profile validation. The final sizing table will be provided after the preliminary BOM and propulsion selection are locked.

7.6 Wind Performance Approach

VectorSight-M is designed to support operation in sustained or gusting wind conditions up to the challenge threshold of 10 m/s, subject to safe flight-test procedures, test-site conditions, and operational risk controls. Wind performance affects flight stability, payload pointing, endurance, target observation, range-measurement consistency, and return margin.

The wind performance approach includes:

- **Propulsion margin:** thrust and control authority sized to support wind hold, correction, and recovery.
- **Flight-control tuning:** controller settings adjusted and validated for payload mass, inertia, and wind response.
- **Payload stabilization:** stabilized payload interface used to reduce image disturbance and pointing error during wind exposure.
- **Mission planning:** route, altitude, loiter/hover time, and return path planned against wind direction and reserve energy.
- **Operator limits:** test cards and go/no-go criteria used to manage safe operation under wind conditions.
- **Data logging:** wind condition, aircraft attitude, power draw, payload stability, and target-measurement behavior logged during testing.

Wind performance will be verified progressively. Initial flights will establish safe handling and payload behavior in low-wind conditions. Later tests will evaluate controlled operation under increasing wind conditions, with attention to endurance reduction, control margin, payload pointing stability, and operator workload.

7.7 Field Maintainability

Field maintainability is included as a platform performance requirement because tactical UAS value depends on repeatable operation, rapid turnaround, inspectability, and controlled configuration. VectorSight-M is designed to support field inspection, modular replacement, and controlled maintenance without compromising safety or test traceability.

The maintainability approach includes:

- **Accessible subsystem layout:** battery, payload, avionics, power distribution, antennas, and connectors positioned for inspection and service access.
- **Modular replacement:** payload, propellers, landing gear, batteries, and selected electronics designed for replacement through defined procedures.
- **Labeled harnessing:** wiring and connectors labeled to support assembly, troubleshooting, inspection, and configuration control.
- **Inspection points:** structural joints, arms, fasteners, payload mounts, battery retention, propellers, connectors, and thermal areas identified for routine checks.
- **Spare parts planning:** critical spare parts identified for field operation and testing continuity.

- **Configuration records:** maintenance events, replaced components, firmware/software versions, payload calibration state, and test configuration recorded.

Maintainability will be verified through assembly review, pre-flight inspection, post-flight inspection, payload access checks, connector inspection, replacement demonstrations, and configuration audit. This ensures that VectorSight-M is supportable as a repeatable prototype system during evaluation and refinement.

8 Resilience, Autonomy, and EM Considerations

8.1 GNSS/RTK Architecture

VectorSight-M uses a layered navigation architecture with GNSS as the baseline positioning source, while preserving a defined path for RTK enhancement and degraded-navigation resilience. Because target geolocation depends on aircraft position, aircraft attitude, payload pointing, and timestamped range data, navigation accuracy and navigation confidence are treated as mission-system requirements.

The baseline navigation architecture includes:

- **GNSS receiver:** primary aircraft position source for mission navigation, target-geolocation reference, and flight-log correlation.
- **Flight-controller state estimation:** onboard estimation using GNSS, IMU, barometer, compass, and flight-control data.
- **RTK enhancement path:** correction pathway for improved position accuracy where operationally available and appropriate for the test environment.
- **Navigation quality monitoring:** review of GNSS fix type, satellite quality, position uncertainty, flight logs, and system status during test events.
- **Target-geolocation association:** timestamped linking of navigation state with range measurements and payload pointing data.

RTK is included to support higher-confidence geolocation testing and future operational variants. The proposal does not depend only on RTK availability; RTK is treated as one enhancement layer within a broader navigation-resilience architecture.

GNSS and RTK performance will be verified through bench configuration checks, ground positioning tests, flight logs, known-location comparison, and target-geolocation validation against surveyed ground references.

8.2 Inertial Fallback

Inertial fallback supports continuity of aircraft state estimation when GNSS quality is reduced, interrupted, or uncertain. VectorSight-M uses inertial sensing as part of the flight-control and state-estimation architecture, with the objective of preserving safe flight behavior and maintaining short-duration continuity for range/geolocation workflow.

The inertial fallback approach includes:

- **IMU-based attitude estimation:** roll, pitch, yaw-rate, acceleration, and motion-state inputs for flight control and pose association.
- **Short-duration navigation continuity:** inertial support during brief GNSS degradation or uncertainty events.
- **Pose association:** use of aircraft attitude and motion state at the range-measurement timestamp.

- **Confidence monitoring:** operator and system awareness of navigation quality, sensor consistency, and degraded-mode state.
- **Failsafe integration:** predefined actions for navigation uncertainty, link degradation, low battery, or operator abort.

Inertial fallback is not presented as a complete GNSS-denied navigation solution by itself. It is one layer in a resilience architecture that also includes mission planning, operator procedures, flight-control logic, visual odometry pathway, and controlled degraded-mode operation.

Verification will include IMU calibration checks, flight-log review, GNSS-quality monitoring, degraded-navigation scenario tests, and comparison of target-geolocation outputs under different navigation-confidence conditions.

8.3 Visual Odometry Pathway

Visual odometry is included as a growth path to improve navigation resilience in GNSS-degraded environments. The purpose is to provide an additional motion-estimation source that can support aircraft state awareness, drift monitoring, and future GNSS-denied or GNSS-degraded mission operation.

The visual odometry pathway may support:

- **Camera-based motion estimation:** onboard visual data used to estimate relative movement over terrain or features.
- **Navigation confidence improvement:** comparison or fusion with inertial and GNSS-based estimates.
- **Drift monitoring:** support for identifying divergence between estimated and observed motion.
- **Degraded-mode support:** improved continuity during partial GNSS degradation or local GNSS uncertainty.
- **Future autonomy growth:** foundation for improved positioning, obstacle awareness, and navigation without sole reliance on GNSS.

For the funded prototype, visual odometry is treated as an architecture and integration pathway unless selected for the final prototype configuration. The design preserves compute, payload, data, and software interface provisions so that visual odometry can be integrated without restructuring the air vehicle or mission system.

Verification of this pathway may include bench software evaluation, dataset review, camera mounting assessment, compute-load evaluation, and comparison of visual-motion estimates against flight logs where implemented.

8.4 Degraded-Mode Operation

Degraded-mode operation defines how VectorSight-M preserves mission safety and useful capability when navigation, communications, payload, or environmental conditions are reduced from nominal performance. Tactical systems must manage uncertainty without becoming unsafe or unusable.

The degraded-mode operation approach includes:

- **Navigation degradation:** monitor GNSS quality, position uncertainty, attitude consistency, and estimator status.
- **Link degradation:** monitor command/control quality, telemetry continuity, video quality, and target-data link status.
- **Payload degradation:** monitor rangefinder status, EO/IR payload state, gimbal/stabilization state, and measurement validity.
- **Energy degradation:** monitor battery voltage, current draw, reserve margin, and return-to-home threshold.
- **Operator decision support:** provide visible status cues and predefined procedures for continue, hold, return, land, or abort decisions.
- **Data integrity:** continue logging system state, degraded-mode triggers, operator actions, and recovery behavior for test review.

The degraded-mode workflow prevents a single subsystem issue from creating uncontrolled mission risk. It also supports final assessment readiness by defining procedures and logs for non-ideal operating cases.

Verification will include bench fault-injection where appropriate, communications-range testing, GNSS-quality scenario testing, low-battery threshold checks, payload-status checks, flight-log review, and operator procedure review.

8.5 Waypoint/Semi-Autonomous Operation

VectorSight-M supports waypoint and semi-autonomous operation for repeatable tactical ISR and rangefinding missions. The intent is to reduce operator workload during transit and observation while preserving operator control over target confirmation, range measurement, cueing, safety decisions, and mission abort.

Waypoint/semi-autonomous operation supports:

- **Planned transit:** navigation to defined observation points or mission areas.
- **Observation positioning:** controlled loiter, hold, or orbit behavior for target search and measurement.
- **Repeatable test profiles:** consistent flight patterns for endurance, radius, rangefinding, geolocation, and moving-target validation.
- **Operator-supervised cueing:** operator confirmation before or during target range measurement and cueing.
- **Failsafe behavior:** return, hold, land, or abort actions based on link, battery, navigation, or operator command conditions.

The semi-autonomous approach balances tactical usefulness with safety and verification control. The system remains operator-supervised, while automation supports repeatability, stability, and reduced workload.

Verification will include waypoint mission testing, hold/loiter evaluation, operator intervention checks, failsafe testing, flight-log review, and comparison of planned versus actual flight profiles.

8.6 Obstacle-Awareness Growth Path

Obstacle awareness is included as a growth path to improve mission safety, operator workload, and autonomous navigation capability. For the initial VectorSight-M prototype, the priority remains precision rangefinding, target geolocation, navigation resilience, and field validation. The architecture preserves growth space for obstacle-awareness sensors and software.

The obstacle-awareness growth path may include:

- **Forward or downward sensing:** range sensors, depth sensing, stereo vision, or other proximity-awareness payloads.
- **Terrain and structure awareness:** support for low-altitude operation, landing-area review, and obstacle-proximity warning.
- **Operator warning:** visual or telemetry alerts when obstacle-proximity thresholds are reached.
- **Autonomy integration:** future path for assisted avoidance, safer waypoint execution, and improved degraded-mode operation.
- **Payload-interface compatibility:** reserved compute, power, data, and mounting provisions for future sensor integration.

Obstacle awareness is treated as a growth capability rather than a primary compliance dependency. This keeps the prototype focused while demonstrating that the architecture can mature toward more autonomous and resilient operation.

Verification of the growth path will include interface review, sensor-placement assessment, compute/power margin review, and future test planning.

8.7 EM-Resilient GCS/Data-Link Options

VectorSight-M includes electromagnetic resilience considerations in the ground-control and data-link architecture. Tactical UAS operation depends on maintaining sufficient command/control, telemetry, video, and target-data communication under realistic field conditions. Link planning is therefore treated as an engineering requirement.

The EM-resilience approach includes:

- **Link-budget review:** assessment of range, antenna gain, transmitter power, receiver sensitivity, frequency band, terrain, and margin.
- **Antenna placement planning:** aircraft and ground antenna placement to reduce shadowing, interference, and self-noise.
- **Remote GCS transmitter option:** ability to separate the operator station from the transmitter/antenna location where useful for terrain, concealment, or link quality.
- **Alternate data-link planning:** evaluation of backup or alternate links for telemetry, video, or target-data transfer where appropriate.
- **Degraded-link procedures:** predefined operator actions and failsafe behavior for weak link, lost video, lost telemetry, or lost command/control events.

- **Data logging continuity:** onboard logging of mission state and target data to preserve verification records during temporary link degradation.
- **Non-EM growth path:** assessment pathway for tethered, fibre, or other non-RF data-transfer concepts where applicable to future variants or specialized test configurations.

The communications and GCS resilience approach will be verified through bench link tests, ground range tests, antenna placement review, flight-link monitoring, telemetry-log review, operational-radius testing, and degraded-link procedure checks.

9 Interoperability and Data Handling

9.1 Real-Time Telemetry

VectorSight-M uses real-time telemetry to provide the operator with aircraft status, navigation state, payload state, communications status, power-system status, and mission-relevant information. Telemetry is both an operational function and a verification function: it supports situational awareness during flight and provides data for post-test analysis.

The real-time telemetry architecture includes:

- **Aircraft state:** position, altitude, speed, attitude, heading, flight mode, and navigation quality.
- **Power state:** battery voltage, current draw, consumed capacity, reserve margin, and power-system alerts.
- **Payload state:** EO/IR payload status, rangefinder status, stabilized mount or gimbal state, and measurement readiness.
- **Communications state:** command/control link quality, telemetry link status, video link status, and target-data link health.
- **Mission state:** active mission mode, waypoint state, target observation state, range-measurement status, and cueing output status.
- **Test state:** logging status, test configuration, event markers, and verification data-capture status.

Telemetry data will be displayed through the ground-control workflow and recorded for verification. Logs will support endurance review, operational-radius validation, navigation-quality assessment, target-geolocation review, payload-status analysis, and final reporting.

9.2 Target Cueing Output

VectorSight-M converts target observation and range measurement into structured target cueing output. The purpose is to help the operator move from visual awareness to actionable target information.

The target cueing output includes:

- Target range measurement.
- Target-location estimate.
- MGRS-compatible coordinate output.
- Measurement timestamp.
- Aircraft position and attitude at the measurement event.
- Payload pointing state or line-of-sight reference.
- Target update status for repeated or moving-target measurements.
- System confidence or quality indicators where implemented.
- Mission log reference for post-test review.

The cueing workflow supports both live operator use and verification. During a mission, the operator can view target range, target-coordinate output, and aircraft/payload status. After the mission, logged data can be reviewed against known target locations, surveyed references, or test-trial ground truth.

The target cueing function will be verified through static target testing, moving-target testing, coordinate-output review, latency measurement, operator workflow review, and mission-log analysis.

9.3 ATAK/COT Integration Path

VectorSight-M includes a defined integration pathway for ATAK/Cursor-on-Target (CoT) style target-message output. The purpose is to support future interoperability with tactical situational-awareness workflows while keeping the primary prototype focused on precision rangefinding, target geolocation, and operator cueing.

The ATAK/CoT integration path includes:

- **Structured target data:** target coordinate, timestamp, target identifier, source platform, and update status organized for message export.
- **MGRS-compatible reporting:** coordinate output pathway aligned with tactical reporting use.
- **Target-event formatting:** conversion path for target cueing output into CoT-style event messages.
- **Ground-system interface:** export pathway from the ground-control workflow to an external tactical display or integration environment.
- **Test-data support:** comparison of exported target messages with internal mission logs and ground-truth records.

For the funded prototype, Zigaton will prioritize reliable internal target cueing, logged output, and an exportable target-data structure. ATAK/CoT integration will be implemented or demonstrated according to final project scope, integration access, test environment, and customer direction. The architecture is structured so that target data can be exported without redesigning the aircraft or payload system.

9.4 STANAG 4586/4609 Alignment Path

VectorSight UAS includes an interoperability growth path aligned with relevant NATO unmanned-system and full-motion-video concepts. The proposal does not claim completed STANAG certification at prototype stage. Instead, it establishes an architecture that can mature toward STANAG-aligned data handling, payload control, and video workflows.

The STANAG alignment path includes:

- **STANAG 4586 alignment path:** future unmanned-system control and payload interface alignment for command/control and payload-related data exchange.
- **STANAG 4609 alignment path:** future full-motion-video metadata and video-handling alignment where applicable.
- **Structured mission data:** consistent formatting of aircraft state, payload state, target cueing output, timestamps, and mission events.

- **Interface documentation:** definition of aircraft, payload, ground-control, telemetry, video, and target-data interfaces.
- **Version and configuration control:** tracking of software, firmware, payload configuration, message formats, and data-export settings.

The intent is to avoid a closed prototype architecture. VectorSight is designed so that target cueing, video, telemetry, and mission logs can evolve toward interoperable formats as operational integration requirements mature.

Verification will include interface review, message-format review, FMV handling review, data-export checks, and comparison of logged internal data with exported data products.

9.5 FMV Handling

Full-motion video (FMV) is central to the operator workflow because EO/IR imagery supports target search, target confirmation, payload pointing, mission awareness, and post-test review. VectorSight-M handles video as part of the mission system, not as a separate accessory feed.

The FMV handling approach includes:

- **Live video display:** EO/IR video presented to the operator through the ground-control workflow.
- **Payload status association:** video associated with payload state, gimbal or stabilized mount state, and rangefinder measurement state.
- **Event association:** target range events, coordinate outputs, and operator actions associated with mission timestamps.
- **Recording pathway:** video and mission logs recorded where appropriate for test review and final reporting.
- **Metadata growth path:** future alignment with structured FMV metadata workflows where required.
- **Data review:** post-test review of video, range measurements, target output, and aircraft logs for verification.

FMV handling will be verified through bench video checks, ground video-link testing, flight video review, target-observation trials, recording checks, and post-test data review.

9.6 ITSP.10.171 Data-Handling Controls

VectorSight-M uses an ITSP.10.171-compliance-oriented data-handling approach for mission data, telemetry, video, test records, configuration files, and technical documentation. The objective is to protect sensitive project and test information while maintaining usable records for engineering verification and government evaluation.

The data-handling approach includes:

- **Data classification awareness:** project data handled according to sensitivity, purpose, and distribution requirement.
- **Access control:** controlled access to mission logs, test results, configuration files, technical data, and proposal records.

- **Storage control:** organized storage of technical data package materials, test reports, vendor records, and mission logs.
- **Transmission control:** controlled transfer of mission data, test results, and deliverables to authorized recipients.
- **Configuration control:** traceability of BOM, firmware, software, payload configuration, calibration state, and test setup.
- **Auditability:** records showing generation, modification, review, or approval of important project data.
- **Data export control:** defined process for exporting logs, target data, video records, and technical reports.

This approach will be documented in the project data-handling procedure and verified through configuration review, access-control review, log-export testing, and deliverable audit.

9.7 Logging, Access Control, Encryption, and Exportable Test Data

Logging and data control are essential for evaluating VectorSight-M against the challenge outcomes. The system must not only perform during demonstration events; it must preserve the evidence required to show how performance was achieved.

The logging architecture is designed to capture:

- Aircraft flight logs.
- Telemetry records.
- Payload status records.
- Range measurement records.
- Target-coordinate output records.
- Timestamp and event markers.
- Operator actions where applicable.
- Power-system data.
- Communications-link status.
- Calibration and configuration references.

Access control will restrict project data, test data, configuration files, and deliverables to authorized project personnel and authorized evaluation personnel. Encryption will be applied or configured where appropriate to the system configuration, test environment, and customer direction.

Exportable test data will be organized so evaluators can review performance against stated requirements. The exported data package may include mission logs, range measurements, target-coordinate outputs, flight logs, payload logs, video records, calibration records, test cards, test reports, and configuration records.

The logging and data-control approach will be verified through bench logging checks, ground-test log review, flight-test log review, export tests, access-control checks, and final deliverable audit.

10 TRL, Prior Experience, and Readiness

10.1 Current TRL Justification

VectorSight-M is positioned as a prototype-oriented integration and validation effort, not as basic scientific research. The proposed system combines available UAS, payload, avionics, communications, navigation, compute, and power-system technologies into a mission-specific tactical UAS architecture for precision rangefinding, target geolocation, and real-time cueing.

The current readiness position is supported by the following factors:

- The primary air vehicle configuration is based on established multirotor UAS architecture.
- The payload architecture uses vendor-supported EO/IR, stabilization, and Class 1 eye-safe laser rangefinding pathways.
- The navigation architecture uses available GNSS, inertial sensing, flight-controller estimation, and RTK enhancement pathways.
- The mission compute layer is structured around defined data association, target-geolocation, logging, and cueing functions.
- The communications approach uses known command/control, telemetry, video, and target-data link categories.
- The power architecture is based on controlled power distribution, regulated rails, current/voltage sensing, protection, and configuration management.
- The verification plan maps each essential and desired outcome to measurable tests, logs, inspections, calibration records, and final reporting.

The main technical effort is disciplined integration and validation: bringing selected subsystems into a coherent tactical mission system, validating range and target-geolocation performance, demonstrating operational-radius and endurance capability, and producing the technical data package required for assessment and follow-on development.

10.2 Relevant UAV Build and Integration Experience

Zigaton's project team brings relevant capability in electrical systems, avionics integration, mechanical design, CAD, FEA, power-system design, technical documentation, vendor coordination, and prototype execution. These capabilities directly support VectorSight-M as an integrated UAS mission system.

Relevant experience areas include:

- **Electrical and avionics integration:** power architecture, regulated rails, current sensing, protection, flight-controller interfacing, payload electronics, telemetry interfaces, and wiring/harness planning.
- **Mechanical design and structures:** CAD modelling, payload-bay design, mounting interfaces, structural review, vibration-aware integration, and maintainability features.
- **Systems engineering:** requirements interpretation, subsystem decomposition, interface control, verification planning, risk tracking, and technical documentation.

- **Prototype development:** component selection, supplier review, bench testing, configuration tracking, integration troubleshooting, and test-readiness preparation.
- **Defence-relevant engineering exposure:** industry experience, structured engineering documentation, technical QA/QC discipline, and awareness of safety-critical design practices.

The VectorSight-M development plan builds on these capabilities through a requirements-driven execution model: define the mission need, select and integrate appropriate subsystems, control interfaces, verify subsystem performance, validate integrated performance, document results, and prepare for final capability assessment.

10.3 Existing Payload Integration Capability

Payload integration is one of the central technical workstreams for VectorSight-M. Zigaton treats the payload as an integrated mission subsystem, not as an accessory mounted after airframe selection.

The payload integration capability includes:

- **Payload-bay definition:** mechanical envelope, mass allocation, center-of-gravity control, mounting points, landing clearance, and field access.
- **Sensor integration:** EO/IR observation pathway, Class 1 eye-safe laser rangefinder, stabilized mount or gimbal, payload health monitoring, and payload state reporting.
- **Electrical integration:** regulated payload power, connector definition, grounding approach, current monitoring, harness routing, and protection strategy.
- **Data integration:** sensor communication, range measurement capture, timestamp association, payload pointing state, mission compute interface, and logging.
- **Calibration and verification:** boresight calibration, range verification, target-location comparison, payload stability review, and configuration records.

This capability directly supports precision rangefinding and target geolocation because payload performance depends on more than the rangefinder specification. It depends on stabilized pointing, boresight alignment, data synchronization, aircraft pose association, mission compute logic, and operator workflow.

10.4 Vendor Engagement Status

The VectorSight-M proposal is based on a vendor-supported integration pathway across the major subsystem categories required for the prototype. Vendor engagement is treated as a program-readiness requirement because subsystem availability, documentation, lead time, country of origin, integration support, and alternate sourcing directly affect delivery confidence.

Vendor engagement and supplier readiness will be managed across the following categories:

- Airframe and structural components.
- Motors, propellers, ESCs, and propulsion accessories.
- Flight batteries, chargers, and battery management accessories.

- Flight controller, GNSS/RTK hardware, telemetry radios, and antennas.
- EO/IR payload, Class 1 eye-safe laser rangefinder, gimbal or stabilized mount, and payload interfaces.
- Mission compute hardware, video link, ground-control equipment, and target-data link equipment.
- Power distribution hardware, regulated power modules, current/voltage sensing, connectors, fuses, transient protection, and harness materials.
- Fabrication, machining, assembly, test support, and specialized integration services where required.

Zigaton will maintain vendor records including datasheets, budgetary pricing, lead-time assumptions, country-of-origin information, interface requirements, alternate supplier options, and integration notes. These records will support the preliminary BOM, country-of-origin matrix, risk register, procurement plan, and cost justification.

10.5 Prototype Readiness

VectorSight-M is structured for prototype execution through a phased development model. The system architecture, compliance mapping, technical workstreams, verification approach, and deliverable structure are organized around the True North Precision challenge outcomes.

Prototype readiness is supported by:

- A defined primary prototype configuration: VectorSight-M.
- A scalable VectorSight architecture that preserves future growth without distracting from the funded prototype.
- A compliance matrix mapping essential and desired outcomes to technical approach and verification method.
- A technical architecture covering air vehicle, payload, navigation, compute, communications, ground control, power, thermal, and maintainability.
- A target-geolocation architecture based on range, aircraft pose, payload pointing, timestamping, and boresight calibration.
- A verification plan covering rangefinding, geolocation, moving targets, endurance, operational radius, navigation degradation, environmental conditions, laser safety, and data handling.
- A supply-chain approach based on vendor documentation, alternate suppliers, lead-time control, and country-of-origin tracking.
- A configuration-control approach for BOM, firmware, software, payload setup, calibration records, test logs, and deliverables.

The prototype-readiness objective is to move from final design and procurement into integration, bench testing, ground testing, flight testing, refinement, and final capability assessment readiness with controlled technical risk.

10.6 Team Roles and Execution Capability

Zigaton's execution model assigns clear ownership across program coordination, electrical and avionics integration, mechanical design, logistics, vendor readiness, documentation, and prototype delivery. The core team will lead the prototype effort while using vendors, advisors, and subcontractors for specialized payload, manufacturing, compliance, or test support where required.

Name	Role	Project Responsibilities
John Oladunjoye Olatomiwa	Co-founder; Electrical, Avionics, and UAV Systems Integration Lead	Program coordination, proposal leadership, UAV systems integration oversight, electrical and avionics architecture, power distribution, flight-controller interfaces, payload electronics, vendor coordination, stakeholder communications, technical documentation, and delivery management.
Benjamin Apori	Co-founder; Mechanical Design and Logistics Lead	Mechanical design ownership, CAD/FEA, payload-bay design, airframe integration, mounting and enclosure design, logistics planning, supplier coordination, parts readiness, maintainability inputs, mechanical BOM support, and mechanical test readiness.

10.7 Key Personnel Summary

John Oladunjoye Olatomiwa — Co-founder / Electrical, Avionics, and UAV Systems Integration Lead

John Oladunjoye Olatomiwa serves as Co-founder and Electrical, Avionics, and UAV Systems Integration Lead for the VectorSight UAS project. He is responsible for system-level integration of the air vehicle, electrical power architecture, avionics stack, flight-controller interfaces, telemetry, payload electronics, mission compute interfaces, vendor coordination, requirements traceability, and verification documentation.

John Oladunjoye Olatomiwa has an Electrical Engineering background from Carleton University and brings engineering experience from Hydro One, independent UAV research, electronics design, power distribution design, technical documentation, and system-integration work. His technical strengths include electrical and electronic systems, avionics architecture, power systems, CAD-supported integration, propulsion and aerodynamic trade studies, PCB-level design thinking, and structured engineering verification.

For this project, John Oladunjoye Olatomiwa will lead electrical and avionics integration, power architecture definition, sensor and payload interface planning, communications integration, mission-system requirements traceability, technical data package development, and coordination of the prototype readiness workflow.

Benjamin Apori — Co-founder / Mechanical Design and Logistics Lead

Benjamin Apori serves as Co-founder and Mechanical Design and Logistics Lead for the VectorSight UAS project. He supports the mechanical architecture of the air vehicle, payload bay, structural layout, mounting interfaces, vibration-aware integration, manufacturability, and prototype assembly planning.

Benjamin Apori is pursuing engineering studies at Carleton University and has relevant industry experience through an internship with Raytheon. His technical strengths include finite element analysis, CAD modelling, structural design, mechanical integration, logistics coordination, and mechanical engineering competencies relevant to UAV airframe and payload integration.

For this project, Benjamin Apori will lead airframe mechanical design, payload-bay layout, CAD model development, FEA-supported structural review, gimbal and sensor mounting design, mechanical interface definition, supplier coordination, component readiness tracking, field maintainability planning, and mechanical subsystem procurement support.

10.8 Capability Gaps Covered by Vendors, Advisors, and Subcontractors

Zigaton will use external vendors, advisors, and subcontractors where specialized support strengthens delivery, reduces schedule risk, or improves final prototype quality. This allows the core team to retain system-level ownership while using specialized partners for subsystem-level support.

Potential external support areas include:

- EO/IR payload and laser rangefinder vendor support.
- Gimbal or stabilized payload integration support.
- Airframe fabrication, machining, or composite/metal manufacturing support.
- Flight-test site, flight operations, or RPAS operations support.
- Software integration support for target cueing, ATAK/CoT pathway, or mission-data export.
- Cybersecurity and data-handling review support.
- Environmental, vibration, or safety testing support.
- Documentation, quality, or configuration-management review support.

All external support will be managed through defined work packages, deliverables, interface requirements, configuration records, and vendor documentation. This ensures that VectorSight-M remains a Zigaton-led integrated mission system while benefiting from specialized technical support where appropriate.

11 Work Plan and Milestones

11.1 Project Duration

Zigaton proposes a phased development, integration, test, and refinement program for the VectorSight-M prototype. The schedule is structured to move from final design and procurement through subsystem integration, bench testing, ground testing, flight testing, accuracy validation, refinement, documentation, and final capability assessment readiness.

The work plan is organized around five major phases:

- **Phase 1: Final design and procurement.**
- **Phase 2: Airframe/payload integration.**
- **Phase 3: Bench and ground testing.**
- **Phase 4: Flight testing and accuracy validation.**
- **Phase 5: Refinement and final capability assessment readiness.**

The schedule is designed to preserve time for procurement, integration issues, test iteration, safety review, configuration control, documentation, and final reporting. Each phase produces measurable outputs that support the next stage of technical maturity and verification.

The project plan also supports participation in interim capability validation and refinement activities, where applicable, and is structured to reach final capability assessment readiness for the May 2027 assessment window.

11.2 Phase 1: Final Design and Procurement

Phase 1 establishes the final technical baseline for VectorSight-M and initiates procurement of critical and long-lead components. The objective is to freeze the prototype configuration sufficiently to begin procurement and integration while maintaining controlled flexibility for vendor-supported substitutions where required.

Phase 1 activities include:

- Finalize VectorSight-M system requirements and compliance mapping.
- Confirm airframe class, payload envelope, propulsion architecture, power architecture, communications architecture, and ground-control workflow.
- Finalize preliminary BOM for air vehicle, propulsion, avionics, payload, communications, power distribution, mission compute, ground equipment, spares, and test equipment.
- Confirm vendor datasheets, lead times, country-of-origin information, interface requirements, integration support, and alternate supplier options.
- Finalize system architecture, payload-bay interface, wiring/interface definition, data-handling approach, and configuration-control structure.
- Develop initial test plan, safety plan, calibration plan, flight-operations plan, and data-handling procedure.

- Identify test-site needs, flight authorization pathway, range safety considerations, and final assessment readiness constraints.
- Procure long-lead components and critical-path subsystems.

Phase 1 deliverables include:

- Finalized prototype requirements matrix.
- Updated essential and desired outcome compliance matrix.
- Preliminary BOM and procurement package.
- Vendor and country-of-origin records.
- System architecture and interface-control baseline.
- Draft verification and validation plan.
- Draft risk register, safety plan, and flight-test readiness framework.

11.3 Phase 2: Airframe/Payload Integration

Phase 2 focuses on physical, electrical, payload, and software integration of the VectorSight-M prototype. The objective is to assemble the air vehicle, integrate the payload bay, install avionics, connect power distribution, and prepare the system for controlled bench and ground testing.

Phase 2 activities include:

- Assemble or fabricate the airframe, landing gear, payload bay, battery mount, avionics bay, and enclosure elements.
- Install propulsion components including motors, propellers, ESCs, wiring, and mechanical mounts.
- Install flight controller, GNSS receiver, inertial sensors, telemetry hardware, mission compute hardware, antennas, and related avionics.
- Integrate the EO/IR payload architecture, Class 1 eye-safe laser rangefinder, stabilized mount or gimbal, and payload interface electronics.
- Install power distribution hardware, regulated rails, current/voltage sensing, connectors, fusing, transient protection, and harnessing.
- Complete cable routing, labeling, strain relief, grounding review, connector inspection, and configuration documentation.
- Establish preliminary boresight alignment and payload-to-airframe reference relationships.
- Record initial mass properties, center-of-gravity condition, payload configuration, and installed hardware configuration.

Phase 2 deliverables include:

- Integrated airframe and payload-bay assembly.

- Installed avionics, payload, communications, mission compute, and power architecture.
- Wiring/interface documentation.
- Initial mass properties and center-of-gravity record.
- Payload installation record.
- Preliminary boresight and calibration records.
- Updated configuration-control record.

11.4 Phase 3: Bench and Ground Testing

Phase 3 verifies subsystem operation before flight. The objective is to reduce integration risk by validating power, avionics, payload, communications, software configuration, logging, operator workflow, and safety procedures under controlled non-flight conditions.

Phase 3 activities include:

- Verify regulated power rails, current/voltage sensing, protection features, and power-load behavior.
- Verify flight-controller configuration, sensor calibration, GNSS reception, telemetry, and failsafe settings.
- Verify EO/IR payload operation, rangefinder communication, payload status reporting, and stabilized mount or gimbal behavior.
- Verify mission compute data paths, timestamping, range association, payload pointing association, target-location processing, and logging.
- Verify ground-control workflow, live telemetry, video display, payload status, target range display, target-coordinate output, and mission logging.
- Conduct ground rangefinder tests against known-distance targets.
- Conduct communications bench tests and ground range tests.
- Review laser safety procedures, battery safety controls, operator checklists, emergency procedures, and test cards.
- Complete pre-flight inspection process and formal flight-test readiness review.

Phase 3 deliverables include:

- Bench test report.
- Power-system validation record.
- Payload and rangefinder ground-test record.
- Communications ground-test record.
- Ground-control workflow validation record.
- Updated risk register.
- Flight-test readiness checklist and review record.

11.5 Phase 4: Flight Testing and Accuracy Validation

Phase 4 validates integrated aircraft and mission-system performance in controlled flight conditions. The objective is to demonstrate safe flight, payload operation, target observation, rangefinding, target geolocation, endurance, operational radius, communications, and data logging.

Phase 4 activities include:

- Conduct initial shakedown flights to verify flight stability, control response, telemetry, battery behavior, and failsafe configuration.
- Conduct hover and handling tests with the installed payload configuration.
- Conduct payload stability tests, EO/IR observation trials, and rangefinder flight-readiness checks.
- Conduct static target rangefinding tests against known-distance targets.
- Conduct target geolocation tests using surveyed or independently measured ground-truth target locations.
- Conduct moving-target trials to evaluate update continuity, latency, positional consistency, and operator cueing workflow.
- Conduct endurance and operational-radius tests under controlled conditions.
- Conduct GNSS-quality monitoring and degraded-navigation scenario evaluation where appropriate.
- Record and review flight logs, range measurements, target-coordinate outputs, payload logs, video records, power data, and operator observations.

Phase 4 deliverables include:

- Flight-test reports.
- Rangefinding accuracy test results.
- Target geolocation validation results.
- Moving-target test results.
- Endurance and operational-radius test results.
- Navigation-resilience test records.
- Updated geolocation error budget.
- Updated configuration and calibration records.

11.6 Phase 5: Refinement and Final Capability Assessment Readiness

Phase 5 focuses on refinement, documentation, repeatability, and readiness for final capability assessment. The objective is to incorporate lessons from testing, close technical issues, stabilize the final prototype configuration, and complete the documentation package.

Phase 5 activities include:

- Review Phase 4 test results and identify required refinements.

- Update payload calibration, boresight records, software configuration, wiring documentation, and operator workflow where required.
- Refine power, communications, navigation, payload, and target-cueing performance based on test data.
- Complete final integrated system weigh-in and configuration audit.
- Complete final verification matrix and traceability review.
- Complete operator documentation, maintenance notes, safety procedures, and final test report.
- Complete final capability assessment readiness package.

Phase 5 deliverables include:

- Refined VectorSight-M prototype configuration.
- Final configuration-control record.
- Final test report and verification summary.
- Final technical data package.
- Operator guide and maintenance documentation.
- Safety documentation and test records.
- Final capability assessment readiness package.

11.7 Milestone Deliverables and Acceptance Criteria

The work plan is controlled through milestone deliverables and acceptance criteria. Each milestone is tied to a measurable output, review event, or test result. This structure allows progress to be evaluated against technical readiness rather than narrative progress alone.

Milestone	Phase	Deliverable	Acceptance Criteria
M1	Final Design and Procurement	Requirements matrix, system architecture, preliminary BOM, vendor records, procurement package, and draft verification plan.	Requirements traced to challenge outcomes; critical components identified; procurement package ready for execution.
M2	Airframe/Payload Integration	Integrated airframe, payload bay, avionics, power distribution, communications, mission compute, and initial configuration records.	Physical integration complete; interfaces documented; initial mass, payload, and configuration records completed.
M3	Bench and Ground Testing	Bench test report, power validation, payload ground test, communications test, GCS workflow check, safety review, and flight-test readiness checklist.	Power, avionics, payload, communications, logging, operator workflow, and safety checks completed before flight.
M4	Flight Testing and Accuracy Validation	Flight-test reports, rangefinding results, target geolocation validation, moving-target results, endurance/radius results, and updated error budget.	Integrated system demonstrates controlled flight and mission-system performance against defined test objectives.
M5	Refinement and Assessment Readiness	Final prototype configuration, final test report, technical data package, operator documentation, safety records, and final assessment readiness package.	Prototype configuration stabilized; verification records complete; final deliverables ready for review and assessment.

The milestone structure provides clear decision points for procurement control, integration readiness, flight-test readiness, performance validation, risk reduction, and final reporting.

12 Verification and Validation Plan

12.1 Requirement-Based Test Matrix

VectorSight-M will be verified using a requirement-based test approach. Each essential and desired outcome will be mapped to one or more verification methods, including analysis, inspection, bench testing, ground testing, flight testing, calibration review, vendor evidence review, and final test reporting.

The verification approach ensures that compliance is supported by evidence rather than narrative claims. Test records will include measured data, system logs, operator observations, configuration records, calibration state, test conditions, and pass/fail assessment against defined objectives.

The verification matrix will track:

- Challenge outcome or internal requirement.
- Verification method.
- Test environment.
- Required equipment.
- Success criteria.
- Data to be recorded.
- Responsible owner.
- Configuration state.
- Result and corrective action, if required.

A detailed test matrix will be maintained in Appendix G. The summary verification structure is shown below.

Requirement Area	Primary Verification Method	Evidence Produced
Precision rangefinding	Static range test, moving-target test, calibration review.	Range test data, calibration record, test report.
Target geolocation	Ground-truth comparison, coordinate-output review, error-budget analysis.	Geolocation report, logs, error budget.
Operational radius	Ground link test and flight mission profile.	Flight logs, link logs, operator record.
Endurance	Hover test, mission-profile flight, power-log review.	Flight logs, power logs, battery records.
GNSS degradation	Controlled degradation scenario and navigation-quality review.	GNSS logs, degraded-mode report.
Environmental operation	Wind, temperature, precipitation/obscurant, and field-maintenance checks.	Environmental test record, operator notes.
Laser safety	Vendor evidence, safety procedure review, operational controls.	Laser safety file, checklist, configuration record.
Data handling	Access-control review, logging/export test, configuration audit.	Data-handling record, export package.

12.2 Rangefinding Accuracy Test

The rangefinding accuracy test verifies VectorSight-M range measurement against known-distance references. The objective is to validate Class 1 laser rangefinder integration, payload pointing stability, measurement repeatability, and range-data logging.

The test approach includes:

- Establish known-distance target locations using surveyed, measured, or independently verified references.
- Use vehicle-sized targets at representative distances, including a 1 km test condition where test-site conditions allow.
- Conduct repeated range measurements from controlled aircraft or ground positions.
- Record measured range, timestamp, aircraft position, aircraft attitude, payload pointing state, target type, weather, visibility, and operator action.
- Compare measured range against ground-truth distance.
- Evaluate repeatability, outliers, target-surface effects, and payload stability effects.

The test will produce a rangefinding accuracy record showing measurement error, repeatability, test conditions, configuration state, and required calibration or integration refinements. Results will feed directly into the geolocation error budget in Appendix D.

12.3 Target Geolocation Test

The target geolocation test verifies that VectorSight-M can convert range, aircraft position, aircraft attitude, payload pointing, timestamp, and calibration data into target-location output. This test is central to the True North Precision mission requirement because the system must provide actionable target information, not only range measurement.

The test approach includes:

- Establish known target locations using surveyed or independently verified ground-truth references.
- Conduct static target trials from selected aircraft positions and payload pointing conditions.
- Generate target-location outputs using the VectorSight target-geolocation workflow.
- Record aircraft position, aircraft attitude, payload pointing angle, measured range, timestamp, target output, operator action, and mission logs.
- Compare generated target-location output against ground-truth target location.
- Quantify geolocation error and identify primary contributors.

The target geolocation test will produce coordinate-output records, comparison results, error measurements, and log files. These results will validate the target-cueing workflow and refine the geolocation error budget.

12.4 Moving Target Test

The moving-target test verifies that VectorSight-M can support repeated target updates for moving targets. The objective is to evaluate update continuity, latency, measurement consistency, target-location output behavior, and operator workflow under controlled target motion.

The test approach includes:

- Use a controlled moving target with known or independently measured movement path.
- Conduct repeated observation and range measurement events during target motion.
- Record target update timing, measured range, aircraft state, payload pointing state, target-location output, and operator actions.
- Evaluate target-update performance up to 10 m/s, subject to test-site safety and control conditions.
- Measure target-update latency, continuity, output stability, and operator readability.
- Review logs to identify timing, pointing, communication, or workflow limitations.

The moving-target test will produce moving-target update records, latency measurements, output consistency results, operator observations, and refinement recommendations.

12.5 GNSS Degradation Test

The GNSS degradation test verifies system response when navigation quality is reduced or uncertain. The objective is to evaluate degraded-mode operation, navigation confidence monitoring, range/geolocation continuity, operator workflow, and failsafe behavior.

The test approach includes:

- Establish nominal GNSS performance baseline through normal ground and flight operation.
- Monitor GNSS fix quality, satellite count, position uncertainty, estimator state, and flight logs.
- Conduct controlled GNSS-degradation scenarios where safe and permitted.
- Evaluate system behavior during reduced navigation confidence.
- Verify operator alerts, degraded-mode procedures, logging, and abort/return decision pathways.
- Compare target-geolocation output quality under nominal and degraded navigation conditions where applicable.

The GNSS degradation test will produce navigation-quality records, degraded-mode behavior logs, operator procedure results, and recommendations for RTK, inertial fallback, or visual odometry refinement.

12.6 Endurance/Radius Test

The endurance and operational-radius test verifies that VectorSight-M can support the required mission duration and operating radius under controlled test conditions. This test links airframe mass, propulsion efficiency, battery energy, payload power draw, communications performance, and mission profile.

The endurance/radius test approach includes:

- Establish final integrated system weight and payload configuration before flight.
- Conduct hover and mission-profile tests with the installed payload.
- Record battery voltage, current draw, consumed capacity, aircraft state, payload state, wind conditions, and flight time.
- Conduct progressive radius expansion using safe test increments.
- Monitor command/control, telemetry, video, and target-data link performance.
- Preserve return and landing reserve according to test safety rules.
- Review power logs, link-quality logs, operator notes, and flight performance.

The endurance/radius test will produce flight logs, battery records, power-consumption data, communications performance data, and final assessment of mission-profile endurance and radius capability.

12.7 Wind, Temperature, Obscurant, and Precipitation Test

Environmental and operational testing evaluates VectorSight-M under field-relevant conditions. The objective is to assess how wind, temperature, visibility, obscurants, precipitation, vibration, and field handling affect flight performance, payload stability, target observation, range measurement, and operator workflow.

The environmental test approach includes:

- Conduct cold-operation checks at 0 °C and above, where test conditions allow.
- Conduct flight and payload checks in increasing wind conditions up to the defined safe test envelope.
- Evaluate payload stability, aircraft control margin, power draw, and operator workload under wind exposure.
- Conduct target-acquisition assessment under dust, smoke, haze, or other obscurant conditions where safe and permitted.
- Evaluate precipitation exposure as a growth-path test condition based on prototype configuration and safety limits.
- Record environmental conditions, aircraft behavior, payload behavior, rangefinding performance, and operator observations.

Environmental testing will produce condition-specific test records, payload stability observations, operational limitations, and refinement recommendations for enclosure design, vibration isolation, flight procedures, and payload protection.

12.8 Laser Safety Verification

Laser safety verification confirms that the VectorSight-M rangefinder installation, procedures, and configuration support safe prototype operation. The rangefinder baseline is based on Class 1 eye-safe laser operation.

Laser safety verification includes:

- Vendor documentation confirming Class 1 eye-safe classification.
- Installed hardware identification and serial/configuration record.
- Review of power, mounting, interface, and operational controls.
- Ground-test and flight-test laser safety procedure.
- Operator checklist for rangefinder readiness and safe use.
- Test-site procedure for personnel location, target area, and rangefinder use.
- Configuration audit to confirm the installed rangefinder matches the documented safety file.

Laser safety records will be included in the technical data package and referenced during test-readiness reviews.

12.9 Flight Safety Test Plan

Flight safety is managed through progressive test readiness, controlled flight cards, operator procedures, emergency actions, and configuration control. The flight safety test plan reduces risk during integration, shakedown flights, payload flights, rangefinding trials, and final assessment preparation.

The flight safety plan includes:

- Pre-flight inspection checklist.

- Battery safety and charging procedure.
- Propulsion and fastener inspection.
- Payload and rangefinder installation check.
- GNSS, telemetry, video, and link-quality check.
- Failsafe configuration check.
- Geofence, return, hold, land, and abort procedures.
- Test-card structure for each flight event.
- Emergency procedures for lost link, low battery, navigation issue, payload issue, flyaway risk, and hard landing.
- Post-flight inspection and log review.

Flight safety will be verified through bench checks, ground checks, pre-flight reviews, shakedown flights, controlled expansion of the flight envelope, incident/issue tracking, and corrective-action closure before higher-risk test events.

13 Risk, Safety, and Configuration Control

13.1 Technical Risk Register

Zigaton will maintain a technical risk register throughout the VectorSight-M development program. The risk register will identify technical uncertainties early, assign mitigation actions, track ownership, and prevent unresolved issues from affecting integration, flight testing, or final capability assessment readiness.

Technical risks will be tracked across the following categories:

- Airframe and payload integration.
- Propulsion, endurance, and operational radius.
- Payload stabilization and boresight alignment.
- Laser rangefinder integration and measurement accuracy.
- Target-geolocation error and timestamp synchronization.
- GNSS degradation and navigation confidence.
- Communications, telemetry, video, and target-data links.
- Power distribution, thermal behavior, and battery management.
- Software, logging, and mission-data export.
- Environmental, wind, vibration, and field-maintenance factors.

The risk register will include risk description, likelihood, consequence, mitigation action, owner, status, verification method, and residual risk. Risks will be reviewed at each major milestone and updated after procurement, integration, bench testing, ground testing, flight testing, and final refinement.

Risk Area	Mitigation Approach	Verification / Control Method
Payload integration	Controlled payload envelope, interface definition, mounting review, vibration-aware design.	CAD review, payload installation record, ground test.
Rangefinding accuracy	Vendor-supported rangefinder selection, boresight calibration, repeated known-distance tests.	Static range test, calibration report.
Target geolocation	Error-budget control, timestamp association, pose/pointing data logging, ground-truth comparison.	Geolocation test, log review, Appendix D.
Endurance/radius	Mass control, propulsion sizing, power logging, reserve-energy rules.	Hover test, mission-profile flight, power-log review.
Communications	Link-budget review, antenna placement, ground range testing, degraded-link procedures.	Ground range test, flight-link log, operator checklist.
Navigation degradation	GNSS quality monitoring, inertial fallback, RTK pathway, degraded-mode procedures.	GNSS degradation test, flight-log review.
Power and thermal	Regulated rails, current/voltage monitoring, fusing, thermal inspection.	Bench power test, thermal review, flight power log.
Configuration drift	BOM control, firmware/software version tracking, calibration records, change log.	Configuration audit, final technical data package.

13.2 Program Risk Register

Program risks will be managed separately from technical risks because schedule, procurement, supplier availability, test access, and documentation readiness can affect delivery even when the technical design is sound.

Program risk categories include:

- Long-lead component availability.
- Vendor documentation or integration-support delay.
- Alternate supplier qualification.
- Flight-test site availability.
- Weather-related test delays.
- Flight authorization and range-safety constraints.
- Subcontractor or advisor availability.

- Documentation and deliverable readiness.
- Final capability assessment preparation.

Zigaton will reduce program risk through early procurement planning, alternate supplier identification, phased testing, controlled readiness gates, documented test cards, and continuous configuration control. Program risks will be reviewed during milestone reviews and updated when supplier, schedule, test, or documentation conditions change.

13.3 Flight Safety

Flight safety is controlled through progressive testing, readiness gates, pre-flight inspections, operator procedures, emergency actions, and post-flight review. VectorSight-M will progress through bench validation, ground validation, shakedown flights, payload flights, rangefinding trials, and mission-profile tests before final assessment readiness.

Flight safety controls include:

- Pre-flight inspection checklist for airframe, fasteners, propellers, battery, payload, antennas, wiring, and connectors.
- Flight-controller configuration review, including failsafe, geofence, return-to-home, low-battery, and link-loss behavior.
- Progressive flight envelope expansion from low-risk shakedown flights to mission-profile trials.
- Test cards defining objective, configuration, location, weather limits, operator roles, abort criteria, and data to be recorded.
- Emergency procedures for lost link, low battery, navigation uncertainty, payload issue, propulsion abnormality, hard landing, or flyaway risk.
- Post-flight inspection, flight-log review, issue tracking, and corrective-action closure.

Flight safety will be documented through checklists, test cards, flight logs, inspection records, operator notes, and issue-resolution records.

13.4 Battery Safety

Battery safety is critical because VectorSight-M depends on high-energy flight batteries and regulated power distribution for propulsion, payload, avionics, communications, and mission compute. Battery safety will be managed through handling procedures, charging controls, inspection, storage practices, and operational reserve rules.

Battery safety controls include:

- Battery selection matched to voltage, capacity, discharge rating, payload load, and mission endurance requirements.
- Charging procedure using appropriate chargers, charging settings, supervision, and fire-safe handling practices.

- Pre-flight inspection for swelling, physical damage, connector condition, wire damage, state of charge, and secure mounting.
- In-flight monitoring of voltage, current draw, consumed capacity, and reserve margin.
- Defined low-battery thresholds for return, landing, and abort.
- Post-flight inspection, cooling period, storage-voltage management, and cycle tracking.
- Battery incident procedure for damage, overheating, abnormal voltage behavior, or hard landing.

Battery records will be maintained as part of the technical data package where relevant to endurance, safety, power-system validation, and final assessment readiness.

13.5 Laser Safety

Laser safety is managed through Class 1 eye-safe rangefinder selection, vendor documentation, installation control, test procedures, operator training, and configuration records. The baseline VectorSight-M rangefinder implementation is based on Class 1 eye-safe operation to support safe prototype development and field testing.

Laser safety controls include:

- Vendor documentation confirming Class 1 eye-safe classification.
- Installed rangefinder identification, model number, serial number, and configuration record.
- Controlled mounting and alignment to prevent unintended beam direction or uncontrolled hardware movement.
- Operator checklist for rangefinder readiness, status confirmation, and safe measurement procedure.
- Test-site procedure for personnel positioning, target-area control, and rangefinder operation.
- Ground-test and flight-test safety review before rangefinder activation.
- Configuration audit confirming the installed rangefinder matches the documented safety file.

Laser safety records will be reviewed during test-readiness checks and included in the final technical data package.

13.6 Cyber/Data Risk

Cyber and data risk will be managed through controlled access, organized data storage, configuration tracking, export control, and an ITSP.10.171-compliance-oriented data-handling process. The objective is to protect project data while preserving the logs and records required for verification.

Cyber and data-risk controls include:

- Controlled access to mission logs, test data, configuration files, vendor records, and technical documentation.
- Defined storage locations for technical data package materials, test reports, calibration files, and flight records.
- Version control for software, firmware, message formats, payload configuration, and calibration state.

- Controlled export process for mission logs, target data, video records, and deliverables.
- Encryption consideration for stored or transmitted data where appropriate to the test environment and customer direction.
- Audit trail for generated, modified, reviewed, or approved technical data.
- Removal or protection of sensitive information before wider distribution of test data.

Data-risk controls will be verified through access-control review, log-export testing, configuration audit, and final deliverable review.

13.7 Configuration Management

Configuration management ensures that the prototype being tested is the same prototype described in the technical data package. This is essential for traceability, safety, repeatability, and final assessment confidence.

The configuration-management approach includes:

- Unique configuration record for the VectorSight-M prototype.
- BOM tracking for airframe, propulsion, payload, avionics, communications, power distribution, mission compute, ground equipment, and spares.
- Firmware and software version tracking for flight controller, mission compute, payload interfaces, logging tools, and ground-control configuration.
- Calibration record tracking for boresight alignment, payload installation, rangefinder configuration, and target-geolocation parameters.
- Wiring, connector, and interface documentation.
- Change log for hardware, software, firmware, calibration, payload, wiring, or structural modifications.
- Test-log linkage to prototype configuration state.

Configuration changes will be reviewed before higher-risk test events. Any change affecting flight safety, rangefinding accuracy, target geolocation, power, communications, or payload mounting will require updated inspection or test records before the next major test activity.

13.8 BOM, Firmware, Software, Calibration, and Test-Log Control

VectorSight-M will use controlled records to link the physical prototype, software configuration, payload calibration, and test evidence. This prevents performance claims from being separated from the specific configuration that produced the result.

Controlled records will include:

- **BOM control:** part number, supplier, country of origin, quantity, revision, cost basis, lead time, and substitute options.
- **Firmware control:** flight-controller firmware version, payload firmware where applicable, radio firmware, and update history.

- **Software control:** mission compute software version, target-geolocation logic version, logging tools, export tools, and ground-control configuration.
- **Calibration control:** boresight calibration, payload-to-airframe reference relationship, rangefinder configuration, and target-geolocation parameters.
- **Test-log control:** flight logs, range logs, payload logs, video records, power logs, target-coordinate outputs, environmental conditions, and operator notes.
- **Issue and corrective-action control:** test issue, root cause, corrective action, verification method, and closure status.

These controls will support repeatable testing, technical review, risk management, final reporting, and future prototype improvement. The final technical data package will include the relevant configuration records required to understand, reproduce, and evaluate VectorSight-M prototype performance.

14 Supply Chain and Manufacturability

14.1 Vendor Readiness

Zigaton will manage the VectorSight-M supply chain through vendor readiness tracking, alternate sourcing, country-of-origin documentation, procurement records, and configuration control. Supply-chain readiness is treated as both a technical and program requirement because component availability, documentation quality, lead time, integration support, and substitution control directly affect prototype delivery and final capability assessment readiness.

Vendor readiness will be managed across the following subsystem categories:

- Airframe, structural components, landing gear, payload-bay hardware, and mechanical mounting hardware.
- Motors, propellers, ESCs, propulsion wiring, and propulsion accessories.
- Flight batteries, chargers, power accessories, and battery-management equipment.
- Flight controller, GNSS/RTK equipment, inertial sensors, telemetry hardware, antennas, and supporting avionics.
- EO/IR payload, Class 1 eye-safe laser rangefinder, stabilized mount or gimbal, and payload interface hardware.
- Mission compute hardware, video link, target-data link, ground-control station equipment, and operator interface equipment.
- Power distribution hardware, regulated power modules, current/voltage sensing, connectors, fuses, transient protection, and harness materials.
- Fabrication, machining, assembly, environmental test support, flight-test support, and specialized integration services where required.

For each critical vendor or supplier, Zigaton will track datasheets, interface requirements, budgetary pricing, lead-time assumptions, availability, country-of-origin information, integration notes, alternate suppliers, and procurement status. This information will support the preliminary BOM, risk register, cost justification, test plan, and final technical data package.

14.2 Critical Component Matrix

A critical component matrix will be maintained to identify parts and subsystems with the greatest effect on performance, schedule, integration risk, safety, or final assessment readiness. Components will be classified as critical when they affect flight safety, rangefinding accuracy, geolocation accuracy, communications reliability, endurance, navigation resilience, laser safety, data handling, or payload integration.

The critical component matrix will track:

- Component or subsystem name.
- Function in the VectorSight-M architecture.
- Supplier or vendor.

- Part number or configuration identifier.
- Country of origin.
- Lead time and procurement status.
- Interface requirements.
- Known integration risks.
- Alternate supplier or substitute option.
- Verification evidence required.

A preliminary critical component structure is shown below.

Component Category	**Criticality Reason**	**Control Method**
EO/IR payload	Supports target observation, operator confirmation, and FMV workflow.	Vendor review, interface control, payload test.
Laser rangefinder	Supports precision rangefinding, target geolocation, and laser safety compliance.	Datasheet review, safety file, range test.
Stabilized mount/gimbal	Affects pointing stability, boresight consistency, and measurement repeatability.	Integration inspection, stability test, calibration record.
Flight controller	Provides aircraft stabilization, navigation, failsafe, telemetry, and flight logs.	Configuration control, bench test, flight-log review.
GNSS/RTK hardware	Supports navigation accuracy, target-geolocation reference, and log correlation.	Ground test, flight-log review, GNSS quality review.
Mission compute	Supports timestamp association, geolocation logic, logging, and cueing output.	Bench test, software version control, log review.
Communications links	Support command/control, telemetry, video, target data, and operational radius.	Link-budget review, ground range test, flight-link logs.
Power distribution	Supports safe power delivery, regulated rails, sensing, protection, and diagnostics.	Bench power test, inspection, flight power logs.
Battery system	Affects endurance, radius, safety, reserve energy, and field operation.	Datasheet review, inspection, power-log review.

The critical component matrix will be updated as the BOM matures and included in the technical data package.

14.3 Country-of-Origin Matrix

Zigaton will maintain a country-of-origin matrix for critical components, software elements, payload systems, communications hardware, navigation hardware, flight-control equipment, power electronics, and ground-control equipment. This supports supply-chain transparency, risk management, procurement planning, and defence evaluation.

The country-of-origin matrix will include:

- Component or subsystem name.
- Supplier or manufacturer.
- Part number or configuration identifier.
- Country of origin.
- Country of final assembly where available.
- Software or firmware origin where relevant.
- Data-link or communications relevance.
- Alternate supplier options.
- Documentation source or vendor evidence.

This matrix will be maintained as a living procurement and configuration document. Any substitution affecting flight control, communications, payload sensing, rangefinding, navigation, power distribution, mission compute, or data handling will require an update to the country-of-origin matrix and configuration-control record.

14.4 Alternate Suppliers

Alternate suppliers will be identified for critical components to reduce schedule, cost, integration, and availability risk. Alternate sourcing is especially important for components with long lead times, export constraints, limited documentation, single-source availability, high cost, or direct effect on final assessment readiness.

Alternate supplier planning will focus on:

- Flight batteries and chargers.
- Motors, propellers, ESCs, and propulsion accessories.
- Flight controller and GNSS/RTK hardware.
- Telemetry, video, and target-data communications hardware.
- EO/IR payload, Class 1 eye-safe laser rangefinder, and stabilized mount or gimbal.
- Mission compute hardware and ground-control equipment.
- Power distribution, regulators, current/voltage sensors, connectors, fuses, and harness materials.
- Airframe structural elements, fabrication services, and mounting hardware.

Alternate components will not be treated as uncontrolled substitutions. Any alternate supplier or replacement part must be reviewed for interface compatibility, mass, power, thermal behavior, software/firmware compatibility, safety, country of origin, cost, and verification impact before use in the prototype.

14.5 Lead-Time Control

Lead-time control will be managed through early identification of critical-path items, staged procurement, supplier follow-up, alternate sourcing, and integration sequencing. The work plan is structured so that long-lead components are identified during Phase 1 and procurement risk is addressed before airframe/payload integration becomes schedule-critical.

Lead-time control measures include:

- Identify long-lead and critical-path components during final BOM development.
- Obtain vendor lead-time estimates and budgetary pricing early.
- Prioritize procurement of payload, rangefinder, flight controller, GNSS/RTK, propulsion, battery, communications, and mission-compute items.
- Maintain alternate supplier options for critical components.
- Track order status, shipping status, received condition, and documentation completeness.
- Sequence bench and ground testing around available subsystems where practical.
- Update schedule risk when procurement conditions change.

Lead-time risk will be reviewed during milestone reviews. Any delayed critical component will trigger a mitigation review covering alternate suppliers, interim test configuration, schedule adjustment, or controlled substitution.

14.6 Assembly and Maintenance Concept

VectorSight-M will be assembled as a maintainable prototype system with controlled subsystem access, labeled wiring, modular payload integration, and documented inspection points. The assembly concept supports repeatable integration, field maintenance, test turnaround, and configuration traceability.

The assembly concept includes:

- Modular airframe assembly with controlled payload, battery, avionics, and power-distribution zones.
- Payload bay designed for installation, removal, inspection, and calibration access.
- Labeled wiring harnesses and connector definitions.
- Accessible avionics and mission-compute installation.
- Power-distribution layout designed for inspection, current/voltage sensing, and protection review.
- Antenna placement with attention to link quality, interference, and service access.
- Mechanical fastening and inspection approach for arms, landing gear, payload mount, battery retention, and propulsion hardware.
- Maintenance records linked to configuration state and test logs.

The maintenance concept includes pre-flight inspection, post-flight inspection, payload inspection, battery inspection, connector inspection, propeller inspection, calibration review, and issue tracking. This supports prototype reliability and reduces the risk of uncontrolled configuration changes during test execution.

14.7 Spare Parts Strategy

A spare parts strategy will reduce downtime during integration, ground testing, flight testing, and final assessment preparation. Spare parts will be prioritized based on likelihood of wear, field replacement need, lead time, safety relevance, and impact on mission readiness.

The spare parts strategy includes:

- Propellers, propeller hardware, and propulsion mounting hardware.
- Landing gear and payload-protection hardware.
- Battery straps, retention hardware, connectors, and harness repair materials.
- Fuses, power connectors, regulated power modules, current/voltage sensing accessories, and transient-protection components.
- Antennas, telemetry accessories, video-link accessories, and cable adapters.
- Payload mounting hardware, vibration-isolation elements, and calibration fixtures where required.
- Fasteners, standoffs, strain-relief materials, labels, and field-repair tools.
- Spare batteries and charging accessories as required by the test plan.

Spare parts will be tracked through the BOM and configuration-control process. Any spare part installed on the prototype will be recorded so that flight logs, test results, calibration records, and final reports remain tied to the actual tested configuration.

14.8 Production Scalability

VectorSight-M is the primary prototype, but the architecture is designed with future production scalability in mind. The objective is to avoid a one-off demonstrator that cannot be repeated, maintained, or matured into follow-on systems.

The production scalability approach includes:

- Use of vendor-supported subsystems where practical.
- Controlled mechanical, electrical, data, and payload interfaces.
- Modular payload bay supporting alternate sensor configurations.
- BOM and country-of-origin tracking for critical components.
- Alternate supplier planning for procurement resilience.
- Assembly documentation, wiring records, calibration procedures, and test procedures.
- Configuration control for hardware, firmware, software, calibration, and test records.
- Growth path from VectorSight-M to heavier-lift and longer-endurance configurations.

Scalability will be evaluated through design review, supplier review, assembly documentation, maintainability assessment, spare-parts planning, and cost-to-capability review. This ensures that VectorSight-M supports immediate prototype evaluation while creating a credible path toward repeatable Canadian tactical UAS production and future mission-system variants.

15 Budget and Cost Justification

15.1 Total Funding Request

Zigaton requests funding to design, procure, integrate, test, refine, document, and prepare the VectorSight-M prototype for final capability assessment. The budget is structured around a phased prototype-development program covering final design, procurement, airframe and payload integration, bench testing, ground testing, flight testing, rangefinding validation, target-geolocation validation, documentation, and final assessment readiness.

The total funding request will be finalized after vendor quotations, labour allocation, test-site requirements, subcontractor needs, and procurement assumptions are locked. The budget will show clear traceability between cost categories, work-plan phases, milestone deliverables, and challenge outcomes.

The funding request will cover the following major cost categories:

- Labour for engineering, integration, documentation, project management, testing, and reporting.
- Air vehicle, propulsion, battery, avionics, communications, payload, power, and mission-compute materials.
- EO/IR payload, Class 1 eye-safe laser rangefinder, stabilized mount or gimbal, and payload interface equipment.
- Ground-control equipment, telemetry, video, target-data link, and operator-interface hardware.
- Subcontractor, vendor, advisor, fabrication, test, and integration-support costs.
- Flight-test preparation, test-site support, travel, field equipment, spares, and safety equipment.
- Contingency for controlled substitutions, integration refinements, procurement changes, and test-identified corrective actions.

The budget supports a complete prototype mission system, not only the purchase of a drone platform. The cost structure reflects the full system required for precision rangefinding, target geolocation, real-time cueing, safe testing, configuration control, and final technical reporting.

15.2 Labour Budget

The labour budget supports the engineering and program execution required to deliver VectorSight-M as an integrated tactical UAS prototype. Labour is allocated across systems engineering, electrical and avionics integration, mechanical design, payload integration, software and mission-data workflow, test planning, flight-test support, documentation, and project management.

Labour activities include:

- System requirements definition, compliance mapping, and interface control.
- Air vehicle configuration, payload-bay design, CAD support, and mechanical integration.
- Electrical architecture, avionics integration, power distribution, wiring, and harness planning.
- Payload integration, EO/IR interface planning, rangefinder integration, boresight calibration, and payload test support.

- Mission compute workflow, timestamping, target-geolocation logic, logging, and target-cueing output.
- Communications integration, telemetry, video, target-data link, antenna placement, and ground-control workflow.
- Verification and validation planning, test cards, safety procedures, data review, and corrective-action tracking.
- Technical data package development, operator documentation, maintenance notes, final reporting, and assessment-readiness preparation.

The labour estimate will be mapped to the five project phases and milestone deliverables so that labour costs are tied directly to measurable outputs.

Labour Category	**Primary Activities**	**Budget Allocation**
Project management and systems engineering	Requirements control, compliance mapping, milestone reviews, risk tracking, reporting, and stakeholder coordination.	TBD
Electrical and avionics integration	Power architecture, avionics interfaces, regulated rails, sensing, wiring, communications, and configuration control.	TBD
Mechanical design and integration	CAD, payload bay, mounting, enclosure, vibration-aware design, maintainability, and assembly support.	TBD
Payload and mission-system integration	EO/IR payload, laser rangefinder, gimbal/stabilization, boresight calibration, payload data flow, and cueing workflow.	TBD
Software, data, and logging workflow	Timestamping, target-geolocation logic, mission logs, data export, GCS workflow, and configuration tracking.	TBD
Testing and documentation	Bench testing, ground testing, flight-test support, test reports, operator guide, maintenance notes, and final data package.	TBD

15.3 Materials and Equipment

Materials and equipment costs cover the physical systems required to build, integrate, test, and operate the VectorSight-M prototype. The materials budget will be supported by vendor quotations, datasheets, supplier records, lead-time assumptions, country-of-origin information, and substitute options.

The materials and equipment budget includes:

- **Airframe and structure:** frame, arms, landing gear, payload bay, mounting hardware, enclosures, fasteners, vibration-isolation elements, and structural accessories.

- **Propulsion and energy:** motors, propellers, ESCs, flight batteries, chargers, battery retention, and propulsion wiring.
- **Avionics and navigation:** flight controller, GNSS/RTK equipment, inertial sensors, telemetry hardware, antennas, and supporting electronics.
- **Payload system:** EO/IR payload, Class 1 eye-safe laser rangefinder, stabilized mount or gimbal, payload interface hardware, and payload protection.
- **Mission compute and communications:** onboard compute hardware, video link, target-data link, ground-control equipment, operator display, and interface equipment.
- **Power distribution:** regulated power modules, current/voltage sensing, fuses, connectors, transient protection, wiring, harness materials, and test points.
- **Test and support equipment:** calibration targets, measurement equipment, field tools, spare parts, safety equipment, and data-storage media.

The final materials budget will be reflected in Appendix E as the preliminary BOM and in Appendix F as the vendor and country-of-origin matrix.

15.4 Subcontractors/Vendors

Zigaton will use vendors, advisors, and subcontractors where specialized support improves delivery quality, reduces risk, or provides capability beyond the core team. This may include payload support, fabrication, flight-test support, software integration support, safety review, cybersecurity/data-handling review, and environmental or vibration testing.

Potential subcontractor and vendor-support areas include:

- EO/IR payload and laser rangefinder vendor support.
- Gimbal or stabilized payload integration support.
- Airframe fabrication, machining, composite, or metalwork support.
- Flight-test site, flight operations, RPAS operations, or range-safety support.
- Mission-data workflow, target-cueing export, ATAK/CoT pathway, or software integration support.
- Cybersecurity, data-handling, or ITSP.10.171-oriented review support.
- Environmental, vibration, or safety testing support.
- Documentation, quality, or configuration-management review support.

Subcontractor and vendor costs will be controlled through defined scopes of work, deliverables, interface requirements, procurement records, and acceptance criteria. External support will strengthen the prototype while preserving Zigaton's system-level ownership and responsibility for integration.

15.5 Test and Travel Costs

Test and travel costs support the field activities required to validate VectorSight-M against the challenge outcomes. These costs may include ground testing, flight testing, rangefinding validation, target-geolocation validation, moving-target trials, environmental test activities, safety preparation, field equipment, transportation, and final assessment readiness.

Test and travel costs may include:

- Test-site access, range support, or field-operation fees.
- Transportation of prototype, ground-control station, batteries, tools, spares, and support equipment.
- Local travel for integration reviews, vendor coordination, field testing, and final assessment activities.
- Field equipment including targets, markers, tripods, measurement tools, safety equipment, shelters, cases, and charging equipment.
- Surveyed or independently verified target-location support where required.
- Flight-test support, observer support, safety support, or RPAS operations support where required.
- Data collection, data storage, video recording, log review, and test-report preparation.

Test and travel costs will be mapped to the verification plan in Section 12 and Appendix G so that each test activity is linked to a required outcome, evidence record, and final reporting need.

15.6 Contingency

A contingency allocation will manage normal prototype-development uncertainty. Contingency is especially important for supply-chain substitutions, integration refinements, test-identified corrective actions, spare parts, additional field-test time, and vendor-supported changes.

Contingency may be used for:

- Controlled replacement of delayed or unavailable components.
- Additional brackets, mounts, wiring, connectors, fasteners, and protection hardware.
- Payload or gimbal integration refinements.
- Power, thermal, communications, or antenna-layout refinements.
- Additional spare parts required after bench or flight testing.
- Additional test-site time, field support, or repeat testing.
- Corrective actions required to close issues before final assessment readiness.

Contingency will not be treated as uncontrolled spending. Use of contingency will be tied to risk-register items, procurement changes, test findings, corrective actions, or milestone-readiness needs. Any contingency-driven hardware or software change will be captured in the configuration-control record.

15.7 Cost-to-Milestone Mapping

The budget will be mapped to the five project phases and milestone deliverables defined in Section 11. This ensures that funding is tied to technical progress, procurement control, integration readiness, test evidence, and final reporting.

Milestone	Phase	Major Cost Drivers	Budget Allocation
M1	Final Design and Procurement	Engineering labour, requirements control, vendor review, preliminary BOM, procurement package, long-lead items.	TBD
M2	Airframe/Payload Integration	Airframe, propulsion, payload bay, avionics, mission compute, power distribution, wiring, mechanical integration.	TBD
M3	Bench and Ground Testing	Bench equipment, power validation, payload ground testing, communications testing, GCS workflow, safety review.	TBD
M4	Flight Testing and Accuracy Validation	Flight testing, rangefinding validation, geolocation testing, moving-target trials, endurance/radius tests, field support.	TBD
M5	Refinement and Assessment Readiness	Corrective actions, final configuration audit, final documentation, final test report, operator guide, assessment package.	TBD

The final budget table will include the total funding request, labour budget, materials and equipment, subcontractors/vendors, test and travel costs, contingency, and milestone mapping. This structure allows evaluators to understand not only what the project costs, but why each cost is necessary for prototype delivery and challenge-outcome verification.

16 Deliverables, Reporting, IP, and Data Rights

16.1 Prototype Deliverables

The primary technical deliverable is **VectorSight-M**, an integrated tactical multirotor UAS mission system configured for precision laser rangefinding, target geolocation, real-time cueing, and final capability assessment readiness.

Prototype deliverables will include:

- Integrated VectorSight-M air vehicle.
- EO/IR payload architecture integrated with a Class 1 eye-safe laser rangefinder.
- Stabilized payload interface or gimbal integration.
- Navigation architecture using GNSS, inertial sensing, flight-controller estimation, and RTK enhancement pathway.
- Mission compute workflow for range association, pose association, timestamping, target-geolocation logic, cueing output, and logging.
- Ground-control configuration for aircraft telemetry, video, payload status, target range, target-coordinate output, and mission-data review.
- Power distribution, regulated subsystem rails, current/voltage sensing, protection features, and harness documentation.
- Configured communications links for command/control, telemetry, video, payload status, and target-data transfer.
- Field support equipment, spare parts, and safety equipment required for prototype testing and demonstration.

The prototype will be delivered with configuration records identifying installed hardware, firmware, software, payload configuration, calibration state, test configuration, and verification status. This ensures that the evaluated system is traceable to the technical data package and test records.

16.2 Technical Data Package

Zigaton will provide a technical data package to support prototype evaluation, repeatability, configuration control, and future development. The package will document system architecture, subsystem interfaces, integration approach, verification results, and final prototype configuration.

The technical data package will include:

- System architecture and subsystem block diagrams.
- Air vehicle configuration summary.
- Payload-bay mechanical interface description.
- EO/IR and laser rangefinder integration notes.
- Electrical power architecture and regulated rail summary.

- Wiring, connector, and harness documentation.
- Communications and ground-control configuration summary.
- Target-geolocation architecture and data-flow description.
- Boresight calibration procedure and calibration records.
- Preliminary and final BOM records.
- Vendor records, datasheets, and country-of-origin matrix.
- Firmware, software, payload, and configuration records.
- Verification and validation test reports.
- Risk register, safety records, and corrective-action records.
- Operator guide and maintenance documentation.

The technical data package will be updated during the project as the design matures from preliminary configuration to final prototype configuration.

16.3 Test Reports

Zigaton will produce test reports documenting bench testing, ground testing, flight testing, rangefinding validation, target-geolocation validation, moving-target testing, endurance/radius testing, navigation degradation assessment, communications testing, environmental checks, laser safety verification, and data-handling checks.

Test reports will include:

- Test objective and requirement reference.
- Test configuration and prototype configuration state.
- Test date, location, personnel, and environmental conditions.
- Equipment used and ground-truth reference method.
- Test procedure or flight card reference.
- Data recorded, including logs, measurements, video, telemetry, and operator notes.
- Results and comparison against success criteria.
- Issues identified, corrective actions, and re-test requirements where applicable.
- Final conclusion and relevance to challenge outcomes.

This structure allows evaluators to trace performance claims to measured evidence, system logs, calibration records, and configuration state.

16.4 User/Maintenance Documentation

VectorSight-M will be supported by user and maintenance documentation suitable for prototype operation, field testing, inspection, and final assessment preparation. The documentation will focus on safe operation, repeatable setup, controlled configuration, and maintainability.

User and maintenance documentation will include:

- Operator setup checklist.
- Pre-flight inspection checklist.
- Battery handling and charging procedure.
- Payload installation and inspection procedure.
- Rangefinder readiness and laser safety checklist.
- Ground-control setup and mission logging procedure.
- Launch, observation, rangefinding, cueing, return, and landing workflow.
- Emergency procedures for lost link, low battery, navigation issue, payload issue, or hard landing.
- Post-flight inspection and log-review procedure.
- Maintenance notes for propellers, landing gear, payload mount, wiring, connectors, antennas, and power distribution.
- Calibration re-check triggers after payload removal, transport, hard landing, maintenance, or configuration change.

This documentation will support safe prototype operation and help ensure that test results are repeatable and traceable.

16.5 Progress Reporting Schedule

Zigaton will use structured progress reporting to communicate technical progress, schedule status, procurement status, risk status, test readiness, budget status, and deliverable maturity. Reporting will align to the project phases and milestone gates described in Section 11.

Progress reporting will cover:

- Work completed during the reporting period.
- Upcoming work and milestone targets.
- Procurement and vendor-readiness status.
- Technical risks, program risks, and mitigation actions.
- Configuration changes and controlled substitutions.
- Test activities completed and evidence generated.
- Issues, corrective actions, and closure status.

- Budget status and cost-to-milestone tracking.
- Final assessment readiness status.

Reports will show measurable progress against the work plan rather than general activity updates. This supports accountability, technical traceability, and early identification of issues that may affect final delivery.

16.6 IP Ownership

Zigaton will retain ownership of background intellectual property, project know-how, integration methods, internal design files, software logic, configuration methods, and technical approaches developed or brought into the project by Zigaton, except where otherwise agreed in the final contract terms.

Project foreground intellectual property developed under the funded work will be managed according to applicable contract requirements. Zigaton will identify which elements are Zigaton-developed, vendor-provided, open-source, commercially licensed, or third-party controlled.

IP-related records will include:

- Zigaton-developed technical materials.
- Vendor-supplied documentation and restrictions.
- Open-source software notices where applicable.
- Third-party software, firmware, payload, or hardware-license conditions.
- Internal software and configuration-control records.
- Deliverable data rights and usage restrictions as required by the contract.

This approach protects Zigaton's core technical capability while ensuring that government evaluators receive the technical information required to assess the prototype and project outcomes.

16.7 Third-Party/Vendor IP

VectorSight-M will use selected vendor-supported subsystems, including airframe components, avionics, payload equipment, laser rangefinder hardware, flight-controller hardware, communications equipment, mission-compute hardware, power electronics, and ground-control equipment. These components may include vendor-owned intellectual property, firmware, software, interface documentation, or license restrictions.

Third-party and vendor IP will be managed through:

- Supplier documentation review.
- License and usage-condition review where applicable.
- Identification of vendor-controlled hardware, software, firmware, and documentation.
- Separation of vendor IP from Zigaton-developed integration records.
- Tracking of restrictions affecting redistribution, modification, export, or disclosure.
- Clear identification of third-party elements in the technical data package.

Vendor IP will not be represented as Zigaton-owned IP. Zigaton's value is in system architecture, integration, configuration control, payload mission workflow, verification approach, and prototype execution.

16.8 Open-Source Software Disclosure

Where open-source software is used, Zigaton will disclose relevant software elements, license types, intended use, modification status, and project relevance. Open-source software may be used for flight-control configuration, mission-data handling, logging, geolocation processing, ground-control workflow, data export, or supporting engineering tools.

Open-source software disclosure will include:

- Software name and version.
- License type.
- Use within the VectorSight-M architecture.
- Whether the software is used unmodified or modified.
- Any distribution or disclosure requirements.
- Security or configuration considerations.
- Relationship to deliverable data, test logs, or operator workflow.

Open-source software use will be controlled through configuration management and data-handling procedures. Any open-source element affecting flight behavior, payload control, target geolocation, logging, or data export will be recorded in the final technical data package.

16.9 Data Rights and Deliverable Use

Zigaton will provide deliverable data needed to evaluate the VectorSight-M prototype, verify challenge-outcome performance, review safety and configuration records, and support final assessment. Deliverable data will be organized so evaluators can understand the tested configuration, test conditions, measurement results, and technical basis for the prototype.

Deliverable data may include:

- Final prototype configuration records.
- System architecture and interface documentation.
- BOM, vendor, and country-of-origin records.
- Calibration records and geolocation error-budget records.
- Rangefinding, geolocation, moving-target, endurance, radius, navigation, and safety test reports.
- Flight logs, payload logs, power logs, communications logs, and target-output records.
- Operator guide, maintenance documentation, and safety checklists.
- Final assessment readiness package.

Data rights, disclosure permissions, and usage limitations will be managed according to final contract terms. Zigaton will maintain clear separation between deliverable project data, Zigaton background IP, vendor-controlled IP, open-source materials, and sensitive test information.

17 Strategic Value To Canada

17.1 Canadian UAV Capability

VectorSight UAS supports the development of a Canadian tactical UAS capability focused on precision rangefinding, target geolocation, real-time cueing, modular payload integration, and field-testable defence innovation. The proposed prototype directly addresses the need for lower-cost UAS systems that can provide useful target information to tactical users without relying only on scarce or higher-cost ISR platforms.

The strategic value of VectorSight-M is not limited to one airframe. The project creates a Canadian integration pathway across air vehicle design, payload integration, Class 1 eye-safe laser rangefinding, target geolocation, mission compute, ground-control workflow, verification testing, configuration control, and supply-chain documentation.

This capability is relevant because Canada requires scalable and adaptable systems that can be developed, tested, improved, and produced with clear understanding of subsystem origin, integration risk, field maintainability, and operational relevance. VectorSight-M provides a prototype foundation for that pathway.

17.2 Defence Industrial Base Contribution

The VectorSight UAS project contributes to Canada's defence industrial base by strengthening domestic capability in tactical UAS systems integration. The project brings together electrical systems, avionics, payload integration, mechanical design, mission-data handling, software workflow, verification planning, supplier coordination, and prototype execution under a Canadian-led development effort.

The defence industrial base contribution includes:

- Development of Canadian expertise in tactical UAS mission-system integration.
- Integration of air vehicle, payload, rangefinder, navigation, communications, mission compute, and ground-control functions into a coherent prototype.
- Creation of controlled technical documentation, test reports, configuration records, and supplier records.
- Establishment of a repeatable prototype-to-test workflow for future UAS capability development.
- Support for Canadian supplier engagement, fabrication support, test support, and specialized subcontractor participation where appropriate.
- Development of a scalable architecture that can mature toward follow-on systems, production variants, and mission-specific payload configurations.

The project therefore supports both near-term prototype delivery and long-term Canadian defence capability growth.

17.3 Scalable Tactical ISR Capability

VectorSight-M is the primary prototype, but the underlying architecture supports a scalable tactical ISR family. The baseline multirotor configuration focuses on the True North Precision requirement for precision laser rangefinding, target geolocation, and real-time cueing. The same mission-system architecture can support future growth toward heavier-lift, longer-endurance, and VTOL/fixed-wing variants.

The scalable capability approach includes:

- **VectorSight-M:** primary multirotor prototype for precision rangefinding, target geolocation, tactical ISR, and final capability assessment.
- **VectorSight-CX:** heavier-lift growth configuration for increased payload margin, larger sensor packages, or longer mission endurance.
- **VectorSight-VTOL:** future VTOL/fixed-wing pathway for longer-radius ISR missions and extended-area coverage.

This architecture allows Zigaton to focus the funded prototype on a testable and relevant mission while preserving a growth path for broader tactical use cases. The result is a platform-family concept that can evolve with mission needs, payload availability, interoperability requirements, and Canadian defence priorities.

17.4 MINERVA Relevance

The VectorSight UAS concept is relevant to Canada's broader defence innovation priorities, including improved sensing, autonomous systems, data-enabled decision support, resilient operations, and scalable tactical capability. The system is designed to move from raw sensor observation to actionable target information through range measurement, target geolocation, operator cueing, data logging, and future interoperability pathways.

VectorSight-M supports these priorities through:

- Tactical sensing using EO/IR observation and laser rangefinding.
- Target-location output generated from range, aircraft pose, payload pointing, timestamping, and calibration data.
- Real-time cueing workflow for operator decision support.
- GNSS-degraded resilience pathway using GNSS, inertial sensing, RTK enhancement, visual odometry growth path, and degraded-mode procedures.
- Structured mission data, logs, test records, and configuration records for evidence-based capability assessment.
- Growth path toward interoperability, autonomy, and modular payload integration.

The project aligns with the need for systems that are not only capable in isolation, but also testable, adaptable, data-aware, and scalable across future defence applications.

17.5 Future Operational Growth Path

VectorSight-M establishes a practical foundation for future operational growth. The initial prototype focuses on precision rangefinding and target geolocation, while the architecture supports continued development in autonomy, interoperability, payload flexibility, communications resilience, and mission-specific variants.

Future growth areas include:

- Improved target-geolocation accuracy through refined calibration, higher-confidence navigation, improved payload stabilization, and enhanced error-budget control.
- ATAK/Cursor-on-Target integration pathway for tactical target-message output.
- STANAG-aligned growth for unmanned-system interfaces and full-motion-video handling.
- Advanced payload options, including improved EO/IR sensors, alternate laser rangefinding payloads, and future target-designation workflows where applicable.
- Improved GNSS-degraded operation through RTK, inertial fusion, visual odometry, obstacle awareness, and degraded-mode autonomy.
- EM-resilient communications growth through antenna optimization, alternate links, remote transmitter placement, and specialized data-transfer concepts.
- Expanded air vehicle configurations for heavier payloads, longer endurance, longer operational radius, and broader mission profiles.

These growth paths provide a credible route from prototype demonstration to operationally relevant capability maturation. They allow the system to evolve based on test data, user feedback, procurement priorities, and Canadian defence requirements.

17.6 Summary of Strategic Value

VectorSight UAS provides strategic value by addressing an immediate tactical capability gap while building a foundation for future Canadian UAS development. The proposed system is lower cost than high-end ISR platforms, modular enough for future payload growth, structured for verification, and designed around practical field use.

The strategic value can be summarized as follows:

- Provides a Canadian-led tactical UAS prototype for precision rangefinding and target geolocation.
- Bridges the gap between commercial observation drones and expensive high-end ISR systems.
- Supports platoon- and company-level ISR, target cueing, and mission-data collection.
- Creates a repeatable technical foundation for payload integration, mission compute, data handling, and verification.
- Strengthens Canadian defence industrial capability in UAS integration and prototype execution.
- Preserves growth paths for interoperability, autonomy, resilient navigation, EM-resilient communications, and future UAS variants.

Zigaton's proposed VectorSight-M prototype therefore delivers both near-term assessment value and long-term strategic value. It provides a focused, testable, and defensible response to the True North Precision challenge while establishing a scalable Canadian tactical UAS mission-system architecture.

Appendices

A Essential Outcomes Compliance Matrix

This appendix provides the detailed essential outcomes compliance matrix for the VectorSight-M prototype. The matrix links each essential outcome to the proposed technical approach, supporting design features, verification method, and expected evidence record.

EO	Outcome Area	VectorSight-M Compliance Approach	Verification Method	Evidence Record
EO1	Precision rangefinding and target cueing	VectorSight-M integrates a Class 1 eye-safe laser rangefinder with a stabilized EO/IR payload architecture, boresight calibration, timestamped range measurements, target-geolocation processing, and MGRS-compatible coordinate output. The system is designed to support range accuracy at 1 km, moving-target updates, low-latency cueing, and target-location reporting.	Datasheet review, boresight calibration, static range test, moving-target test, latency measurement, geolocation comparison to surveyed or independently verified ground truth.	Range test report, calibration record, geolocation logs, target-output records.
EO2	Platform performance, endurance, and radius	VectorSight-M is configured as a tactical multirotor UAS designed around a 3–4 km operational radius and minimum 30-minute endurance target under standard operating conditions. The platform performance approach includes mass control, propulsion sizing, battery-energy modelling, communications assessment, and mission-profile testing.	Weight budget, propulsion sizing analysis, hover test, mission-profile flight test, operational-radius test, power-log review.	Mass record, propulsion analysis, flight logs, power logs, endurance/radius report.

EO	Outcome Area	VectorSight-M Compliance Approach	Verification Method	Evidence Record
EO3	Navigation resilience and GNSS-degraded operation	The navigation architecture combines GNSS, inertial sensing, flight-controller state estimation, RTK enhancement pathway, and degraded-mode procedures. The system is designed to preserve mission safety and maintain range/geolocation workflow when GNSS quality is reduced or uncertain.	GNSS-quality monitoring, inertial fallback review, controlled degraded-navigation scenario testing, telemetry/log review, degraded-mode procedure check.	GNSS logs, degraded-mode test record, flight logs, operator procedure record.
EO4	Environmental and operational robustness	The airframe, payload bay, stabilized sensor interface, enclosure approach, flight-control tuning, and test plan are structured to support operation at 0 °C and above, wind conditions up to 10 m/s subject to safe test limits, and target-acquisition assessment in dust, smoke, haze, or other obscurant conditions where permitted.	Cold-operation check, wind flight test, payload stability review, obscurant target-acquisition test, environmental condition logging.	Environmental test record, operator notes, payload stability review, flight logs.
EO5	Safety and data handling	VectorSight-M uses a Class 1 eye-safe laser rangefinder baseline, flight-safety procedures, battery-safety controls, controlled mission logging, access-control review, configuration tracking, and an ITSP.10.171-compliance-oriented data-handling approach.	Laser classification evidence, safety procedure review, data-handling checklist, access-control review, logging/export test, configuration audit.	Laser safety file, safety checklists, data-handling record, export package, configuration audit.
EO6	Integrated system weight	VectorSight-M is designed to remain below the ≤ 25 kg integrated drone-plus-payload weight limit through mass budgeting, component selection, payload-bay control, power-system sizing, and configuration management.	Component weigh-in, mass properties review, final integrated system weigh-in, configuration-control review.	Mass budget, final weigh-in record, configuration record.

EO	Outcome Area	VectorSight-M Compliance Approach	Verification Method	Evidence Record
EO7	Final capability assessment readiness	The work plan is structured to deliver a prototype ready for the May 2027 final capability assessment window. Activities include final design, procurement, integration, bench testing, ground testing, flight testing, refinement, documentation, and final assessment readiness preparation.	Milestone reviews, test-readiness review, flight-test records, final verification review, final deliverable audit.	Milestone records, test reports, final technical data package, assessment readiness package.

Appendix A Summary

The VectorSight-M compliance approach is based on controlled integration and measurable verification. Each essential outcome is supported by a technical design path, test method, and evidence record. Detailed verification methods are described in Section 12 and Appendix G.

B Desired Outcomes Alignment Matrix

This appendix provides the desired outcomes alignment matrix for VectorSight-M. Desired outcomes are treated as differentiators and growth-path capabilities that strengthen the prototype and support future operational maturity.

DO	Outcome Area	VectorSight Alignment / Growth Path	Verification or Evidence Path	Status
DO1	Target designation growth path	The VectorSight architecture preserves payload, power, compute, gimbal, software, and data-interface pathways for future target-designation payload integration. The funded baseline remains focused on Class 1 eye-safe rangefinding while maintaining growth space for advanced payload variants.	Payload-bay design review, power/interface budget, vendor data, growth-path assessment.	Growth path
DO2	Advanced rangefinding accuracy	The architecture uses payload stabilization, boresight calibration, timestamp synchronization, range-data logging, and geolocation error-budget control to support improvement toward higher-precision rangefinding objectives.	Static range test, target-size sensitivity review, calibration report, geolocation error-budget update.	Prototype/growth
DO3	Enhanced environmental robustness	The design supports growth toward lower-temperature operation, higher wind tolerance, precipitation exposure, and obscurant performance evaluation through airframe hardening, payload protection, mounting design, and environmental test planning.	Environmental growth test plan, enclosure review, payload protection review, field-test data.	Growth path

DO	Outcome Area	VectorSight Alignment / Growth Path	Verification or Evidence Path	Status
DO4	Sensor-to-shooter / interoperability pathway	VectorSight is designed to output structured target cueing information through the ground-control workflow. The data architecture supports ATAK/Cursor-on-Target integration pathway, STANAG 4586/4609 alignment path, FMV handling, and future fires-adjustment workflow support.	GCS integration test, target-message export test, FMV handling review, interoperability demonstration where implemented.	Prototype/growth
DO5	Higher-speed target tracking growth	The moving-target architecture uses timestamped range measurements, update-rate control, target update logs, and operator-facing cueing. The baseline supports the challenge target-speed requirement and preserves a path toward higher-speed target tracking.	Moving-target test, update-rate measurement, latency measurement, track-continuity review.	Prototype/growth
DO6	Modular payload design	The platform uses a modular payload bay with defined mechanical, electrical, power, data, calibration, and maintainability interfaces to support payload replacement or upgrade without major redesign.	CAD/interface review, payload swap demonstration, connector/interface checklist, maintainability review.	Prototype
DO7	Autonomous navigation growth	VectorSight supports waypoint and semi-autonomous operation with a growth path toward improved GNSS-degraded operation using inertial fusion, visual odometry, obstacle-awareness features, and degraded-mode procedures.	Waypoint mission test, degraded-navigation assessment, visual-odometry integration review, flight-log analysis.	Prototype/growth
DO8	Supply-chain and origin transparency	Zigaton will maintain a country-of-origin matrix for critical systems including flight controller, data transmission devices, onboard computers, software elements, payload sensors, radios, and power electronics.	Vendor declarations, datasheets, procurement records, country-of-origin matrix.	Prototype

DO	Outcome Area	VectorSight Alignment / Growth Path	Verification or Evidence Path	Status
DO9	EM resilience and alternate communications pathway	The communications architecture includes EM-resilience considerations such as link-budget review, antenna placement planning, remote GCS transmitter options, alternate data-link planning, degraded-link procedures, and non-RF growth-path assessment where applicable.	Link-budget analysis, range test, antenna placement review, remote transmitter configuration test.	Prototype/growth

Appendix B Summary

The desired outcomes strengthen the VectorSight-M proposal by showing that the prototype is not a one-off airframe. The architecture is designed to support modular payload growth, interoperability, navigation resilience, supply-chain transparency, and future operational variants while keeping the funded prototype focused and testable.

C System Architecture Diagram

This appendix provides a text-based system architecture diagram for the VectorSight-M prototype. The diagram summarizes subsystem boundaries, data paths, power paths, command/control paths, payload sensing paths, target-geolocation processing, ground-control functions, and verification logging.

VectorSight-M System Architecture

Layer	**Major Functions and Interfaces**
Air Vehicle Layer	Multirotor frame, propulsion, landing gear, payload bay, battery bay, avionics bay, mechanical mounting, center-of-gravity control, field-maintenance access.
Propulsion and Energy Layer	Motors, propellers, ESCs, main flight battery, battery retention, power input protection, current/voltage sensing, reserve-energy monitoring.
Power Distribution Layer	Regulated rails, payload power, avionics power, mission-compute power, communications power, fusing, transient protection, grounding strategy, harnessing, connectors.
Payload Layer	EO/IR sensor, Class 1 eye-safe laser rangefinder, stabilized mount or gimbal, payload status, payload pointing state, boresight references, range measurement output.
Navigation Layer	GNSS, RTK enhancement pathway, IMU, barometer, compass, flight-controller state estimation, navigation-quality monitoring, degraded-mode state.
Mission Compute Layer	Range association, aircraft pose association, payload pointing association, timestamping, target-geolocation logic, cueing output, quality indicators, mission logging.
Communications Layer	Command/control link, telemetry link, video link, target-data link, antenna placement, link-quality monitoring, degraded-link procedures.
Ground-Control Layer	Aircraft telemetry, live video, payload status, target range, target coordinates, MGRS-compatible output, mission logs, data export, operator workflow.
Verification Data Layer	Flight logs, range logs, payload logs, power logs, communications logs, calibration records, configuration records, test reports, exportable evidence package.

Figure 2: VectorSight-M system architecture layer summary.

Primary Architecture Flows

Architecture Flow **Evidence / Output**	**Description**
Command and control flow	Operator commands are transmitted from the ground-control system to the flight controller, mission compute layer, and payload control interfaces where applicable.
Command logs, telemetry logs, operator records.	
Payload sensing flow	EO/IR imagery and rangefinder measurements are collected through the stabilized payload architecture and passed to the operator workflow and mission compute layer.
Video records, payload logs, range logs.	
Navigation and pose flow	GNSS, inertial, flight-controller, and payload pointing data are associated with range measurements to support target geolocation.
Flight logs, pose records, timestamp records.	

Architecture Flow Evidence / Output	Description
Target cueing flow	Measured range, aircraft pose, payload pointing, timestamp, and calibration data are processed into target-location and cueing outputs.
Target-coordinate logs, MGRS-compatible output, cueing records.	
Power and health-monitoring flow	Battery, current, voltage, regulated rails, payload state, communications state, and aircraft health are monitored for safety and verification.
Power logs, health telemetry, inspection records.	
Verification flow	Mission logs, test records, calibration files, configuration records, and output data are preserved for post-test analysis and final reporting.
Technical data package, test reports, final assessment package.	

Architecture Control Notes

The architecture will be controlled through interface documentation, BOM records, wiring and connector records, software/firmware version records, calibration records, and test logs. Any change affecting flight safety, payload pointing, rangefinding accuracy, target geolocation, communications, power, or data handling will be captured in the configuration-control record before further major testing.

D Geolocation Error Budget

This appendix provides the preliminary geolocation error budget for VectorSight-M. The purpose of the error budget is to identify the major contributors to target-location error and connect each contributor to a mitigation and verification method.

VectorSight treats target geolocation as a system-level performance outcome. Accuracy depends not only on rangefinder performance, but also on aircraft position, aircraft attitude, payload pointing, boresight alignment, timestamp association, target definition, and environmental conditions.

Geolocation Data Inputs

The target-geolocation calculation uses the following primary data inputs:

- Laser-measured range to the selected target.
- Aircraft GNSS or RTK-enhanced position.
- Aircraft attitude from flight-controller and inertial sensors.
- Payload pointing angle or stabilized mount/gimbal orientation.
- Timestamp associated with range measurement, aircraft pose, and payload pointing state.
- Boresight calibration data between EO/IR sensor, laser rangefinder, payload frame, and aircraft reference frame.
- Target observation and operator confirmation record.

Preliminary Error Contributors

Error Contributor	Effect on Target Location	Mitigation / Control Approach	Verification Method
Range measurement error	Incorrect measured distance shifts the target estimate along the payload line of sight.	Vendor-supported Class 1 rangefinder selection, repeated known-distance measurements, measurement filtering, target-surface awareness.	Static range test, datasheet review, range repeatability test.

Error Contributor	Effect on Target Location	Mitigation / Control Approach	Verification Method
Aircraft position error	Uncertainty in aircraft position shifts the starting point of the target-location projection.	GNSS quality monitoring, RTK enhancement pathway, flight-log review, known-position comparison.	Ground truth comparison, GNSS log review, RTK validation where implemented.
Aircraft attitude error	Roll, pitch, or yaw uncertainty affects the line-of-sight direction used for target projection.	Flight-controller calibration, IMU checks, stable measurement profile, attitude log review.	Flight-log review, hover/stability test, attitude consistency check.
Payload pointing error	Incorrect gimbal or stabilized mount pointing state causes angular projection error.	Stabilized mount/gimbal feedback, mechanical interface control, payload stability review.	Payload pointing review, stability test, gimbal log review where available.

Error Contributor	**Effect on Target Location**	**Mitigation / Control Approach**	**Verification Method**
Boresight alignment error	Misalignment between camera, rangefinder, payload frame, and aircraft frame causes systematic target-location error.	Boresight calibration, alignment references, post-maintenance re-check, calibration records.	Boresight calibration report, known-target verification.
Timestamp synchronization error	Mismatch between range time, aircraft pose time, and payload pointing time causes error during aircraft motion or target motion.	Timestamped measurement events, sensor-state association, log review, software timing checks.	Bench timing test, flight-log review, moving-target test.
Target definition error	Target size, surface geometry, reflectivity, obscuration, or operator selection may affect the measured point on the target.	Operator confirmation, target-size test cases, repeated observations, target-surface documentation.	Static target trials, moving-target trials, video review.
Environmental error	Wind, vibration, visibility, precipitation, dust, smoke, haze, or thermal conditions can affect flight stability, pointing, sensing, and range consistency.	Environmental condition logging, vibration-aware mounting, payload stabilization, test-condition limits.	

Error Contributor	Effect on Target Location	Mitigation / Control Approach	Verification Method
			Environmental test report, payload stability review, flight-log review.
Data handling / processing error	Incorrect coordinate conversion, data association, or export formatting can affect reported target output.	Software version control, test datasets, coordinate-output review, configuration control.	Bench software test, coordinate-output comparison, log-export test.

Preliminary Error-Budget Structure

The detailed numerical error budget will be refined during integration and testing. The preliminary structure below will be used to capture measured and estimated contributors.

Error Source	**Pre-Test Estimate**	**Measured / Updated Value**	**Notes**
Laser range error	TBD	TBD	Updated after known-distance range tests.
Aircraft position error	TBD	TBD	Updated using GNSS/RTK logs and ground-truth comparison.
Aircraft attitude error	TBD	TBD	Updated using flight-controller logs and stability review.
Payload pointing error	TBD	TBD	Updated after payload stability and pointing review.
Boresight alignment error	TBD	TBD	Updated after boresight calibration and known-target checks.
Timestamp association error	TBD	TBD	Updated after bench timing and flight-log review.
Target definition error	TBD	TBD	Updated from target trials and video review.
Environmental contribution	TBD	TBD	Updated based on wind, vibration, visibility, and field conditions.
Coordinate conversion / processing error	TBD	TBD	Updated through software test and coordinate-output validation.

Verification Approach

The geolocation error budget will be verified and refined using the following activities:

- Static range testing against known-distance targets.
- Target geolocation testing against surveyed or independently measured target locations.
- Moving-target testing to evaluate update continuity and timestamp effects.
- Boresight calibration and calibration re-checks.
- GNSS/RTK quality review and aircraft position comparison.
- Payload stability review using video, pointing logs, or operator observations.
- Flight-log review for aircraft attitude, vibration, navigation state, and timing.
- Coordinate-output comparison against known target references.

- Environmental condition logging during each test event.

Appendix D Summary

The geolocation error budget will become more precise as VectorSight-M progresses through integration, calibration, bench testing, ground testing, and flight testing. The final error budget will be included in the technical data package and will support the final verification summary.

E Preliminary BOM

This appendix provides the preliminary bill of materials structure for the VectorSight-M prototype. The BOM will be refined during final design, vendor quotation, procurement, integration, and configuration-control activities.

The preliminary BOM is organized by subsystem so that cost, lead time, supplier readiness, country of origin, integration risk, and verification evidence can be tracked against the prototype configuration.

Preliminary BOM Structure

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
Airframe and structure	Multirotor frame / structural kit	Primary aircraft structure, payload carriage, landing gear interface, avionics and battery mounting.	TBD	Candidate selection	Datasheet, CAD data, mass record.
Airframe and structure	Landing gear	Ground clearance, payload protection, launch/recovery support.	TBD	Candidate selection	Datasheet, CAD review, integration inspection.

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
Airframe and structure	Payload bay hardware	Modular payload support, alignment references, access provisions, mounting interface.	TBD	Design/procure	CAD record, interface drawing, installation record.
Propulsion	Motors	Lift and control authority for integrated payload configuration.	TBD	Candidate selection	Datasheet, thrust data, supplier record.
Propulsion	Propellers	Thrust generation and efficiency support for endurance/radius objectives.	TBD	Candidate selection	Datasheet, thrust data, spare-parts record.

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
Propulsion	ESCs	Motor drive control and propulsion power interface.	TBD	Candidate selection	Datasheet, current rating, thermal data.
Energy storage	Flight battery pack	Primary energy source for propulsion, payload, avionics, mission compute, and communications.	TBD	Candidate selection	Datasheet, capacity/discharge rating, battery record.
Energy storage	Battery charger and accessories	Safe battery charging, field support, and test readiness.	TBD	Candidate selection	Charger documentation, safety procedure.

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
Avionics/navigation	Flight controller	Flight stabilization, navigation, failsafe behavior, telemetry, and flight logging.	1	Candidate selection	Datasheet, firmware record, configuration file.
Avionics/navigation	GNSS/RTK receiver	Navigation position source, target-geolocation reference, and log correlation.	TBD	Candidate selection	Datasheet, GNSS/RTK capability evidence.
Avionics/navigation	Telemetry radio / data radio	Aircraft telemetry and command/control or supporting data-link function.	TBD	Candidate selection	Datasheet, range data, frequency/interface record.

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
Avionics/navigation	Antennas and RF accessories	Link quality, telemetry, video, and target-data communication support.	TBD	Candidate selection	Datasheet, antenna placement record.
Payload	EO/IR payload	Target search, observation, operator confirmation, and FMV workflow.	1	Candidate selection	Datasheet, video interface, payload record.
Payload	Class 1 laser rangefinder	Precision range measurement for target geolocation and cueing.	1	Candidate selection	Datasheet, Class 1 laser safety evidence.
Payload	Gimbal / stabilized mount	Payload pointing stability, observation quality, range-measurement repeatability.	1	Candidate selection	

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
					Datasheet, payload mass/range compatibility.
Payload	Payload interface hardware	Mechanical, electrical, and data connection between payload and aircraft system.	TBD	Design/procure	Interface record, wiring record, installation record.
Mission compute	Onboard mission computer	Range association, timestamping, target-geolocation logic, logging, and cueing output.	1	Candidate selection	Datasheet, software/configuration record.
Mission compute	Storage media	Mission logs, video records, target outputs, and verification data.	TBD	Candidate selection	

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
					Capacity record, data-handling record.
Communications/GCS	Video link hardware	EO/IR video transmission to ground-control workflow.	TBD	Candidate selection	Datasheet, link test record.
Communications/GCS	Ground-control station equipment	Operator interface, telemetry, video, payload status, target range, target coordinates, and logs.	TBD	Candidate selection	Equipment record, GCS configuration.
Power distribution	Power distribution hardware	Controlled distribution to propulsion, avionics, payload, compute, and communications.	TBD	Design/procure	Schematic, inspection record, bench test.

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
Power distribution	Regulated power modules	Stable voltage rails for avionics, payload, compute, and communications.	TBD	Candidate selection	Datasheet, bench power test.
Power distribution	Current/voltage sensors	Power monitoring, endurance validation, diagnostics, and safety review.	TBD	Candidate selection	Datasheet, calibration/test record.
Power distribution	Connectors, fuses, wiring, harness materials	Safe electrical interconnection, protection, serviceability, and configuration control.	TBD	Procure	Wiring record, inspection record.
Test/support	Calibration targets and measurement tools	Rangefinding, geolocation, boresight, and test verification support.	TBD	Procure	

Subsystem	Item / Component	Function	Qty.	Status	Evidence Required
					Test equipment record.
Test/support	Field tools and safety equipment	Integration, field repair, battery safety, laser safety, and flight-test readiness.	TBD	Procure	Safety checklist, tool record.
Spares	Propellers, connectors, fasteners, landing gear parts, payload-mounting parts	Downtime reduction during integration, field testing, and final assessment preparation.	TBD	Procure	Spare-parts record, BOM update.

BOM Control Notes

The BOM will be controlled as a living configuration document. Each item will be updated with supplier, manufacturer, part number, quantity, cost basis, lead time, country of origin, substitute option, received status, and verification evidence. Any substitution affecting flight safety, payload performance, rangefinding, target geolocation, communications, power distribution, or data handling will require configuration review before installation.

Appendix E Summary

The preliminary BOM supports procurement planning, cost justification, supply-chain transparency, configuration control, and final technical reporting. The finalized BOM will be included in the technical data package and linked to the tested VectorSight-M prototype configuration.

F Vendor and Country-of-Origin Matrix

This appendix provides the vendor and country-of-origin matrix structure for critical VectorSight-M components and subsystems. The matrix supports supply-chain transparency, procurement control, risk management, and defence evaluation.

Vendor and Origin Tracking Approach

For each critical component, Zigaton will track supplier information, manufacturer information, part number, country of origin, country of final assembly where available, lead time, documentation source, alternate supplier options, and configuration-control relevance.

Subsystem / Item	Supplier / Vendor	Manufacturer / Part No.	Country of Origin	Lead-Time Status	Alternate Source
Airframe structure	TBD	TBD	TBD	TBD	TBD
Landing gear	TBD	TBD	TBD	TBD	TBD
Payload bay hardware	TBD	TBD	TBD	TBD	TBD
Motors	TBD	TBD	TBD	TBD	TBD
Propellers	TBD	TBD	TBD	TBD	TBD
ESCs	TBD	TBD	TBD	TBD	TBD
Flight battery	TBD	TBD	TBD	TBD	TBD
Battery charger	TBD	TBD	TBD	TBD	TBD
Flight controller	TBD	TBD	TBD	TBD	TBD
GNSS/RTK receiver	TBD	TBD	TBD	TBD	TBD
Telemetry radio	TBD	TBD	TBD	TBD	TBD
Video link hardware	TBD	TBD	TBD	TBD	TBD
EO/IR payload	TBD	TBD	TBD	TBD	TBD
Class 1 laser rangefinder	TBD	TBD	TBD	TBD	TBD
Gimbal / stabilized mount	TBD	TBD	TBD	TBD	TBD
Mission compute hardware	TBD	TBD	TBD	TBD	TBD
Ground-control station equipment	TBD	TBD	TBD	TBD	TBD

Subsystem / Item	Supplier / Vendor	Manufacturer / Part No.	Country of Origin	Lead-Time Status	Alternate Source
Regulated power modules	TBD	TBD	TBD	TBD	TBD
Current/voltage sensors	TBD	TBD	TBD	TBD	TBD
Connectors, fuses, and harness materials	TBD	TBD	TBD	TBD	TBD
Calibration and test equipment	TBD	TBD	TBD	TBD	TBD

Critical Supplier Documentation

Vendor and supplier documentation will be collected to support technical review, procurement readiness, country-of-origin tracking, configuration control, and final reporting.

Documentation Type	Purpose	Applies To
Datasheet	Confirms electrical, mechanical, interface, environmental, and performance characteristics.	Critical hardware and payloads.
Vendor quotation	Supports budget, procurement, lead-time, and cost justification.	Major purchased items.
Country-of-origin evidence	Supports supply-chain transparency and defence evaluation.	Critical components and subsystems.
Interface documentation	Supports mechanical, electrical, data, power, and software integration.	Payload, avionics, communications, compute.
Laser safety documentation	Confirms Class 1 eye-safe laser classification and safe-use requirements.	Laser rangefinder.
Software/firmware license information	Identifies open-source, commercial, vendor-controlled, or Zigaton-developed software elements.	Flight controller, mission compute, GCS, payload interfaces.
Alternate supplier record	Reduces procurement, schedule, and availability risk.	Critical and long-lead items.

Origin and Substitution Control

Any substitution affecting flight control, rangefinding, target geolocation, payload pointing, communications, power distribution, data handling, or safety will trigger a review of interface compatibility, country of origin, documentation, cost, lead time, software/firmware compatibility, and verification impact. Approved substitutions will be added to the BOM, country-of-origin matrix, and configuration-control record.

Appendix F Summary

The vendor and country-of-origin matrix supports transparency and procurement discipline. It ensures that critical subsystem choices remain traceable from supplier records to prototype configuration and final test evidence.

G Test Matrix

This appendix provides the detailed test matrix structure for the VectorSight-M verification and validation program. The test matrix links challenge outcomes, technical requirements, test methods, success criteria, data records, and evidence products.

Verification Test Matrix

Test ID Phase	Requirement Area	Test Method	Success Criteria / Objective	Evidence
T-01 Phase 3	Power-system validation	Bench test	Regulated rails, current/voltage sensing, fusing, transient protection, and harnessing operate within expected limits before flight.	Bench power record, inspection checklist.
T-02 Phase 3	Flight-controller configuration	Bench and ground test	Flight controller, sensors, GNSS, telemetry, failsafe, geofence, and return/land settings are configured and reviewed.	Configuration file, checklist, log record.

Test ID	Requirement Area	Test Method	Success Criteria / Objective	Evidence	Phase
T-03	Payload communication	Bench and ground test	EO/IR payload, rangefinder, gimbal/stabilized mount, and mission compute interface communicate correctly.	Payload test record, communication logs.	Phase 3
T-04	Ground-control workflow	Bench and ground test	Operator can view telemetry, video, payload status, range status, target data, and logging status.	GCS test record, screenshots where appropriate, operator checklist.	Phase 3
T-05	Laser safety verification	Inspection and documentation review	Installed rangefinder matches documented Class 1 safety file and safe-use procedure.	Laser safety file, checklist, configuration record.	Phase 3

Test ID	Requirement Area	Test Method	Success Criteria / Objective	Evidence	Phase
T-06	Boresight calibration	Ground test	EO/IR line of sight, rangefinder axis, payload frame, and aircraft reference frame are calibrated and recorded.	Calibration report, target images/logs.	Phase 3/4
T-07	Static rangefinding	Ground/flight test	Range measurements are compared against known-distance targets, including representative longer-range conditions where test-site conditions allow.	Range test report, logs, ground-truth record.	Phase 4
T-08	Target geolocation	Flight test	Target-location output is generated and compared against surveyed or independently verified target locations.	Geolocation report, target-output logs.	Phase 4

Test ID	Requirement Area	Test Method	Success Criteria / Objective	Evidence	Phase
T-09	Moving-target update	Flight/controlled trial	System supports repeated target updates, latency review, and output continuity for controlled moving-target conditions.	Moving-target report, latency logs.	Phase 4
T-10	Endurance	Hover and mission-profile flight	Flight time and energy usage are measured with installed payload and mission configuration.	Flight logs, battery records, power logs.	Phase 4
T-11	Operational radius	Ground link test and flight test	Command/control, telemetry, video, and target-data links support progressive mission-radius expansion under safe conditions.	Link logs, flight logs, operator notes.	Phase 4

Test ID	Requirement Area	Test Method	Success Criteria / Objective	Evidence	Phase
T-12	GNSS quality and degradation	Ground/flight scenario test	System monitors GNSS quality and demonstrates defined degraded-mode behavior where safe and permitted.	GNSS logs, degraded-mode report.	Phase 4
T-13	Wind performance	Progressive flight test	Aircraft control, payload stability, power draw, and operator workload are evaluated under increasing wind conditions within safe limits.	Environmental record, flight logs.	Phase 4
T-14	Temperature operation	Ground/flight check	Prototype operation is checked at 0 °C and above where environmental conditions allow.	Cold-operation record, inspection notes.	Phase 4

Test ID	Requirement Area	Test Method	Success Criteria / Objective	Evidence	Phase
T-15	Obscurant target acquisition	Controlled field test	Target observation and rangefinding workflow are assessed in dust, smoke, haze, or obscurant conditions where safe and permitted.	Environmental test record, video/log review.	Phase 4
T-16	Data handling and export	Bench/ground/flight data review	Mission logs, range records, target outputs, video, and configuration records are exported and reviewed.	Export package, data-handling record.	Phase 3/4/5
T-17	Configuration audit	Inspection and document review	Final prototype configuration matches BOM, firmware/software records, calibration records, and test logs.	Configuration audit record.	Phase 5

Test ID	Requirement Area	Test Method	Success Criteria / Objective	Evidence	Phase
T-18	Final readiness review	Review event	Prototype, test records, safety records, operator guide, maintenance notes, and technical data package are ready for assessment.	Final readiness package.	Phase 5

Test Record Fields

Each test record will include:

- Test ID and requirement reference.
- Prototype configuration state.
- Date, location, personnel, and test environment.
- Test procedure, test card, or checklist reference.
- Equipment used and ground-truth reference method where applicable.
- Data recorded, including logs, measurements, video, telemetry, and operator notes.
- Pass/fail or objective completion assessment.
- Issues, corrective actions, and re-test requirement where applicable.

Appendix G Summary

The test matrix provides the evidence structure for final verification. Each test is linked to a requirement area, test method, success objective, evidence record, and project phase.

H Risk Register

This appendix provides the preliminary risk register for the VectorSight-M project. Risks will be reviewed and updated throughout the project as procurement, integration, testing, and final assessment preparation progress.

Risk Rating Approach

Risks will be assessed using likelihood, consequence, mitigation status, owner, and residual risk. The register will be updated during milestone reviews and after major test events.

Risk Field	Description
Risk ID	Unique identifier for tracking.
Risk description	Clear statement of the risk condition and potential impact.
Likelihood	Low, Medium, or High probability of occurrence.
Consequence	Low, Medium, or High impact on safety, schedule, performance, or cost.
Mitigation action	Planned or active action to reduce likelihood or consequence.
Owner	Responsible team member, vendor, or support role.
Status	Open, monitoring, mitigated, closed, or transferred.
Residual risk	Remaining risk after mitigation.

Preliminary Risk Register

ID	Risk Description	Lik.	Cons.	Mitigation Action	Status
R-01	Payload mass exceeds allocation and reduces endurance or handling margin.	Med.	High	Maintain mass budget, payload envelope, CAD review, component weigh-in, and contingency mass allocation.	Open

ID	Risk Description	Lik.	Cons.	Mitigation Action	Status
R-02	Rangefinder integration does not achieve required measurement repeatability.	Med.	High	Use vendor-supported Class 1 rangefinder, controlled mounting, boresight calibration, and repeated known-distance testing.	Open
R-03	Boresight alignment error affects target-geolocation accuracy.	Med.	High	Define alignment references, calibration procedure, re-check triggers, and known-target verification.	Open
R-04	Timestamp mismatch causes target-location error during motion.	Med.	High	Timestamp range, pose, and pointing data; review logs; conduct bench timing and moving-target tests.	Open

ID	Risk Description	Lik.	Cons.	Mitigation Action	Status
R-05	Endurance target is reduced by mass growth, payload power, or wind.	Med.	High	Control mass, validate power model, use flight power logs, preserve reserve margin, refine mission profile.	Open
R-06	Operational-radius performance limited by communications link margin.	Med.	High	Conduct link-budget review, antenna placement review, ground range testing, and progressive radius testing.	Open
R-07	GNSS degradation reduces geolocation confidence.	Med.	Med.	Monitor GNSS quality, preserve RTK path, use inertial fallback, define degraded-mode procedures.	Open

ID	Risk Description	Lik.	Cons.	Mitigation Action	Status
R-08	Battery or power-system issue affects flight safety.	Low/Med.	High	Use battery procedures, power bench testing, current/voltage sensing, fusing, inspection, and reserve thresholds.	Open
R-09	Vendor lead time delays critical subsystem delivery.	Med.	Med.	Identify long-lead items early, request quotations, track procurement, maintain alternate suppliers.	Open
R-10	Payload stabilization insufficient under wind or vibration.	Med.	Med.	Use stabilized mount/gimbal, vibration-aware mounting, hover tests, payload stability review, tuning refinement.	Open

ID	Risk Description	Lik.	Cons.	Mitigation Action	Status
R-11	Test-site or flight authorization constraints delay flight validation.	Med.	Med.	Identify test-site requirements early, plan progressive tests, prepare flight operations and safety documentation.	Open
R-12	Data logs incomplete or not traceable to configuration state.	Low/Med.	Med.	Implement log checklist, configuration-control linkage, export test, and final data audit.	Open
R-13	Software or mission-data workflow error affects coordinate output.	Med.	High	Use version control, bench software tests, known-target comparison, coordinate-output review, and issue tracking.	Open

ID	Risk Description	Lik.	Cons.	Mitigation Action	Status
R-14	Weather delays wind, endurance, or environmental testing.	Med.	Med.	Preserve schedule margin, use phased testing, capture available condition data, plan alternate test windows.	Open
R-15	Configuration changes occur without sufficient re-test.	Low/Med.	High	Enforce change log, configuration review, inspection, and re-test rule for safety or performance-impacting changes.	Open

Risk Review Process

The risk register will be reviewed during:

- Final design and procurement review.
- Airframe/payload integration review.
- Bench and ground test review.
- Flight-test readiness review.
- Post-flight test review.
- Final refinement and assessment readiness review.

Risks will remain open until mitigation actions are completed and evidence confirms that the residual risk is acceptable for the next project phase.

Appendix H Summary

The risk register supports safe and disciplined prototype execution. It links technical and program risks to mitigation actions, owners, status, and verification evidence.

Direct recommendation: paste this immediately after Appendix H. This creates the final Appendix I–K and completes the appendix set.

I Budget Detail

This appendix provides the detailed budget structure for the VectorSight-M project. The final dollar values will be completed after vendor quotations, labour allocation, subcontractor estimates, test-site assumptions, and procurement decisions are finalized.

The budget detail is organized to show traceability between project cost categories, work-plan phases, milestone deliverables, and verification outcomes.

Budget Summary Structure

Budget Category	Description	Basis of Estimate	Amount
Labour	Engineering, systems integration, project management, documentation, testing, reporting, and final assessment readiness.	Labour-hour estimate by project phase and role.	TBD
Airframe and structure	Multirotor frame, landing gear, payload bay, mounting hardware, enclosures, fasteners, vibration-isolation hardware, and structural accessories.	Vendor quotation, supplier pricing, CAD/material estimate.	

Budget Category	Description	Basis of Estimate	Amount
Propulsion and energy	Motors, propellers, ESCs, flight batteries, chargers, battery retention, propulsion wiring, and related accessories.	Vendor quotation, datasheet review, propulsion sizing.	TBD
Avionics and navigation	Flight controller, GNSS/RTK hardware, inertial sensors, telemetry hardware, antennas, and avionics accessories.	Vendor quotation, supplier records, integration requirement.	TBD
Payload system	EO/IR payload, Class 1 eye-safe laser rangefinder, stabilized mount or gimbal, payload interface hardware, and payload protection.	Vendor quotation, payload integration estimate.	TBD
Mission compute and communications	Onboard compute, video link, target-data link, ground-control station equipment, operator interface, storage media, and related accessories.	Vendor quotation, interface requirement, GCS configuration.	TBD

Budget Category	Description	Basis of Estimate	Amount
Power distribution and harnessing	Power distribution hardware, regulated power modules, current/voltage sensing, connectors, fuses, transient protection, wiring, labels, and harness materials.	Supplier pricing, schematic/BOM estimate, integration estimate.	TBD
Test and support equipment	Calibration targets, measurement equipment, field tools, safety equipment, cases, charging support, and data-storage media.	Test-plan estimate and supplier pricing.	TBD
Subcontractors/vendors/advisors	Specialized support for payload integration, fabrication, software, cybersecurity/data handling, flight testing, safety, or environmental testing.	Vendor quotation or scope-of-work estimate.	TBD
Travel and field testing	Test-site support, field travel, transport of prototype and equipment, observer support, target setup, and final assessment preparation.	Test-plan estimate and travel assumption.	TBD

Budget Category	**Description**	**Basis of Estimate**
Amount		
TBD Spare parts	Propellers, connectors, landing gear parts, payload-mount hardware, fuses, fasteners, harness materials, antennas, batteries, and field-repair items.	BOM spare allocation and test-readiness estimate.
TBD Contingency	Controlled allowance for procurement changes, integration refinements, repeat testing, corrective actions, and schedule-risk mitigation.	Percentage or risk-based allocation.
TBD		

Labour Detail

Labour will be estimated by project role and phase. The final labour budget will tie each cost element to a work-plan phase, milestone deliverable, or verification output.

Labour Role Amount	Primary Activities	Hours	Rate
Project management and systems engineering	Requirements control, compliance mapping, milestone reviews, risk tracking, reporting, and stakeholder coordination.	TBD	TBD
TBD Electrical and avionics integration	Power architecture, flight-controller interfaces, regulated rails, sensing, telemetry, wiring, harnessing, and configuration control.	TBD	TBD
TBD Mechanical design and integration	CAD, payload bay, mounting, enclosure, structural review, maintainability, and assembly support.	TBD	TBD

Labour Role	Primary Activities	Hours	Rate	Amount
Payload and mission-system integration	EO/IR payload, laser rangefinder, stabilized mount, boresight calibration, payload data flow, and cueing workflow.	TBD	TBD	TBD
Software, logging, and data workflow	Timestamping, target-geolocation logic, mission logs, data export, GCS workflow, and configuration tracking.	TBD	TBD	TBD
Testing and documentation	Bench testing, ground testing, flight-test support, test reports, operator guide, maintenance notes, and final data package.	TBD	TBD	TBD

Materials and Equipment Detail

The materials and equipment budget will be supported by the preliminary BOM in Appendix E and vendor/country-of-origin records in Appendix F.

Material Category	**Included Items**	**Procurement Status**	**Amount**
Airframe and structure	Frame, arms, landing gear, payload bay, enclosures, fasteners, mounting hardware, and vibration-isolation hardware.	TBD	TBD
Propulsion and energy	Motors, propellers, ESCs, batteries, chargers, battery retention, wiring, and propulsion accessories.	TBD	TBD
Avionics and navigation	Flight controller, GNSS/RTK receiver, inertial sensors, telemetry hardware, antennas, and supporting electronics.	TBD	TBD
Payload system	EO/IR payload, Class 1 laser rangefinder, gimbal/stabilized mount, payload interface hardware, and payload protection.	TBD	

Material Category **Amount**	**Included Items**	**Procurement Status**
TBD Mission compute and communications	Onboard computer, storage media, video link, target-data link, ground-control equipment, operator display, and accessories.	TBD
TBD Power distribution	Power distribution hardware, regulated rails, current/voltage sensing, fuses, connectors, transient protection, harness materials, and test points.	TBD
TBD Test and support equipment	Calibration targets, measurement tools, field tools, cases, chargers, safety equipment, and data-storage media.	TBD

Material Category	Included Items	Procurement Status
Amount		
TBD Spare parts	Propellers, connectors, fasteners, antennas, landing gear parts, payload-mount parts, fuses, harness materials, and field-repair items.	TBD
TBD		

Cost-to-Milestone Budget Mapping

The budget will be mapped to the project milestones defined in Section 11. This ensures that funding supports measurable technical progress.

Milestone	Phase	Cost Basis	Amount
			M1
Final Design and Procurement	Engineering labour, requirements control, vendor review, preliminary BOM, procurement package, and long-lead items.	TBD M2	Airframe/Payload Integration
Airframe, propulsion, payload bay, avionics, mission compute, power distribution, wiring, and mechanical integration.	TBD M3	Bench and Ground Testing	Bench equipment, power validation, payload ground testing, communications testing, GCS workflow, and safety review.
TBD M4	Flight Testing and Accuracy Validation	Flight testing, rangefinding validation, geolocation testing, moving-target trials, endurance/radius tests, and field support.	TBD M5
Refinement and Assessment Readiness	Corrective actions, final configuration audit, final documentation, final test report, operator guide, and assessment package.	TBD	

Budget Control Notes

Budget control will be maintained through vendor quotations, procurement records, BOM updates, configuration-control records, and cost-to-milestone tracking. Any contingency usage will be linked to procurement changes, risk-register items, test findings, corrective actions, or final readiness needs.

Appendix I Summary

The budget detail appendix provides the structure required to complete the final cost proposal once vendor pricing and labour assumptions are locked. It ensures that the funding request is traceable to the technical work plan, prototype deliverables, verification activities, and final assessment readiness.

J Key Personnel Bios

This appendix provides key personnel summaries for the VectorSight-M project. The core team will retain system-level ownership and use vendors, advisors, and subcontractors where specialized support improves delivery quality, reduces schedule risk, or strengthens final prototype readiness.

John Oladunjoye Olatomiwa

Name: John Oladunjoye Olatomiwa **Role:** Co-founder; Electrical, Avionics, and UAV Systems Integration Lead **Organization:** Zigaton Consulting Ltd **Email:** johnoladunjoye@zigaton.com

John Oladunjoye Olatomiwa serves as Co-founder and Electrical, Avionics, and UAV Systems Integration Lead for the VectorSight UAS project. He is responsible for system-level integration of the air vehicle, electrical power architecture, avionics stack, flight-controller interfaces, telemetry, payload electronics, mission compute interfaces, vendor coordination, requirements traceability, verification documentation, and technical data package development.

John Oladunjoye Olatomiwa has an Electrical Engineering background from Carleton University and engineering experience from Hydro One, independent UAV research, electronics design, power distribution design, technical documentation, and system-integration work. His technical strengths include electrical and electronic systems, avionics architecture, power systems, CAD-supported integration, propulsion and aerodynamic trade studies, PCB-level design thinking, configuration control, and structured engineering verification.

For VectorSight-M, John Oladunjoye Olatomiwa will lead:

- UAV systems integration and project technical coordination.
- Electrical and avionics architecture.
- Power distribution and regulated subsystem rails.
- Flight-controller, telemetry, payload, and mission-compute interfaces.
- Sensor and payload interface planning.
- Communications integration and ground-control workflow support.
- Requirements traceability and verification planning.
- Technical proposal leadership, documentation, and final data package development.

Benjamin Apori

Name: Benjamin Apori **Role:** Co-founder; Mechanical Design and Logistics Lead **Organization:** Zigaton Consulting Ltd **Email:** benjaminapori@zigaton.com

Benjamin Apori serves as Co-founder and Mechanical Design and Logistics Lead for the VectorSight UAS project. He supports the mechanical architecture of the air vehicle, payload bay, structural layout, mounting interfaces, vibration-aware integration, manufacturability, logistics planning, and prototype assembly readiness.

Benjamin Apori is pursuing engineering studies at Carleton University and has relevant industry experience through an internship with Raytheon. His technical strengths include finite element analysis, CAD modelling, structural design, mechanical integration, logistics coordination, supplier readiness, and mechanical engineering competencies relevant to UAV airframe and payload integration.

For VectorSight-M, Benjamin Apori will lead or support:

- Airframe mechanical design and payload-bay layout.
- CAD model development and FEA-supported structural review.
- Gimbal, payload, and sensor mounting design.
- Mechanical interface definition and maintainability planning.
- Supplier coordination and component readiness tracking.
- Mechanical BOM support and procurement readiness.
- Field-maintenance planning and mechanical test-readiness support.
- Logistics planning for integration, testing, and final assessment preparation.

External Support Roles

Zigaton may use vendors, advisors, or subcontractors for specialized work packages where external support improves delivery quality, reduces risk, or strengthens final prototype readiness.

Potential external support roles include:

- EO/IR payload and laser rangefinder vendor support.
- Gimbal or stabilized payload integration support.
- Airframe fabrication, machining, composite, or metalwork support.
- Flight-test site, RPAS operations, or range-safety support.
- Mission-data workflow, target-cueing export, ATAK/CoT pathway, or software integration support.
- Cybersecurity, data-handling, or ITSP.10.171-oriented review support.
- Environmental, vibration, or safety testing support.
- Documentation, quality, or configuration-management review support.

External support will be managed through defined work packages, deliverables, interface requirements, procurement records, and acceptance criteria. Zigaton will retain overall system-level ownership of the VectorSight-M prototype.

Appendix J Summary

The VectorSight-M team combines electrical and avionics integration, mechanical design, systems engineering, prototype execution, vendor coordination, and documentation capability. The personnel structure supports a focused prototype effort while preserving the ability to bring in specialized support where needed.

K References, Standards, and Datasheets

This appendix provides the reference structure for standards, guidance documents, vendor datasheets, interface documents, safety records, and technical materials used to support the VectorSight-M proposal. Final references will be completed as the BOM, vendor selections, payload configuration, software stack, and verification plan are finalized.

Challenge and Program References

Reference Type	Description	Status
True North Precision challenge documentation	Solicitation, challenge statement, essential outcomes, desired outcomes, eligibility, submission requirements, and assessment expectations.	To be included
IDEaS program guidance	Program-level guidance, reporting expectations, proposal requirements, and contract-related documentation where applicable.	To be included
Submission and contract documents	Solicitation number, submission identifiers, forms, contract terms, data-rights requirements, reporting requirements, and deliverable requirements.	To be included

Standards and Guidance References

The following standards and guidance categories will be reviewed or referenced where applicable to the final prototype configuration, test environment, and contract requirements.

Standard / Guidance Area	**Project Relevance**
Use in Proposal	
Laser safety classification documentation	Supports Class 1 eye-safe laser rangefinder selection, installation control, and safe test procedures.
Laser safety file, test readiness. RPAS operating requirements and flight safety guidance	Supports safe flight-test planning, operational controls, test cards, emergency procedures, and flight authorization pathway.
Flight safety plan. ITSP.10.171-oriented data-handling guidance	Supports controlled handling of mission data, logs, test records, access control, storage, transfer, and exportable evidence.
Data-handling approach. STANAG 4586 alignment path	Supports future unmanned-system control and interface interoperability pathway.
Growth path reference. STANAG 4609 alignment path	Supports future full-motion-video metadata and FMV interoperability pathway.

Standard / Guidance Area	**Project Relevance**	**Use in Proposal**
		Growth path reference.
ATAK / Cursor-on-Target message concepts	Supports future structured target-message export and tactical situational-awareness integration pathway.	Interoperability pathway.
MGRS coordinate reporting references	Supports tactical coordinate output formatting and target-location reporting pathway.	Target cueing output.
Battery safety and lithium-polymer handling guidance	Supports battery handling, charging, storage, inspection, transport, and incident procedures.	Battery safety procedure.
Electrical power and wiring best practices	Supports power distribution, regulated rails, fusing, grounding, connectors, harnessing, inspection, and configuration control.	Power architecture and safety.
Configuration management and test documentation practices	Supports BOM control, firmware/software tracking, calibration records, issue tracking, and final technical data package.	Configuration control.

Vendor Datasheets and Technical Documents

Vendor datasheets and technical documents will be collected for critical components and included or referenced in the technical data package. The final list will be completed after vendor selection and procurement.

Document Category	Expected Documents	Status
Airframe and structure	Frame datasheet, mechanical drawings, payload-mount documentation, landing-gear documentation, material data where available.	TBD
Propulsion	Motor datasheets, propeller data, ESC datasheets, thrust data, thermal/current limits, integration notes.	TBD
Battery and charging	Battery datasheet, discharge rating, charger documentation, battery safety information, charging settings, storage guidance.	

Document Category **Status**	**Expected Documents**
TBD Flight controller and avionics	Flight-controller datasheet, GNSS/RTK datasheet, telemetry hardware documentation, antenna documentation, firmware/configuration references.
TBD EO/IR payload	EO/IR sensor datasheet, video interface documentation, payload mounting notes, environmental limits, software/interface documentation.
TBD Laser rangefinder	Laser rangefinder datasheet, Class 1 eye-safe documentation, electrical interface, mechanical interface, range performance, operating limits.
TBD Gimbal or stabilized mount	Stabilized mount datasheet, payload mass capacity, control interface, power requirements, pointing accuracy where available.

Document Category **Status**	**Expected Documents**
TBD Mission compute	Onboard compute datasheet, storage documentation, interface documentation, software environment, power and thermal limits.
TBD Communications and GCS	Radio datasheets, video-link documentation, ground-control station equipment documentation, antenna specifications, link-range data.
TBD Power distribution	Regulator datasheets, current/voltage sensor datasheets, fuse data, connector data, wiring specifications, transient-protection data.
TBD Test and measurement equipment	Calibration target documentation, measurement tool specifications, survey/ground-truth reference equipment where used.
TBD	

Software and Open-Source References

Software references will be maintained for all software, firmware, open-source elements, vendor tools, configuration files, mission-data tools, and export tools used in the project.

- Flight-controller firmware and configuration references.
- Ground-control software references.
- Mission-compute operating environment and software dependencies.
- Target-geolocation logic version records.
- Logging and data-export tool records.
- Open-source software names, versions, license types, and modification status.
- Vendor-provided software, firmware, and licensing restrictions.

Reference Control Notes

References, standards, and datasheets will be controlled through the project configuration-management process. Each cited datasheet or technical reference will be linked to the relevant BOM item, supplier record, country-of-origin record, software/firmware record, calibration record, or test procedure.

Any component substitution, software change, firmware update, or payload configuration change will require review of the affected references and update of the technical data package.

Appendix K Summary

The final reference package will support technical traceability, vendor readiness, safety review, configuration control, verification, and final assessment readiness. References will be finalized after component selection, procurement, integration, and test planning are complete.

L References, Standards, and Datasheets

This appendix provides the reference structure for standards, guidance documents, vendor datasheets, interface documents, safety records, and technical materials used to support the VectorSight-M proposal. Final references will be completed as the BOM, vendor selections, payload configuration, software stack, and verification plan are finalized.

Challenge and Program References

Reference Type	Description	Status
True North Precision challenge documentation	Solicitation, challenge statement, essential outcomes, desired outcomes, eligibility, submission requirements, and assessment expectations.	To be included
IDEaS program guidance	Program-level guidance, reporting expectations, proposal requirements, contracting requirements, and deliverable expectations where applicable.	To be included

Reference Type	Description	Status
Submission and contract documents	Solicitation number, submission identifiers, forms, contract terms, data-rights requirements, reporting requirements, and deliverable requirements.	To be included
Final capability assessment guidance	Assessment timing, prototype-readiness expectations, test participation requirements, reporting requirements, and final demonstration constraints.	To be included

Standards and Guidance References

The following standards and guidance categories will be reviewed or referenced where applicable to the final prototype configuration, test environment, and contract requirements.

Standard / Guidance Area	Project Relevance	Use in Proposal
Laser safety classification documentation	Supports Class 1 eye-safe laser rangefinder selection, installation control, operating limits, and safe test procedures.	

Standard / Guidance Area	Project Relevance	Use in Proposal
		Laser safety file, test readiness
RPAS operating requirements and flight safety guidance	Supports safe flight-test planning, operational controls, flight-test cards, emergency procedures, and flight authorization pathway.	Flight safety plan
ITSP.10.171-oriented data-handling guidance	Supports controlled handling of mission data, logs, test records, access control, storage, transfer, exportable evidence, and technical records.	Data-handling approach
STANAG 4586 alignment path	Supports future unmanned-system control and interface interoperability pathway.	Growth path reference
STANAG 4609 alignment path	Supports future full-motion-video metadata and FMV interoperability pathway.	Growth path reference

Standard / Guidance Area	Project Relevance	Use in Proposal
ATAK / Cursor-on-Target message concepts	Supports future structured target-message export and tactical situational-awareness integration pathway.	Interoperability pathway
MGRS coordinate reporting references	Supports tactical coordinate output formatting and target-location reporting pathway.	Target cueing output
Battery safety and lithium-polymer handling guidance	Supports battery handling, charging, storage, inspection, transport, and incident procedures.	Battery safety procedure
Electrical power and wiring best practices	Supports power distribution, regulated rails, fusing, grounding, connectors, harnessing, inspection, and configuration control.	Power architecture and safety

Standard / Guidance Area	Project Relevance
Use in Proposal	
Configuration management and test documentation practices	Supports BOM control, firmware/software tracking, calibration records, issue tracking, corrective-action tracking, and final technical data package.
Configuration control	

Vendor Datasheets and Technical Documents

Vendor datasheets and technical documents will be collected for critical components and included or referenced in the technical data package. The final list will be completed after vendor selection and procurement.

Document Category	Expected Documents
Status	
Airframe and structure	Frame datasheet, mechanical drawings, payload-mount documentation, landing-gear documentation, material data where available, and structural integration notes.
TBD	

Document Category	Expected Documents	Status
Propulsion	Motor datasheets, propeller data, ESC datasheets, thrust data, thermal/current limits, integration notes, and supplier recommendations.	TBD
Battery and charging	Battery datasheet, discharge rating, charger documentation, battery safety information, charging settings, storage guidance, and inspection procedure.	TBD
Flight controller and avionics	Flight-controller datasheet, GNSS/RTK datasheet, telemetry hardware documentation, antenna documentation, firmware references, and configuration records.	TBD

Document Category	**Expected Documents**	**Status**
EO/IR payload	EO/IR sensor datasheet, video interface documentation, payload mounting notes, environmental limits, software/interface documentation, and payload status interface details.	TBD
Laser rangefinder	Laser rangefinder datasheet, Class 1 eye-safe documentation, electrical interface, mechanical interface, range performance, operating limits, and safe-use notes.	TBD
Gimbal or stabilized mount	Stabilized mount datasheet, payload mass capacity, control interface, power requirements, pointing accuracy where available, and installation requirements.	TBD

Document Category	Expected Documents	Status
Mission compute	Onboard compute datasheet, storage documentation, interface documentation, software environment, power requirements, thermal limits, and configuration records.	TBD
Communications and GCS	Radio datasheets, video-link documentation, ground-control station equipment documentation, antenna specifications, link-range data, and configuration records.	TBD
Power distribution	Regulator datasheets, current/voltage sensor datasheets, fuse data, connector data, wiring specifications, transient-protection data, and grounding notes.	TBD

Document Category	Expected Documents	Status
Test and measurement equipment	Calibration target documentation, measurement tool specifications, survey/ground-truth reference equipment where used, and field-test support equipment.	TBD

Software and Open-Source References

Software references will be maintained for all software, firmware, open-source elements, vendor tools, configuration files, mission-data tools, and export tools used in the project.

- Flight-controller firmware and configuration references.
- Ground-control software references.
- Mission-compute operating environment and software dependencies.
- Target-geolocation logic version records.
- Logging and data-export tool records.
- Open-source software names, versions, license types, and modification status.
- Vendor-provided software, firmware, and licensing restrictions.

Reference Control Notes

References, standards, and datasheets will be controlled through the project configuration-management process. Each cited datasheet or technical reference will be linked to the relevant BOM item, supplier record, country-of-origin record, software/firmware record, calibration record, or test procedure.

Any component substitution, software change, firmware update, or payload configuration change will require review of the affected references and update of the technical data package.

Appendix I Summary

The final reference package will support technical traceability, vendor readiness, safety review, configuration control, verification, and final assessment readiness. References will be finalized after component selection, procurement, integration, and test planning are complete.